ESPAIS LATENTS

Xa Manzanares

LAB & TECH XPERIMENTS

```
#include <stdio.h>
int main() { printf("Aloha!\n"); return 0; }
```

. ..具.. .

Espais Latents

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#include <stdio.h>
int main() { printf("Aloha!\n"); return 0; }
```

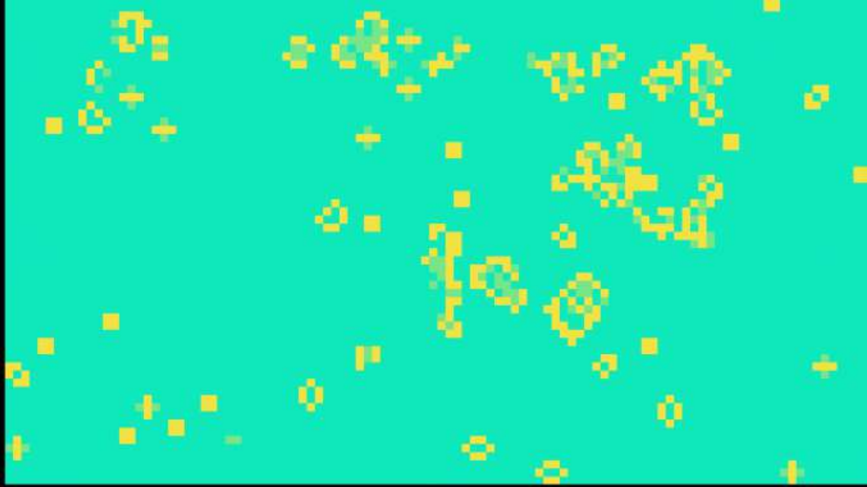
LAB Xperiments
& Tutorials

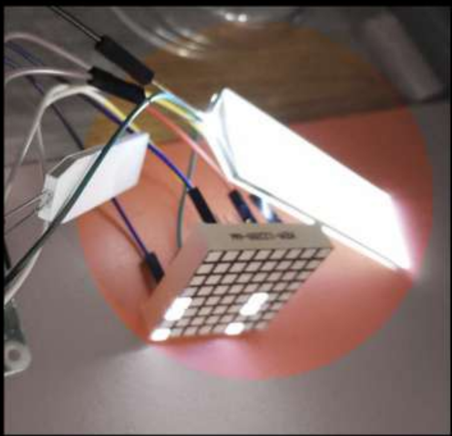
Documentació dels experiments i prototips de procés generats durant la recerca

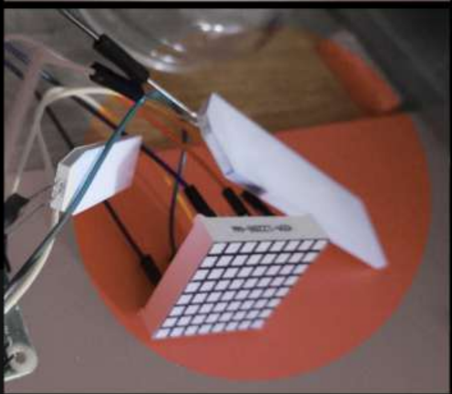
Xavi.Manzanares

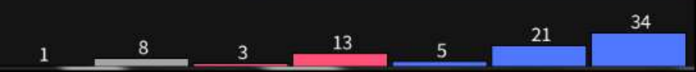
2025-26

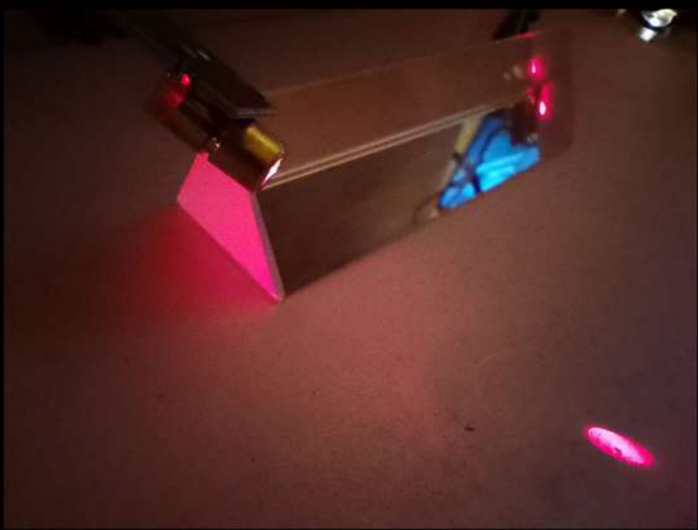

ONES ELECTROMAGNÈTIQUES
BOBINA / COIL
AUDIO OUTPUT
SONIFICATION

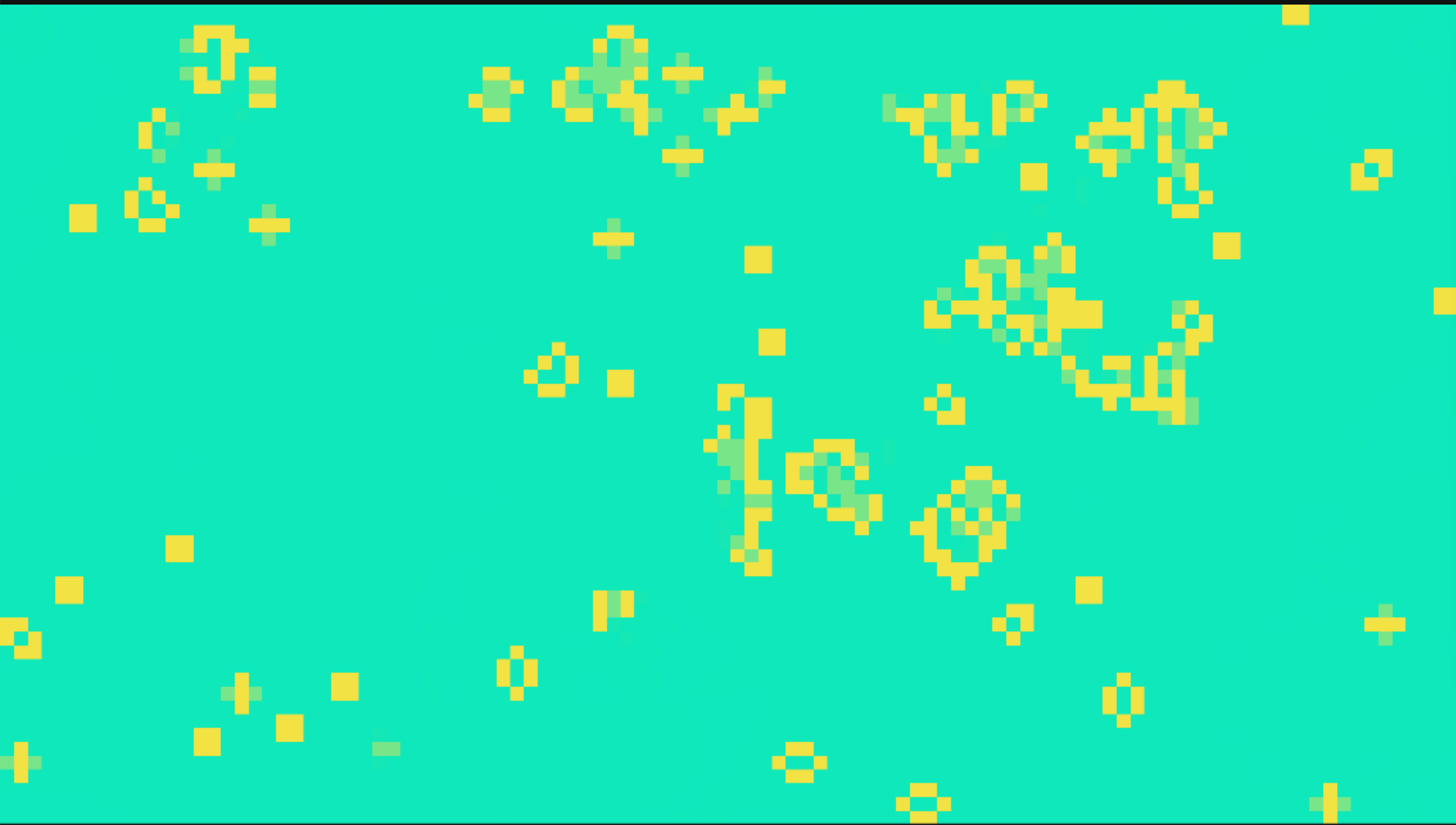







1
8
3
13
5
21
34





Espais Latents

LAB & TUTORIALS

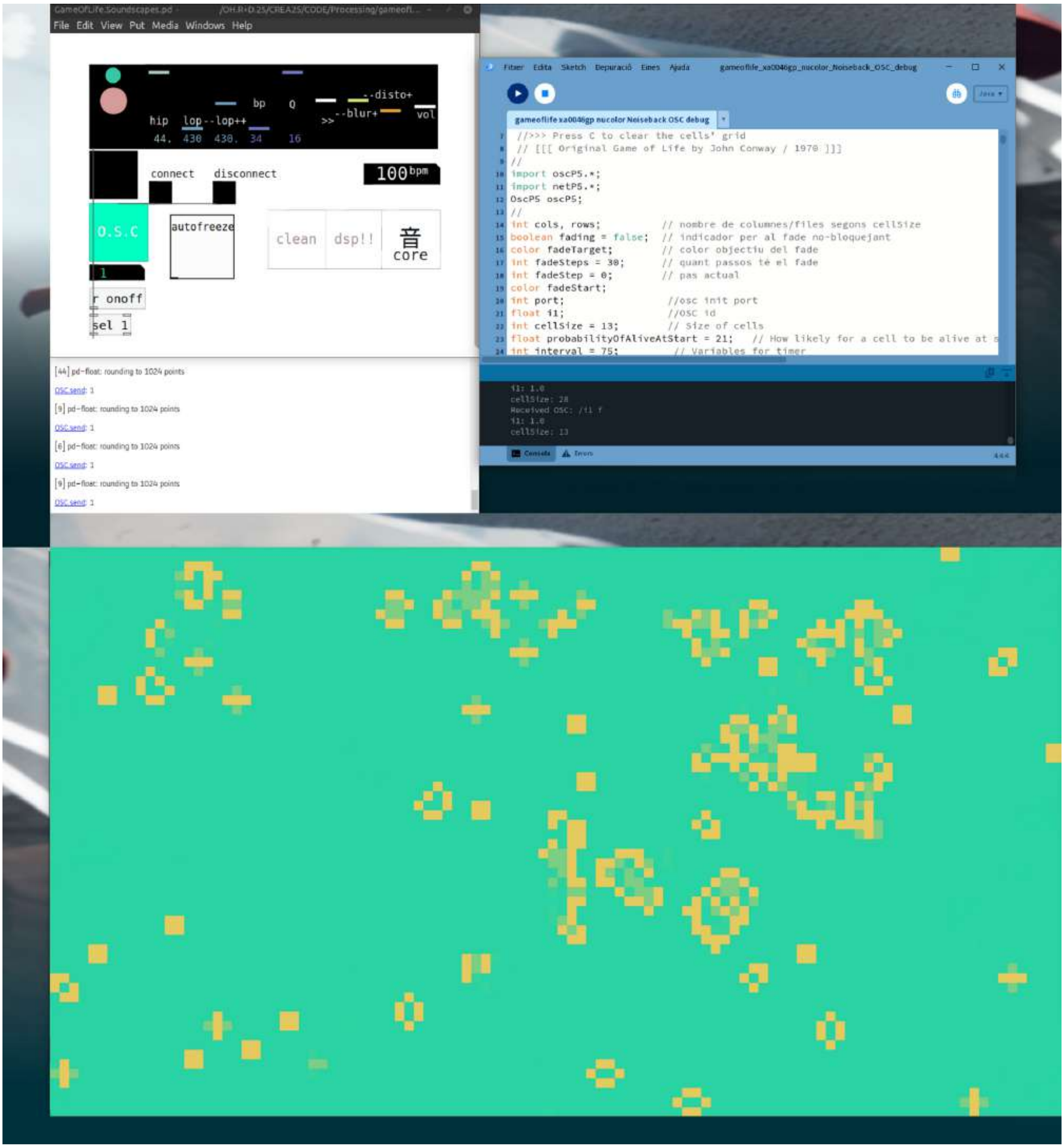
Li Quen

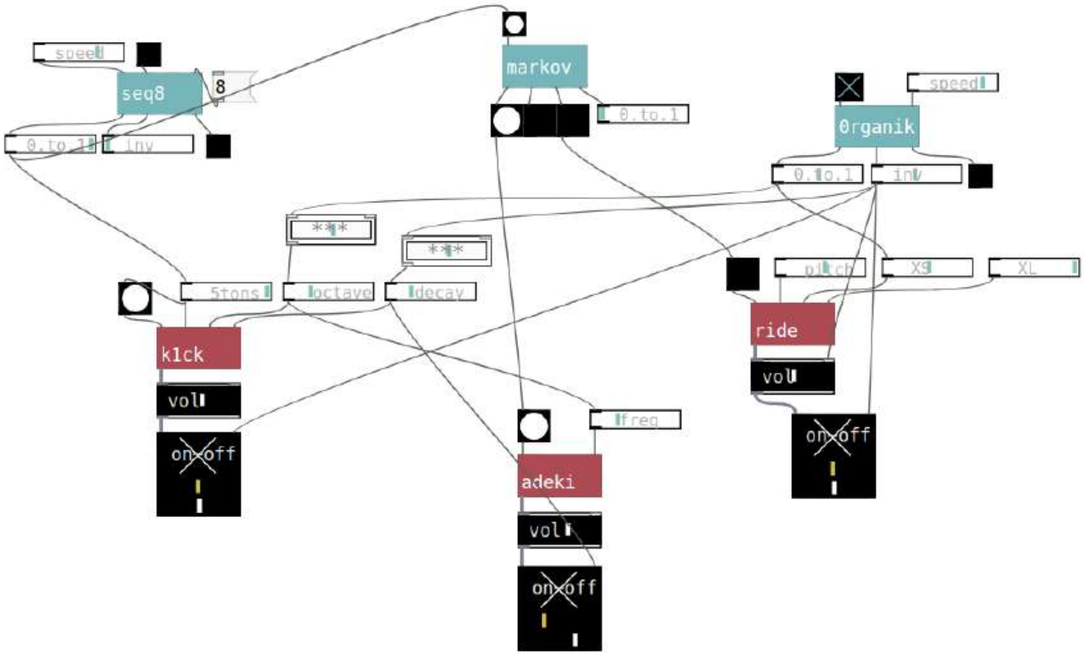
Dual Simbiotic Algorythm

Processing implementation of Game of Life
reacting back and forth
into a sonic algorythm.

: // Lí·Quen (aka Zero Playing Game)

Liquen es una re interpretació del clàssic game of life generador d autòmates cel.lulars.
Dos algoritmes s’entrecreuen fent un creixement evolutiu simbiòtic.
L algoritme de l autòmata cel.lular (programat en processing) és el que genera la capa visual.
L algoritme sonor (programat en pd/cells) és el que genera la capa sonora.
Diversos paràmetres s’envien entre l’un i l altre de manera que la evolució visual i sònica està en continua oscil.lació interactiva.





speed
seq8
8
0.to.1
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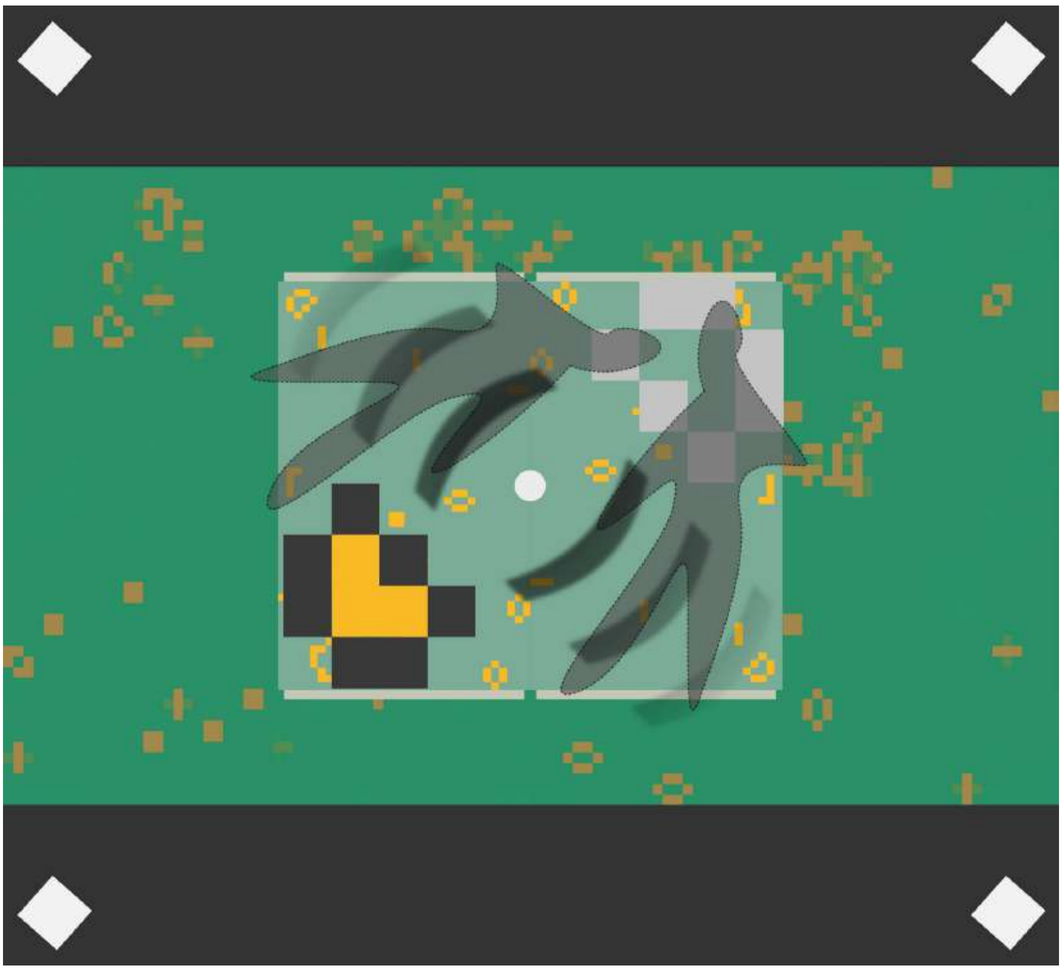


processing code >

```
//-_- Processing implementation of Game of Life -_-
// original code By Joan Soler-Adillon
// fork and OSC interaction by Xa Manzanares / debug tweaks with Generative Pretrained Transformer assistant
 //>>> Press SPACE BAR to pause and change the cell's values with the mouse
 //>>> On pause, click to activate/deactivate cells
 //>>> Press R to randomly reset the cells' grid
 //>>> Press C to clear the cells' grid
 // [[[ Original Game of Life by John Conway / 1970 ]]]
//
import oscP5.*;
import netP5.*;
OscP5 oscP5;
PGraphics noiseLayer;
//
int cols, rows;          // nombre de columnes/files segons cellSize
boolean fading = false;  // indicador per al fade no-bloquejant
color fadeTarget;        // color objectiu del fade
int fadeSteps = 30;      // quant passos té el fade
int fadeStep = 0;        // pas actual
color fadeStart;
int port;                //osc init port
float i1;                //OSC id
int cellSize = 13;       // Size of cells
float probabilityOfAliveAtStart = 21;   // How likely for a cell to be alive at start (in percentage)
int interval = 75;        // Variables for timer
int lastRecordedTime = 0;   // Variables for timer
color alive = color(222, 234, 222);   // Colors for active cells // Acid  color(255, 0, 115);
color dead = color(0, 255, 195);    // Colors for inactive cells // Acid color(0, 255, 195);
int[][] cells;            // Array of cells
// Buffer to record the state of the cells and use this while changing the others in the interations
int[][] cellsBuffer;
boolean pause = false;     // Pause

        //:::::.:.:::.....:·······::::···:::.:...::.:..:.::...·.··..·:::.·..

void setup() {
  size (1600, 900);
  noiseLayer = createGraphics(width, height);
  generateNoiseBackground (color ( int(random(0, 84)), int(random(144, 256)), int(random(166, 233)) )); // inicial
  // ·········· Instantiate arrays
  cells = new int[width/cellSize][height/cellSize];
  cellsBuffer = new int[width/cellSize][height/cellSize];
  //stroke(10);    // This stroke will draw the background grid
  noSmooth();
  // ··········· Initialization of cells
  for (int x=0; x<width/cellSize; x++) {
    for (int y=0; y<height/cellSize; y++) {
      float state = random (random (100));
      if (state > probabilityOfAliveAtStart) {
        state = 0;
      } else {
        state = 1;
      }
      cells[x][y] = int(state); // Save state of each cell
    }
  }
  //background(0, 255, 195);   // Fill in black in case cells don't cover all the windows
  //stroke (0, 255, 195);     // Fill in black in case cells don't cover all the windows
  fadeStart = get(0,0);
  cols = width / cellSize;
  rows = height / cellSize;
  ////OSCs
  port = 11111; /////////////! edit port
  oscP5 = new OscP5(this, port);
}

        //:::::.:.:::.....:·······::::···:::.:...::.:..:.::...·.··..·:::.·..

void draw() {
  // --- Fons de soroll (sense fade) ---
  generateNoiseBackground(fadeTarget);

  // Assegura sincronització amb arrays
```

```
  cols = min(width / cellSize, cells.length);
  rows = min(height / cellSize, cells[0].length);

  // Dibuixa les cèl·lules
  noStroke();
  for (int x = 0; x < cols; x++) {
    for (int y = 0; y < rows; y++) {
      fill(cells[x][y] == 1 ? alive : dead);
      rect(x * cellSize, y * cellSize, cellSize, cellSize);
    }
  }

  // Iteració temporal
  if (millis() - lastRecordedTime > interval) {
    if (!pause) {
      iteration();
      lastRecordedTime = millis();
    }
  }

  // Mode pausa amb edició manual
  if (pause && mousePressed) {
    int xCellOver = constrain(int(mouseX / cellSize), 0, cols - 1);
    int yCellOver = constrain(int(mouseY / cellSize), 0, rows - 1);
    cells[xCellOver][yCellOver] = 1 - cells[xCellOver][yCellOver];
  } else if (pause) {
    arrayCopy(cells, cellsBuffer);
  }
}


      //:::::.:.:::.....:·······::::···:::.:..::.:..:.::....·..··..·:::.·..

void iteration() { // When the clock ticks
// Evita error si arrays no estan sincronitzats
  cols = min(cols, cells.length);
  rows = min(rows, cells[0].length);
  // Copiem l'estat actual a buffer
  for (int x = 0; x < cols; x++) {
    for (int y = 0; y < rows; y++) {
      cellsBuffer[x][y] = cells[x][y];
    }
  }
  // Visitem cada cel·la:
  for (int x = 0; x < cols; x++) {
    for (int y = 0; y < rows; y++) {
      // Iniciem el comptador de veïns per a aquesta cel·la
      int neighbours = 0;
      // Mirem tots els veïns
      for (int xx = x - 1; xx <= x + 1; xx++) {
        for (int yy = y - 1; yy <= y + 1; yy++) {
          // comprovem límits
          if (xx >= 0 && xx < cols && yy >= 0 && yy < rows) {
            // excloem la cel·la mateixa
            if (!(xx == x && yy == y)) {
              if (cellsBuffer[xx][yy] == 1) {
                neighbours++;
              }
            }
          }
        }
      }
      // Apliquem les regles del Game of Life
      if (cellsBuffer[x][y] == 1) { // si està viva
        if (neighbours < 2 || neighbours > 3) {
          cells[x][y] = 0; // mor
        } else {
          cells[x][y] = 1; // sobreviu (opcional per claredat)
        }
      } else { // si està mort
        if (neighbours == 3) {
          cells[x][y] = 1; // neix
        } else {
          cells[x][y] = 0; // segueix mort (opcional per claredat)
        }
      }
    } // end y
```

```
  } // end x
}

          //::::.:.:::.....:·······::::···:::.:..::.:..:.::....·.··..·:::.·..

void oscEvent(OscMessage theOscMessage) {
  println("Received OSC: " + theOscMessage.addrPattern() + " " + theOscMessage.typetag());
  // Comprova que hi ha almenys un argument
  if (theOscMessage.arguments().length > 0) {
    // Detecta si l'adreça és la que volem
    if (theOscMessage.checkAddrPattern("/i1")) {
      // Mira si el tipus del primer argument és integer
      char type = theOscMessage.typetag().charAt(0);
      if (type == 'f') {
        i1 = theOscMessage.get(0).floatValue();
        println("i1: " + i1);
        if (i1 == 1) {
          restartCells();
        //if (i1 == 0) {
          //suspendCells();
        }
      } else {
        println("!!! Argument type not integer: " + type);
      }
    } else {
      println("!!! Different OSC address: " + theOscMessage.addrPattern());
    }
  } else {
    println("!!! OSC message without arguments!");
  }
}
          //::::.:.:::.....:·······::::···:::.:..::.:..:.::....·.··..·:::.·..

void keyPressed() {
  if (key=='r' || key == 'R') {
    // Restart: reinitialization of cells
    for (int x=0; x<width/cellSize; x++) {
      for (int y=0; y<height/cellSize; y++) {
        float state = random (100);
        if (state > probabilityOfAliveAtStart) {
          state = 0;
        }
        else {
          state = 1;
        }
        cells[x][y] = int(state); // Save state of each cell
      }
    }
  }
  if (key==' ') { // On/off of pause
    pause = !pause;
  }
  if (key=='c' || key == 'C') { // Clear all
    for (int x=0; x<width/cellSize; x++) {
      for (int y=0; y<height/cellSize; y++) {
        cells[x][y] = 0; // Save all to zero
      }
    }
  }
}

          //::::.:.:::.....:·······::::···:::.:..::.:..:.::....·.··..·:::.·..

void generateNoiseBackground(color base) {
  noiseLayer.beginDraw();
  noiseLayer.noStroke();
  for (int x = 0; x < width; x += cellSize) {
    for (int y = 0; y < height; y += cellSize) {
      float r = red(base) + random(-10, 10);
      float g = green(base) + random(-10, 10);
      float b = blue(base) + random(-10, 10);
      noiseLayer.fill(constrain(r, 0, 255), constrain(g, 0, 255), constrain(b, 0, 255));
      noiseLayer.rect(x, y, cellSize, cellSize);
    }
  }
  noiseLayer.endDraw();
}
```

```
        //:::::.:.:::.....:·······::::····:::.:..::.:..:.::.....·.··..·:::.·..

void restartCells() {
  // Interval fix i selecció robusta de cellSize
  interval = 135; // interval = 135; 111bpm based      // interval = 150; 100bpm based // 150 OK
  //int[] fibSizes = {13, 20};  // més estables visualment {13, 28, 20};
  //cellSize = fibSizes[int(random(fibSizes.length))];
  cellSize = 15; /// 13 ok render
  //println("cellSize: " + cellSize);

  // Recalcula i recrea arrays
  cols = max(1, width / cellSize);
  rows = max(1, height / cellSize);
  cells = new int[cols][rows];
  cellsBuffer = new int[cols][rows];

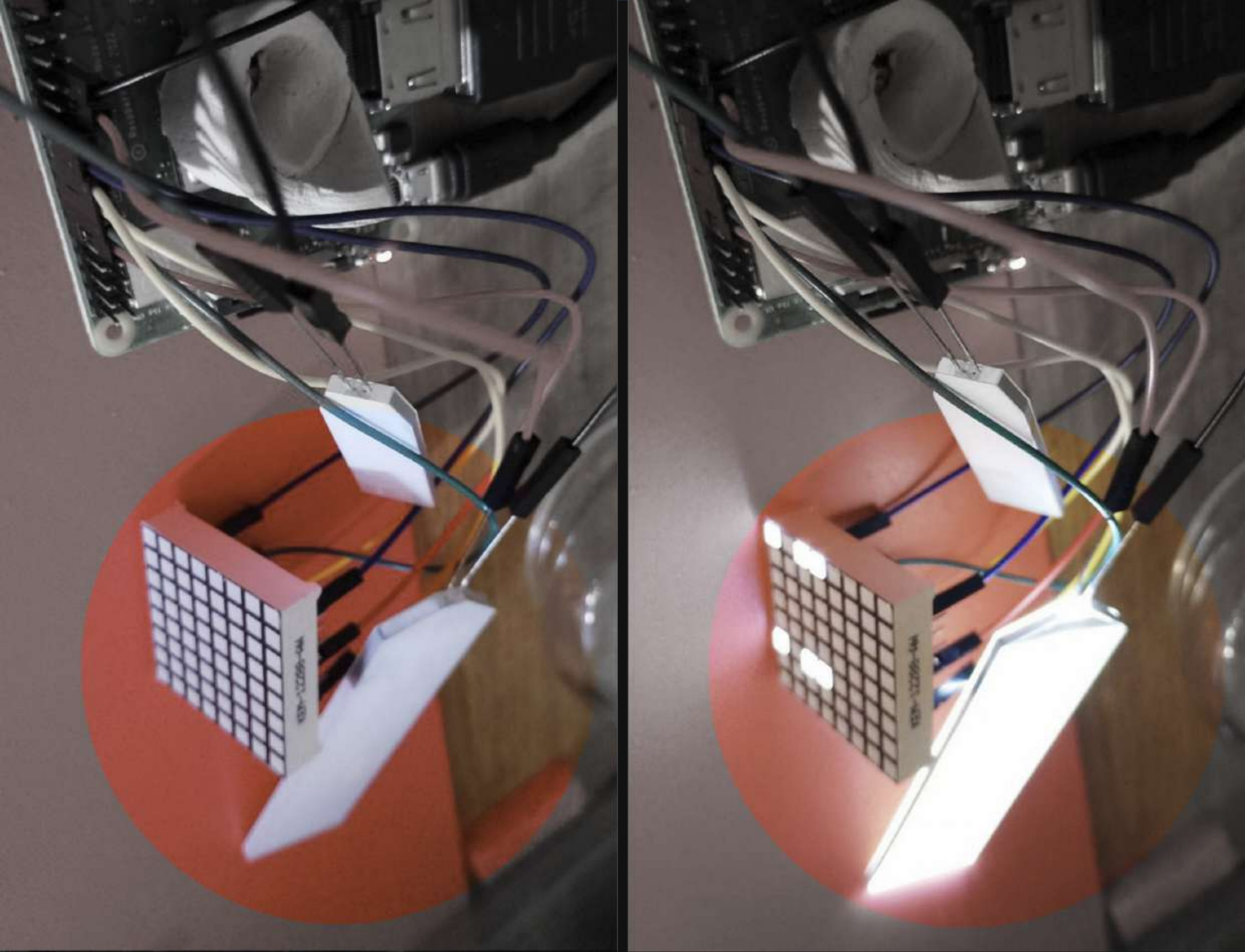
  // Inicialitza estat inicial
  for (int x = 0; x < cols; x++) {
    for (int y = 0; y < rows; y++) {
      cells[x][y] = random(100) > probabilityOfAliveAtStart ? 0 : 1;
    }
  }

  // Colors nous instantanis (sense fades) // Acid color(0, 255, 195);
    fadeTarget = color(int(random(5, 34)), int(random(181, 232)), int(random(155, 166))); // fadeTarget = color(int(random(0, 84)), int(random(144, 256)), int(random(166, 233)));
   alive = color( int(random(200, 232)), int(random(200, 232)), int(random(0, 83))); // alive = color(int(random(166, 234)), int(random(0, 56)), int(random(55, 145)));
   dead = color(int(random(5, 34)), int(random(181, 232)), int(random(155, 166)), int(random(83, 98))); // dead = color(int(random(166, 256)), int(random(55, 145)), int(random(0, 56)), int(random(13, 253)));
    //fadeTarget = color(int(random(5, 34)), int(random(181, 232)), int(random(155, 166))); // fadeTarget = color(int(random(0, 84)), int(random(144, 256)), int(random(166, 233)));
   //alive = color( int(random(13, 34)), int(random(144, 202)), int(random(55, 144))); // alive = color(int(random(166, 234)), int(random(0, 56)), int(random(55, 145)));
   //dead = color(int(random(5, 34)), int(random(181, 232)), int(random(155, 166)), int(random(83, 98))); // dead = color(int(random(166, 256)), int(random(55, 145)), int(random(0, 56)), int(random(13, 253)));

  // Reinicia rellotge
  lastRecordedTime = millis();
}


      //:::::.:.:::.....:·······::::····:::.:..::.:..:.::.....·.··..·:::.·..

void fadeToBackground(color targetColor, int steps) {
  //color startColor = get(0, 0); // Agafa el color actual del canvas
  for (int i = 0; i <= steps; i++) {
    //float amt = map(i, 0, steps, 0, 1);
    //color c = lerpColor(startColor, targetColor, amt);
    color c = color(int(random(5, 21)), int(random(34, 55)), int(random(34, 83)), int(random(55, 83)));
    fill(c);
    noStroke();
    rect(0, 0, width, height);
    delay(270); // controla la durada total del fade // 150 OK
    //interval = 135; 111bpm based      // interval = 150; 100bpm based
  }
}
```



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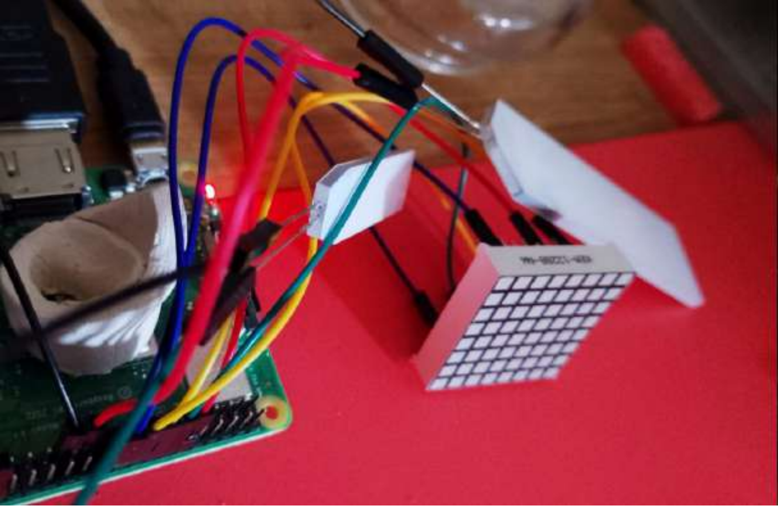
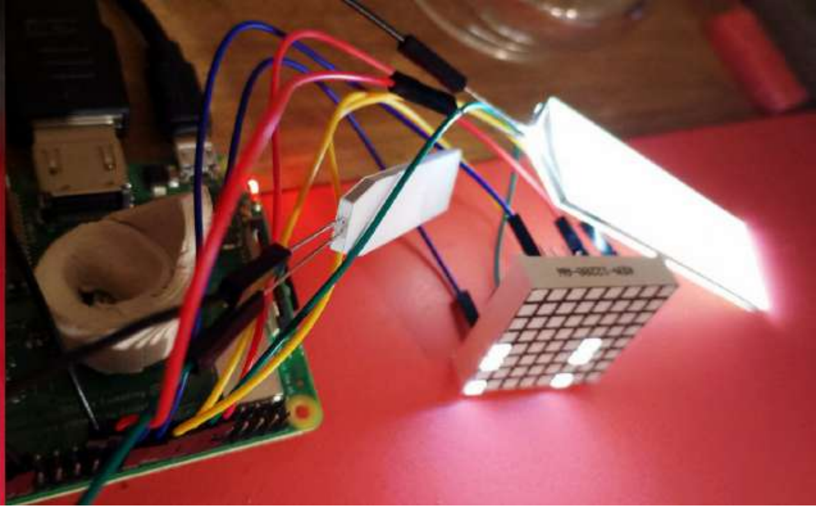
Espais Latents

LAB & TUTORIALS

How to manage GPIO ports through pd in Raspberry Pi

How to manage GPIO ports through pd in Raspberry Pi

by Xavi·Manzanares // Github & IG @xamanza // 2025

Several projects ago i did connect the GPIO ports of RaspberryPi directly from Pd (puredata), with the library WiringPi. Due to the fact that this library was written for Raspberry Versions 2 and 3 and is now deprecated, there is another method very useful which will be described below.

With this method you can control leds or other output triggers (such as motors or other actuators through GPIO ports. In this case GPIo ports are digital which means that allows a 0 / 1 value and therefore you cannot control the velocity of a motor for instance.
<u>Instead, is very useful to just trigger light events related to the Audio Generative Algorythms that are running</u>. Therefore is interesting to create standalone instruments and other luthieristic applications where we don't want the Visualization/monitor/projector layer (avoiding computation for this purpose and focusing it for audio render computation).

This means that the Raspberry will become an **Automat, and a Sonic·Automat or Autonomous instrument if the app is a pd patch.**

In this sense we need several steps, such as :

0 // Installing Raspbian OS
1 // Installing puredata
2// Compiling gpio external (which will control GPIO ports)
3 // Making your pd application.
Check gnrtv.cells in order to build faster your sonic algorythms.
4 // Making an autostart bash script for your pd app.

0

Installing Raspbian OS

Download Raspbian OS

Raspbian OS > https://downloads.raspberrypi.com/raspios_arm64/images/raspios_arm64-2025-
(recomended Software Imager to directly format Raspbian OS .img into a micro SDcard 16gb or 32gb

note : in case RPI 3A > 32 bits // In case Rpi 4 or 5 > 64 bits

Imager >

from linux OS https://downloads.raspberrypi.org/imager/imager_latest_amd64.deb

from win OS https://downloads.raspberrypi.org/imager/imager_latest.exe

from mac OS https://downloads.raspberrypi.org/imager/imager_latest.dmg

1

Installing puredata

Launch the Rpi with a keyboard mouse and monitor.
Setup first the last updates :
open a terminal and run

sudo apt update

Installing puredata (Pd vanilla version)
Obrir terminal i executar
open a terminal and run

sudo apt-get install puredata

open a terminal and run

puredata

Then check out if the system recognized the audio output > set in the OS bar (volume) if the outpout is in the minijackport, not the HDMI port. Swith off this port to avoid undesired setting ups.

In pd Go to menú media and setup the IO audio ports
input device PCH hardware channels 1
output device PCH hardware channels 2

Then make **apply** and **save all settings**.
After that go to ***menú media > test audio and midi.***
In the left part of the patch that will appear there is a Radio button selector with 3 'steps'.
By default is at 'off' position'. Cick on the 80 db step.
If all setting up was fine, then an audio SineWave of 440hz will appear, which means Pd is running ok in the system.
note remind connect your Rpi's minijack to a speaker or headphone.

2

Compiling gpio external (which will control GPIO ports)

After pd is being installed, lets compile a C script in order to build a pd external.
This means that the C code will be compiled into a .pd_linux file which will be our external library to use it within pd to control GPIO ports directly*. This will require an additional description of the pd library(external) and its path, but lets make step by step.
Before start all this process, lets make the context: the OS library pigpio is required for this method.

*controlling GPIO ports directly from pd is a very powerful method to avoid overengineering among different applications and therefore more simplicity in a compact way, which will be ligther from the computational perspective.

2 . 0

Install different libs and tools to setup the OS
Open a terminal and run :

sudo apt install pigpio

after this process has ended, run
sudo apt install puredata-dev

2 . 1

Create a directory called *gpio_out*

You can create it for example on the desktop directly via mouse with right button > create dir
or via terminal with :

mkdir /home/pi/Desktop/gpio_out

2 . 2

Open an editor (on Raspberry mousepad or leafpad) and write the following C code :

```
#include "m_pd.h"
#include <pigpio.h>

static t_class *gpio_out_class;

typedef struct _gpio_out {
    t_object  x_obj;
    int gpio_pin;
} t_gpio_out;

void gpio_out_float(t_gpio_out *x, t_floatarg f) {
    if (f != 0)
        gpioWrite(x->gpio_pin, 1);
    else
        gpioWrite(x->gpio_pin, 0);
}

void *gpio_out_new(t_floatarg f) {
    t_gpio_out *x = (t_gpio_out *)pd_new(gpio_out_class);
    x->gpio_pin = (int)f;
    gpioSetMode(x->gpio_pin, PI_OUTPUT);
    return (void *)x;
}

void gpio_out_free(t_gpio_out *x) {
    gpioWrite(x->gpio_pin, 0);  // Turn off pin on free
}

void gpio_out_setup(void) {
    if (gpioInitialise() < 0) {
        post("pigpio init failed");
        return;
    }
    gpio_out_class = class_new(gensym("gpio_out"),
                           (t_newmethod)gpio_out_new,
                           (t_method)gpio_out_free,
                           sizeof(t_gpio_out),
                           CLASS_DEFAULT,
                           A_DEFFLOAT, 0);
    class_addfloat(gpio_out_class, gpio_out_float);
}
```

save it in /home/pi/Desktop/gpio_out as
gpio_out.c

2 . 3

Open an editor (in Raspberry mousepad or leafpad) and write the following code:

```
PD_INCLUDE = /usr/include/pd
PD_LIB = /usr/lib/pd
CFLAGS = -Wall -Wextra -O2 -fPIC -I$(PD_INCLUDE)
LDFLAGS = -shared -lpigpio -lrt -lpthread

all: gpio_out.pd_linux

gpio_out.pd_linux: gpio_out.c
	$(CC) $(CFLAGS) -o $@ $^ $(LDFLAGS)

clean:
	rm -f *.o *.pd_linux
```

in the same folder save it as
Makefile
Makefile will be responsible for compiling the previous one with the 'headers' and other internal elements.

Close the editors
Check that the files have execution permissions (you can check directly on each file with the right button 'permissions'). Make sure all users has read and write permissions and the execution check is enabled.

Once these files are created,
open a terminal and go to their location, in this case:

cd /home/pi/Desktop/gpio_out

If you do a list (ls) you should see the two files we created previously. (or you can check it via OS GUI directly via mouse).

ls
gpio_out.c Makefile

At this point we are ready to compile. Type in the terminal

make

If the compilation went ok, it will create the file :

gpio_out.pd_linux

2 . 4

Note: Since we are controlling the hardware, we will probably have to start Pd with sudo (administrative permissions)
Open puredata and include the library >

sudo puredata

open File menu and edit (edit preferences) the library paths in ‘path’ and ‘startup’
Save changes.

Create a new patch (Ctrl+N) with the following ‘object’ > (ctrl+1)

[_] toggle
|
[gpio_out 17] object

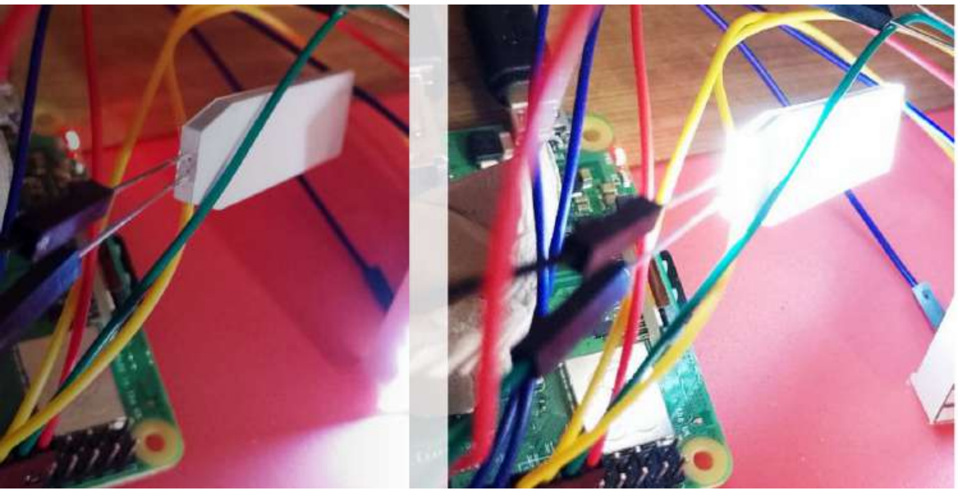
gpio_out and the GPIO port number (note that this ID does not correspond explicitly for architectural and design reasons but you can check it with the following table.

Therefore the (external) object that we have written
[gpio_out 17]
corresponds to GPIOs physical pin 11
(GPIO diagram for Rpi 3A / 3b / 4 / 5)

Lets test an example :

Create a new patch and write the following pd code*.

[_] toggle
 |
[metro 1000] object
 |
[_] toggle
 |
[gpio_out 17] object



Connect a LED to
ports 9 (GND) and 11

LED GND is the shorter leg

In case all your system compiled ok gpio_out and is properly set up in the Pd path library...

....we will have a blinking led every 1000ms

*blinking example pd code : copy this code in an empty file and save it as blink.pd

```
#N canvas 585 309 732 554.125 12;
#X obj 86 34 tgl 31 0 empty empty empty 17 7 0 10 #fcfcfc #000000 #000000
0 1;
#X obj 86 73 metro 1000;
#X obj 86 140 gpio_out 17;
#X obj 86 101 tgl 31 0 empty empty empty 17 7 0 10 #fcfcfc #000000
#000000 0 1;
#X connect 0 0 1 0;
#X connect 1 0 3 0;
#X connect 3 0 2 0;
```

3

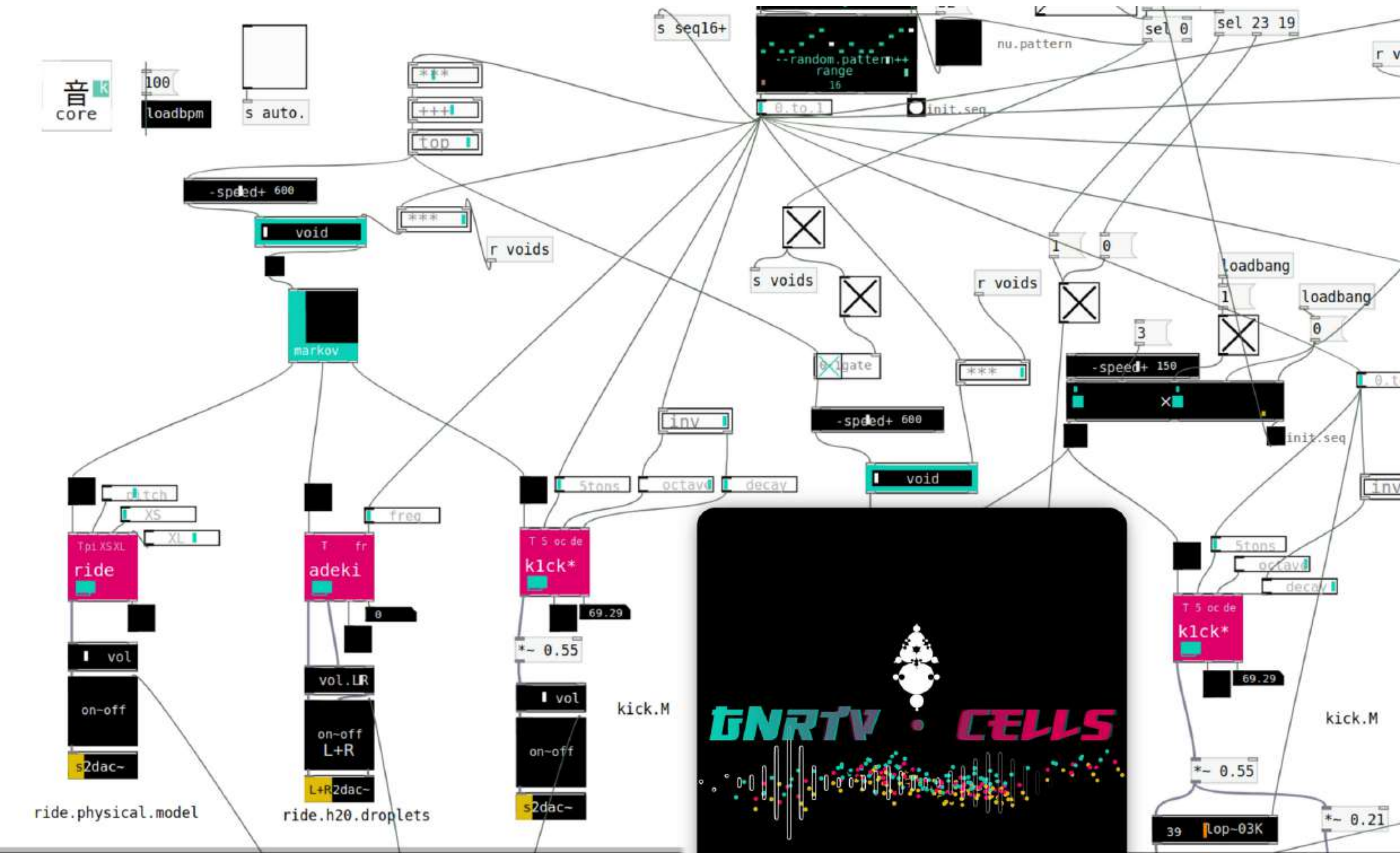
Making your pd application.

This point requires pd programming skills in order to make your desired instrument.
Loadbang objects will be very useful to setup all values you want to launch at startup.

copy your patch (for example via pendrive) to a specific location, for example the desktop
If you do not change the name of the machine by default it is called pi
/home/pi/Desktop/mypatch.pd

I remind that you can use gnrtv.cells* toolkit which is a set of modules designed in Pd in order to easily build generative algorythms.

*https://github.com/xamanza/gnrtv.cells.25_v.edu



4

Making an autostart bash script for your pd app.

4 . 1

/home/pi/.config/autostart/pd.desktop

(assuming pi is your main pi user (can be changed, this is the default name.)
In case is not created, create the autostart directory at /home/pi/.config/
note .config is a hidden directory, you can view it with Control+H (hide)
By default the autostart folder is already created, otherwise you can create it.

Once autostart is created, create a file called pd.desktop at /home/pi/.config/autostart
Open a text editor (in the Raspbian system menu you will find it for example 'leafpad' 'mousepad' or other editors) and copy the following content:

```
[Desktop Entry]
Type=Application
Name=pd.desktop
Exec=/home/AoT/startpd.sh
Terminal=true
```

As expected youll have the file Pd.app that appears in the following path :
/home/pi/.config/autostart/pd.desktop

4 . 2

/home/pi/startpd.sh

Create a bash script to launch the application. Due to the fact that managing GPIo ports usually requires sudo (admin) permissions, the previous script at autostart redirects into this script in order to upboost the sudo / admin privileges.
Create the bash script file at /home/pi you can create it within the same GUI via mouse > right button create file.
Copy the following content :

```
#!/bin/bash
sleep 5
lxterminal -e "sudo /usr/bin/puredata /home/AoT/Desktop/AoT__TinyBugsGPIO.TST.001.pd"
```

Save it as startpd.sh
It must appear the file in this location :
/home/pi/startpd.sh

4 . 3

make the previous files executable, in terminal with the folllowing commands :

```
chmod +x /home/pi/startpd.sh
```

and :

```
chmod +x /home/pi/.config/autostart/pd.desktop
```

5

Open the patch and setup the soundcard (internal or USB card) in Pd.
In case you want to use an external sound card, Rasbian will always require a ClassCompliant SoundCard.*

**(for example SoundBlaster Play3! which is a nice and affordable option to improve the sound render of your algorythms).*

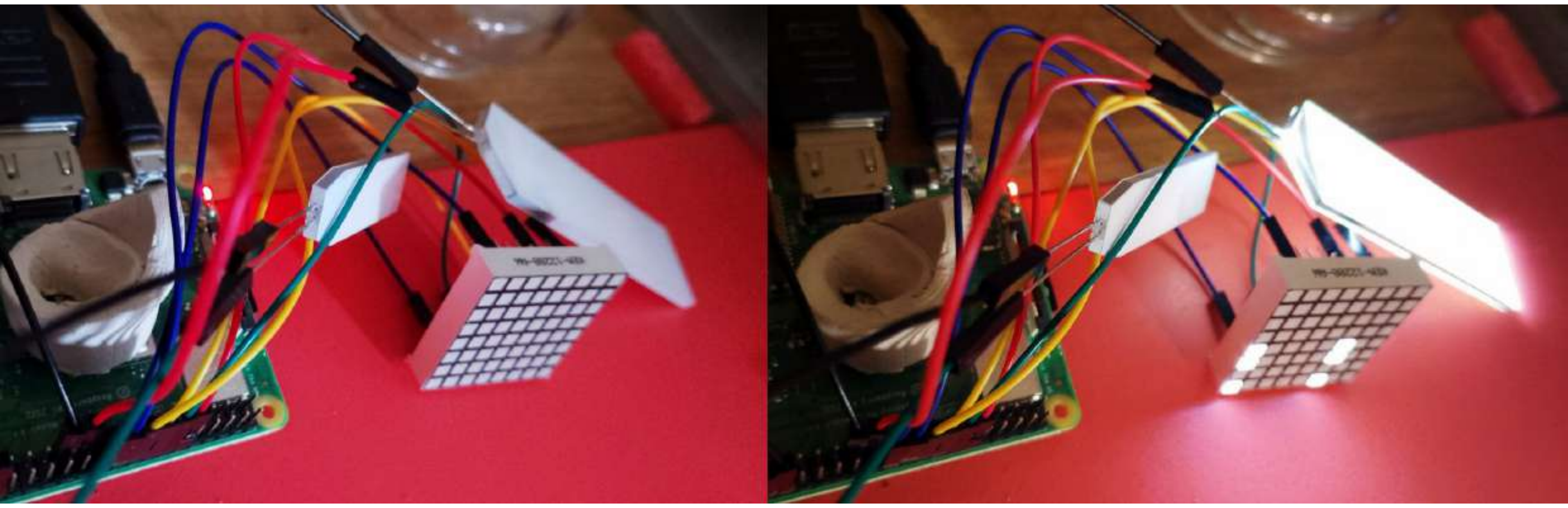
Pd SoundBlasterPlay3's Setting up needs one channel (as a input) and the two channels as output.

Once configured, Save to ensure that it always finds the card when it starts automatically.

Make reboot of the system and check that all is running.

In case not, go back to the point 4 of this tutorial and check all steps are ok.

*But in case is running...you will have an automat working ^_^*



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Espais Latents

LAB & TUTORIALS

Hum electromagètic :
Sonificant ones electromagètiques

: // Hum Electromagnètic

Dispositiu per a captar sons electromagnetics, i poder-los integrar als teus directes o instal.lacions. Inspiració i referència > Soma Ether https://somasynths.com/ether/

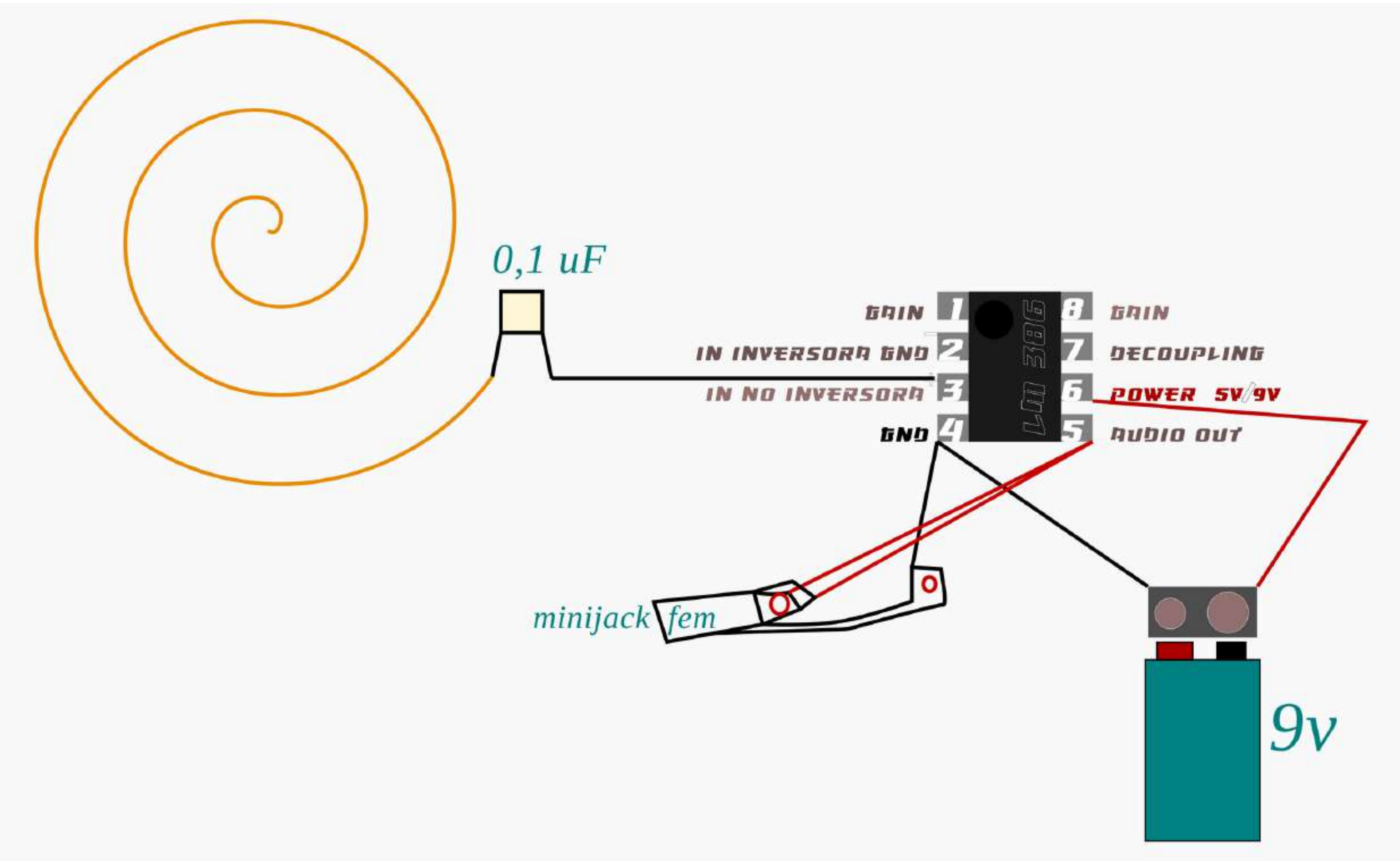
Components necessaris:

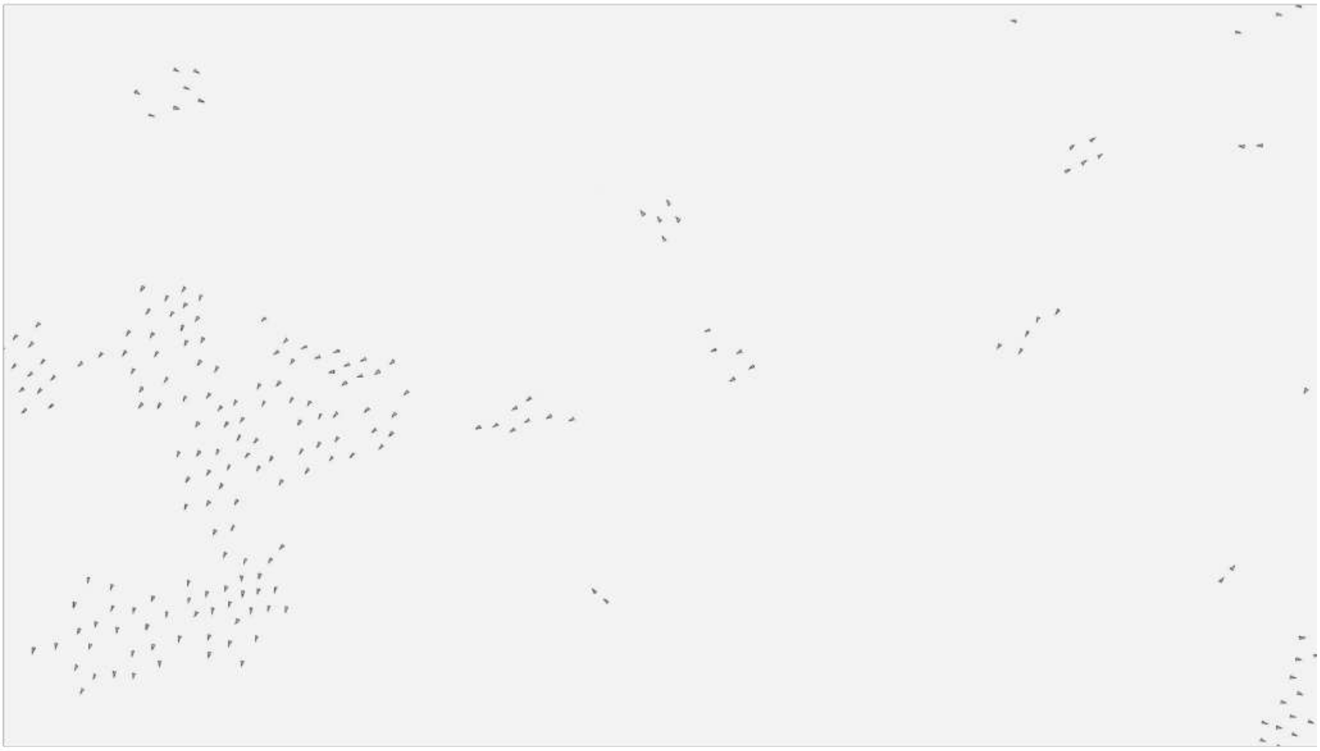
1. Antena:
   - Fil de coure esmaltat (0.2-0.5 mm de diàmetre, 5-10 m) per fer una bobina o una antena AM comercial.
2. Amplificador:
   - IC LM386 + condensadors per al guany.
3. Filtratge:
   - Condensadors: 0.1 µF i 10 µF.
   - Resistències: 10 kΩ i 1 kΩ.
4. Sortida d'àudio:
   - mini Jack 3.5 mm TRS.
5. Alimentació:
   - Porta-piles (9V o 3 piles AA amb suport) o un mòdul USB de 5V.
6. Caixa i altres:
   - Placa perforada per muntar components.
   - Carcassa metàl·lica petita o plàstica amb folre conductor.
   - Cablejat (flexible).

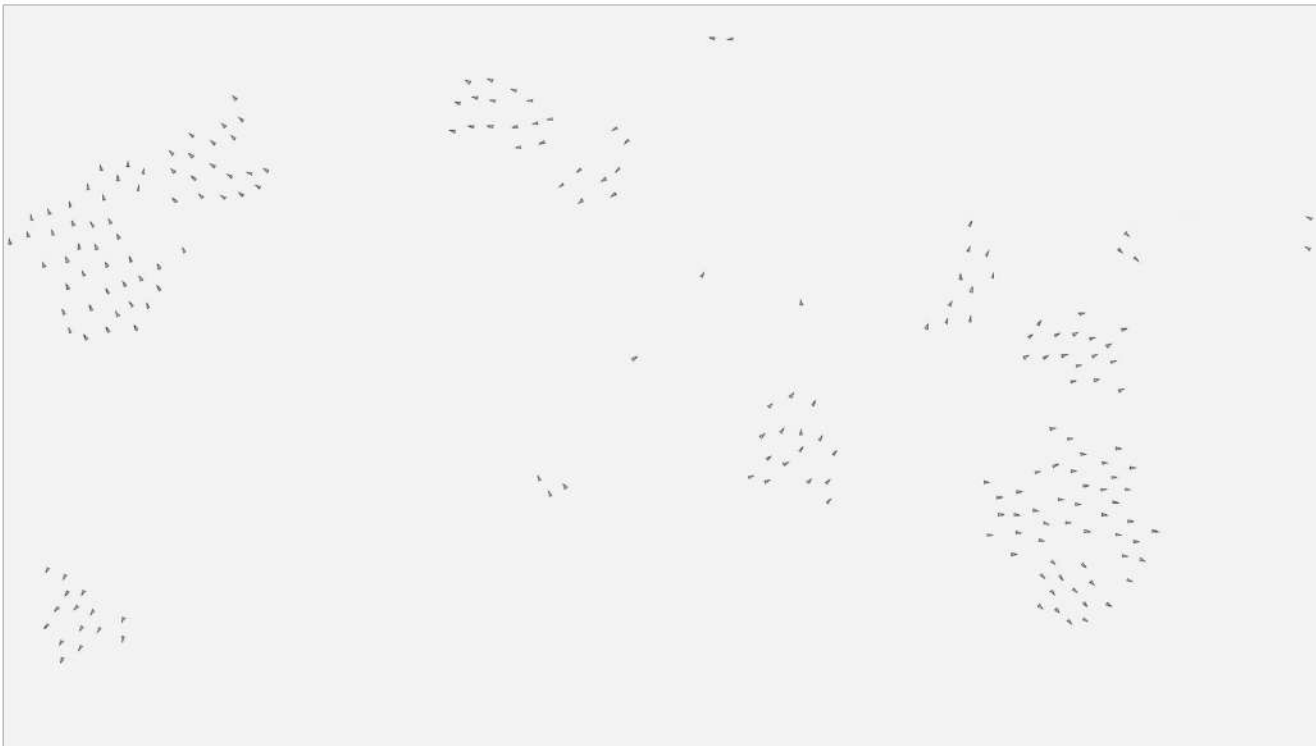

BOBINA / COIL
ONES ELECTROMAGNÈTIQUES
AUDIO OUTPUT
SONIFICATION

Aquí tens un esquema senzill del circuit per a captar sons electromagnètics. Les connexions principals són:

1. **Antena** (en blau): Connectada a l'entrada del LM386 (**pin 3**).
2. **Alimentació** (en vermell): 9V connectats al **pin 6** del LM386.
3. **Terra/GND** (en marró): **Pin 4** connectat a terra.
4. **Sortida a l'àudio** (en verd): **Pin 5** connectat al condensador de 0.1 µF i després al jack d'àudio.
5. **Condensador de guany** (en lila): **10 µF entre els pins 1 i 8** per ajustar el guany.







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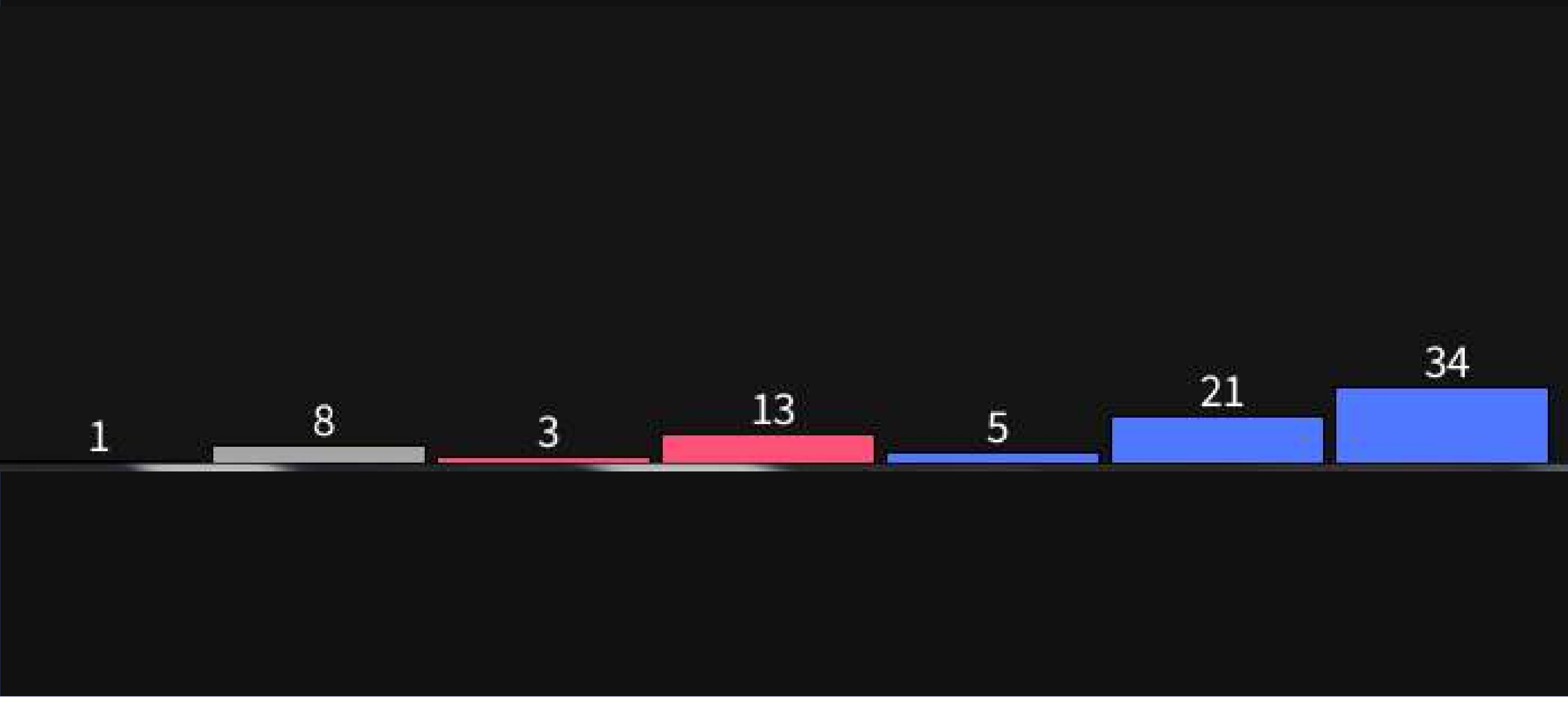
Espais Latents

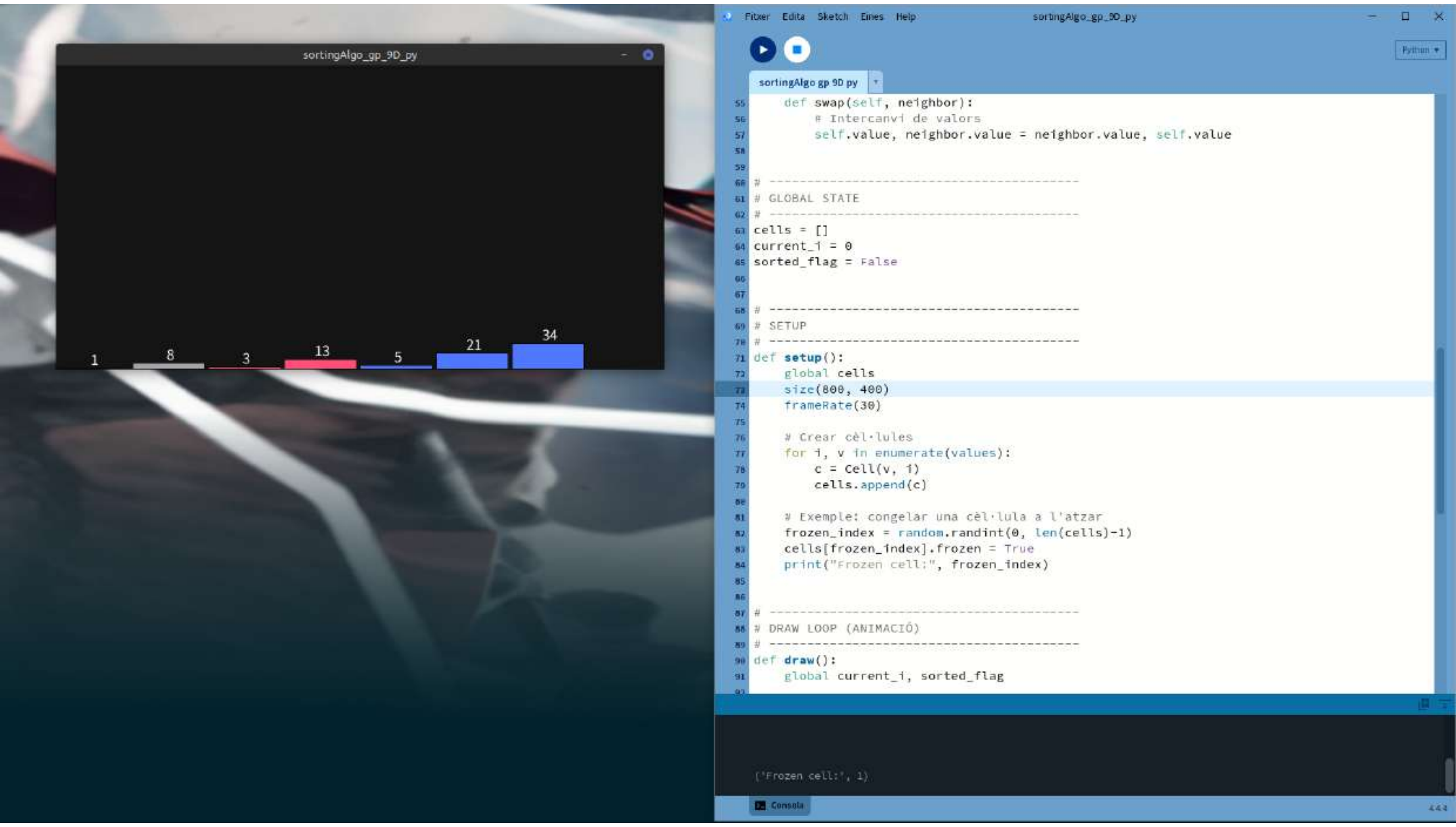
LAB & TUTORIALS

Sorting Algorythms Emergence Behaviour



Aquest és el report d'unes interessants suggerències que el Biòleg evolucionista i Michael Levin i Drs associades en vers a algortimes de 'sorting'.
En aquests experiments ens mostra com fins i tot en un algoritme clàssic (de fa unes dècades) i molt senzill com és el 'sorting algorythm' que simplement ordena un seguit de números aleatoris com a input i els endreça de manera ascendent de menor a major.
En l'experiment plantegen un comportament que en dècades havia passat desapercebut, on si una de les caselles quedava limitada o cancel·lada (es a dir que una de les posicions no feien cas de l algoritme) sembla succeir un comportament emergent sobre la nova estructura tot i que no hi ha un codi suplementari per a fer aquest update.

És com si el mateix algoritme 'tingues una computació autònoma sobre quin rol tindrà en el futur i que recodifica uns nous objectius. Per tant el que suggereixen és que fins i tot en un algoritme ínfim



Llista d'enllaços

Paper original (preprint) de Levin, Zhang & Goldstein
https://arxiv.org/abs/2401.05375

Article divulgatiu de Thoughtforms (Michael Levin Lab)
https://thoughtforms.life/what-do-algorithms-want-a-new-paper-on-the-emergence-of-surprising-behavior-in-the-most-unexpected-places/

Article d'Andrea Morris (Forbes)
https://www.forbes.com/sites/andreamorris/2025/11/13/the-secret-life-of-algorithms/

Reflexió expandida sobre cognició basal i computació distribuïda
https://lfyadda.com/cognitive-kin-and-emergent-reasoningfrom-bubble-sort-arrays-to-billion-parameter-language-models/

code >
copy paste the following code and name it as a py project for example ***sorting_algorythms.py***
in processing python mode

```
# ----------------------------------------
# Distributed Sorting – Processing (Python mode)
# Visual slow-motion + frozen cells + algotypes
# ----------------------------------------

import random

# CONFIG
values = [28, 34, 6, 20, 7, 89, 34, 18, 29, 51]
cell_w = 50
slow_motion = 300   # ms entre passos
algotypes = ["bubble", "alt"]  # dos tipus només per jugar

# ----------------------------------------
# CELL CLASS
# ----------------------------------------
class Cell:
    def __init__(self, value, index):
        self.value = value
        self.index = index
        self.x = index * cell_w
        self.y = height/2
        self.frozen = False
        self.algotype = random.choice(algotypes)

    def draw(self):
        # Color segons algotype
        if self.frozen:
            fill(150)  # gris per a cèl·lula congelada
        else:
            if self.algotype == "bubble":
                fill(0, 120, 255)
            else:
                fill(255, 80, 120)

        rect(self.x, self.y, cell_w-5, -self.value)

        fill(255)
        textSize(14)
        textAlign(CENTER, BOTTOM)
        text(str(self.value), self.x + cell_w/2, self.y - self.value)

    def should_swap(self, neighbor):
        if self.frozen or neighbor.frozen:
            return False

        # Regles locals segons "algotype"
        if self.algotype == "bubble":
            return self.value > neighbor.value
        else:
            # un variant aleatòria per veure fenòmens emergents
            return random.random() < 0.1  # swap ocasional

    def swap(self, neighbor):
        # Intercanvi de valors
        self.value, neighbor.value = neighbor.value, self.value


# ----------------------------------------
# GLOBAL STATE
# ----------------------------------------
cells = []
```

```
current_i = 0
sorted_flag = False


# ----------------------------------------
# SETUP
# ----------------------------------------
def setup():
    global cells
    size(800, 400)
    frameRate(30)

    # Crear cèl·lules
    for i, v in enumerate(values):
        c = Cell(v, i)
        cells.append(c)

    # Exemple: congelar una cèl·lula a l’atzar
    frozen_index = random.randint(0, len(cells)-1)
    cells[frozen_index].frozen = True
    print("Frozen cell:", frozen_index)


# ----------------------------------------
# DRAW LOOP (ANIMACIÓ)
# ----------------------------------------
def draw():
    global current_i, sorted_flag

    background(20)

    # Dibuixar totes les cèl·lules
    for c in cells:
        c.draw()

    if sorted_flag:
        fill(0,255,0)
        textSize(32)
        textAlign(CENTER, CENTER)
        text("SORTED (more or less...) ", width/2, 50)
        return

    # Cada frame → 1 comparació
    if current_i < len(cells)-1:
        a = cells[current_i]
        b = cells[current_i+1]

        if a.should_swap(b):
            a.swap(b)

        current_i += 1
        delay(slow_motion)

    else:
        # Un cicle complet → revisar si està ordenat
        current_i = 0
        if is_sorted():
            sorted_flag = True


# ----------------------------------------
# CHECK SORT STATE
# ----------------------------------------
def is_sorted():
    arr = [c.value for c in cells]
    return all(arr[i] <= arr[i+1] for i in range(len(arr)-1))
```

frases interessants que ha generat el GPT

1. El “jo” com a resultat de regles locals

L’experiment mostra que una col·lecció d’agents que només “miren el veí” pot exhibir comportament que sembla global, coherent i “intencional”.
Filosòficament:
L’ordre no ve d’un director d’orquestra, sinó de la dansa col·lectiva d’interaccions micro.
Això qüestiona la idea d’un “jo” centralitzat — també en biologia, cognició i fins i tot en IA.

2. Intel·ligència sense cervell

Quan un sistema resol un problema (tancar desordre, resistir fallades, reorganitzar-se davant un obstacle) sense cap controlador, queda difusa la frontera entre “algoritme” i “cognició”.
Això dona suport a la idea de Michael Levin:
La cognició és una propietat de l’organització, no del substrat.
És a dir, pot emergir en teixits, colònies, bots, o cèl·lules numèriques.

3. Errors i bloquejos com a fonts de novetat

Aquí, un error (cèl·lula congelada) provoca reconfiguracions creatives.
Les patologies del sistema revelen la seva arquitectura invisible.
Això recorda la cibernètica de Von Foerster: són els desajustos els que exposen les regles profundes.

4. Reducció de dimensions: el sistema decideix què és rellevant

Els agents només comparen valors i decideixen micro-moviments.
Però el sistema global sembla “saber” com arribar a l'ordre.
Això reflecteix el que molts filòsofs cognitivistes defensen:
La intel·ligència és la capacitat de reduir un espai immens de possibilitats a unes poques decisions locals significatives.

5. Emergentisme fort i absència d’intencionalitat

El sistema té comportament coherent sense tenir “intencions”.
Això toca un punt profund:
¿És necessari tenir intencions per tenir comportament funcionalment intel·ligent?
El model desafia la intuïció que “intenció → comportament”;
pot ser a la inversa: “comportament → aparença d’intenció”.

6. Convergència sense supervisió

Cap agent veu la llista sencera, però tots convergeixen.

Això s'assembla a:

formigueig

somites embrionaris

reagrupació cel·lular

escoles de peixos

Tot això recolza la idea que:

L'ordre global no és una característica, sinó una conseqüència.

És el fenomen de **"teleonomia"**: apareix finalitat sense finalisme.

7. **Heterogeneïtat com a motor de clustering**

La presència de diversos "algotypes" crea dinàmiques de grup inesperades:

segregació espontània

fronteres mòbils

sistemes semi-estables

És molt semblant a models de Schelling (segregació urbana) però en un espai conceptual "d'algoritmes".

Aquí hi ha una lectura preciosa:

Les diferències minúscules en les regles locals amplifiquen diferències globals.

<u>És comparable a com petites diferències de cultura, biomecànica o fisiologia creen dinàmiques socials enormes.</u>

8. **Equilibris dinàmics sense solució final**

Quan alguns algotypes tenen objectius oposats (p. ex. ordenar al revés), el sistema crea un equilibri oscil·latori.

No hi ha "solució final", però sí:

homeòstasi

estabilitat en moviment

un estat "viu"

Això toca conceptes de Deleuze i Guattari:

Estructures que no s'estanquen, sinó que s'auto-mantenen en flux.

Una mena de "territori en vibració".

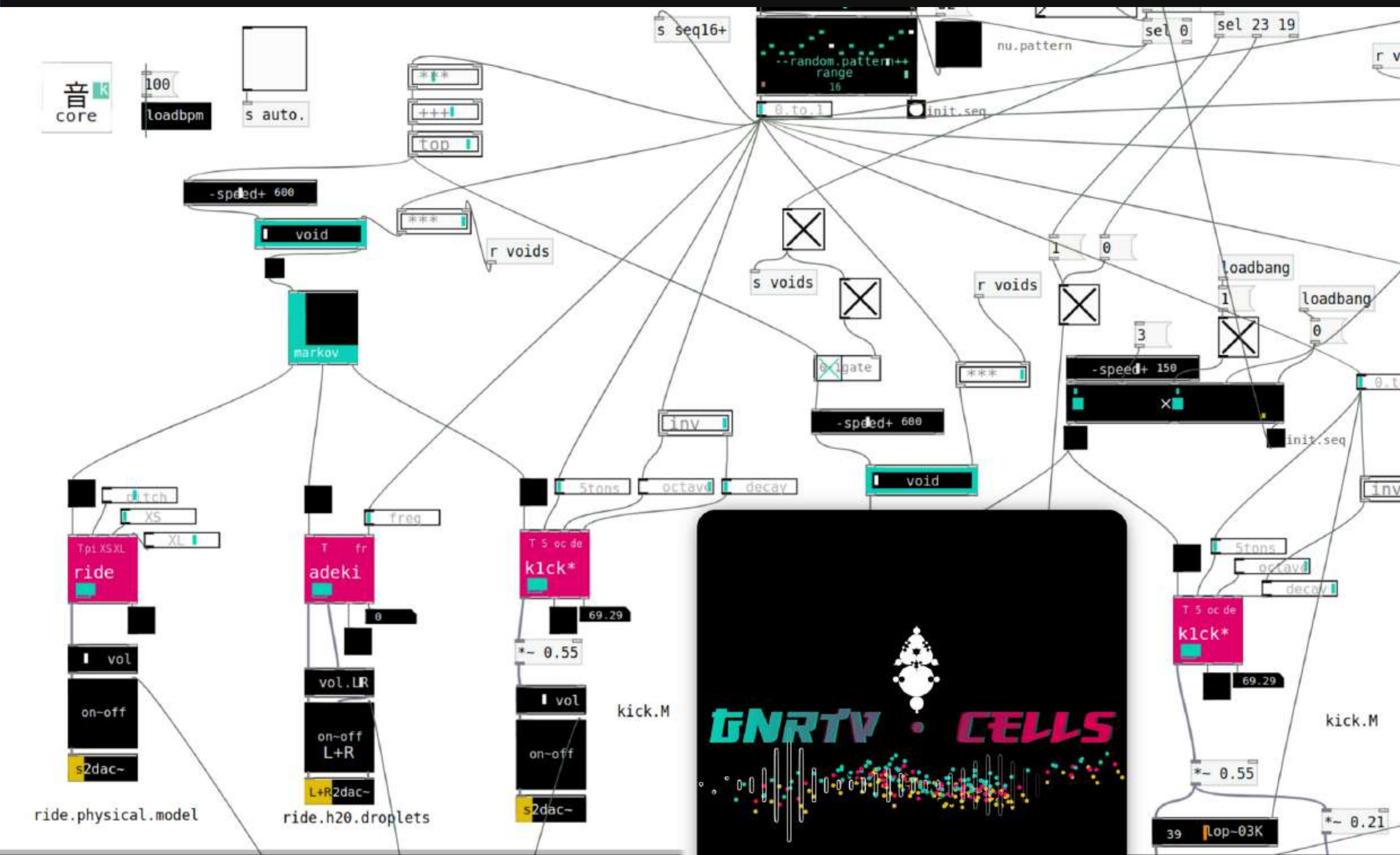
9. L'emergència com a fenomen estètic

. ..具.. .

Espais Latents

Update Repo

gnrtv.cells 26

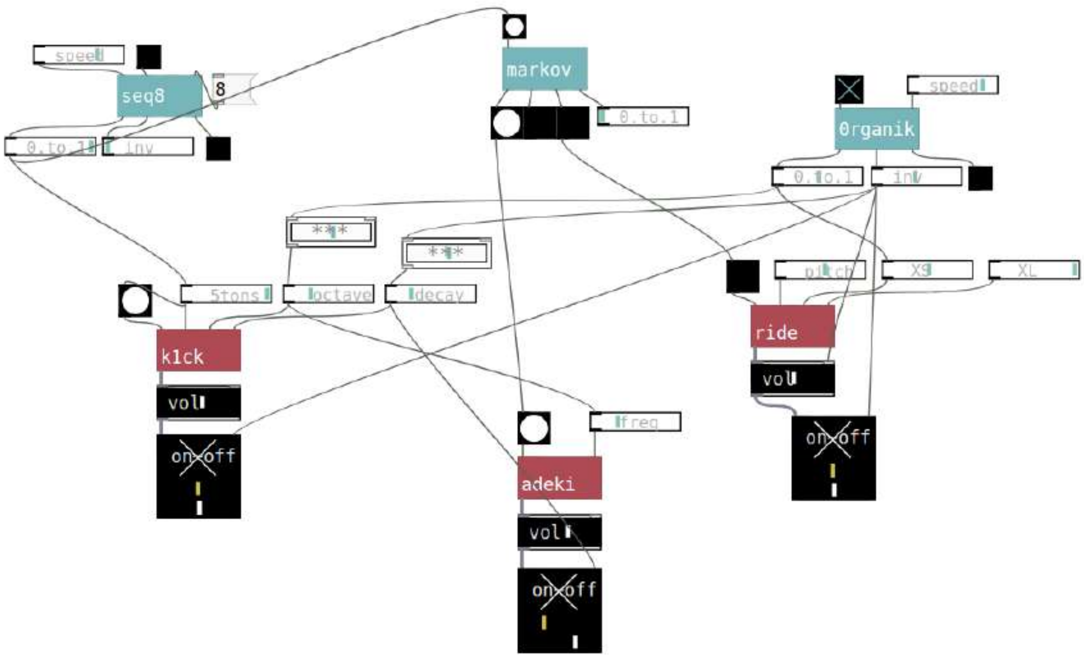


experiments & prototips

: // Gnrtv.Cells'26

Update generative sonic algorythms toolkit

https://github.com/xamanza/gnrtv.cells.26



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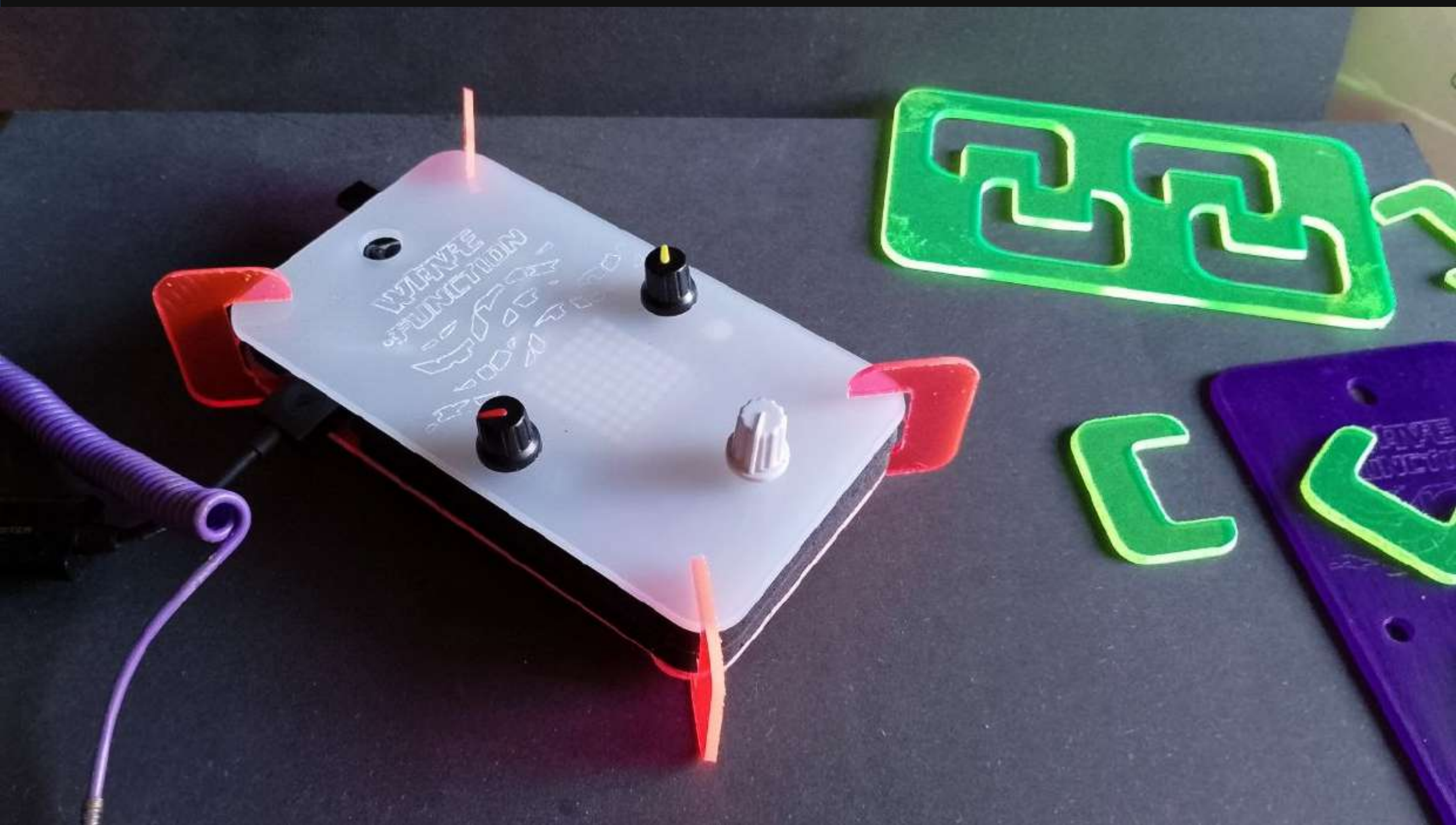
Espais Latents

Prototype

WaveFunction

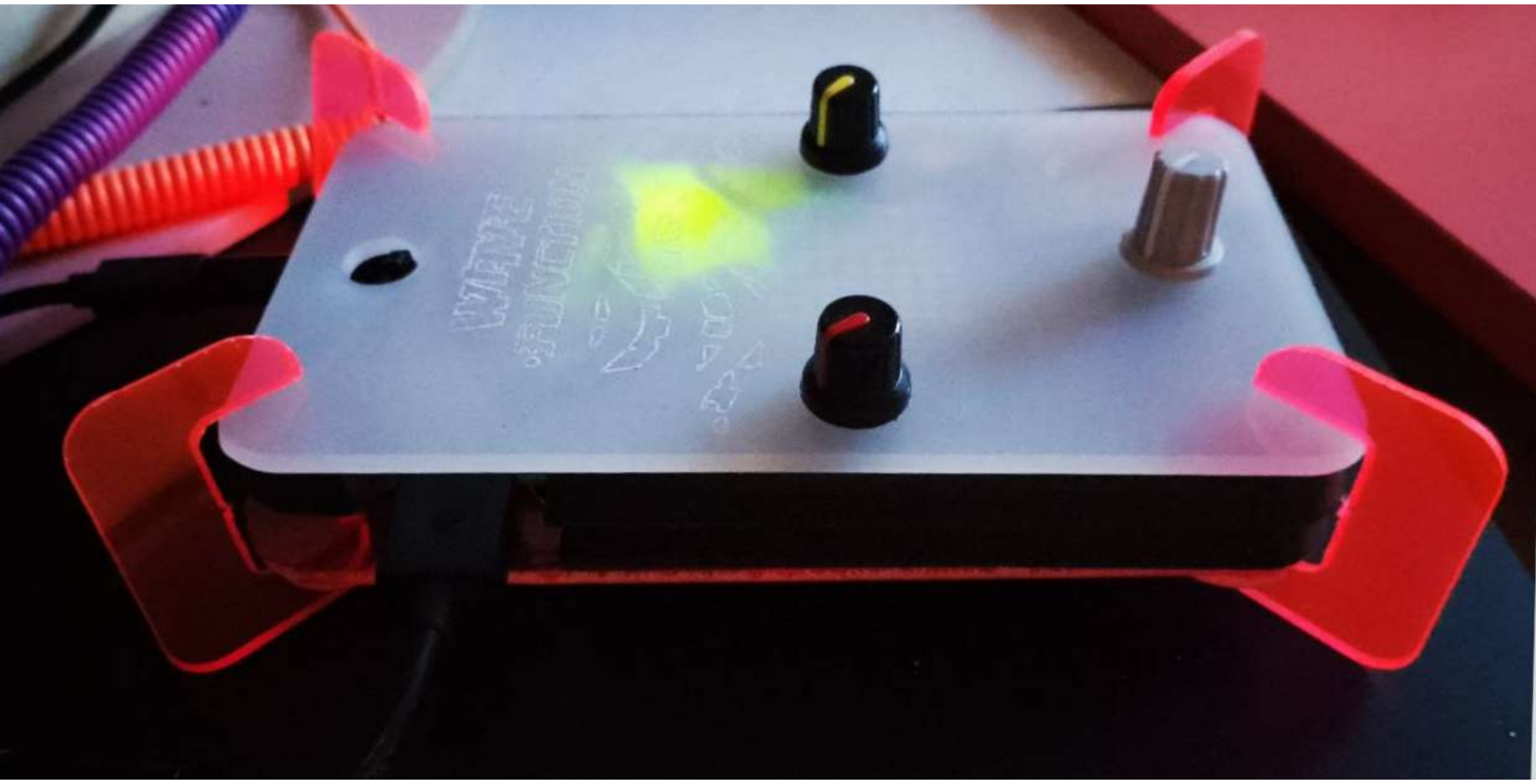


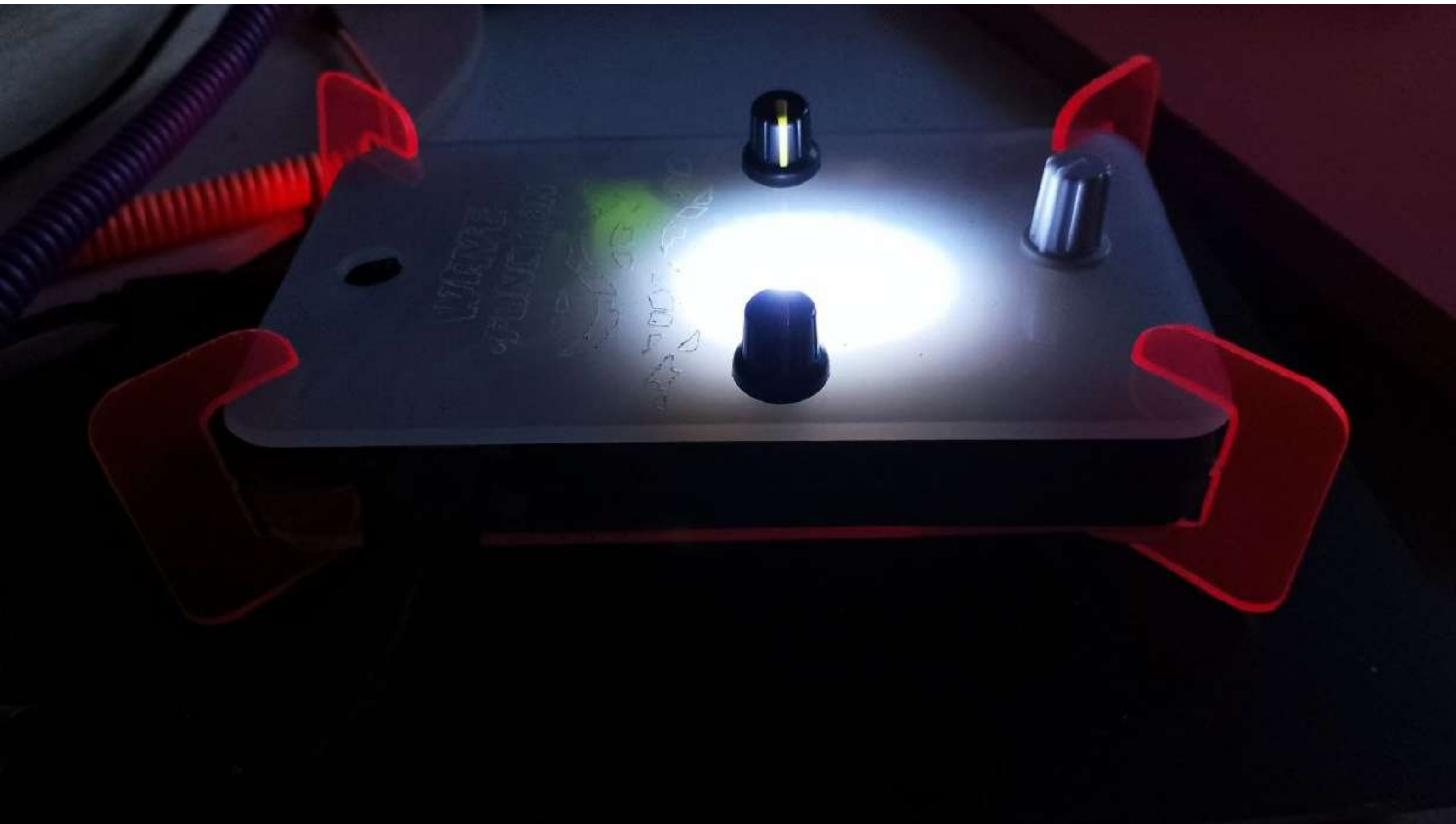
experiments & prototips

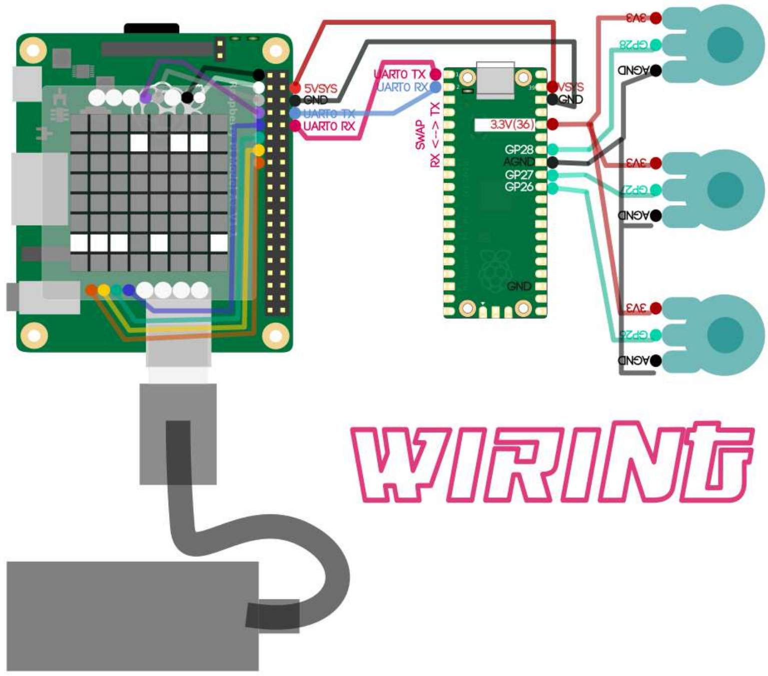
: // wave function

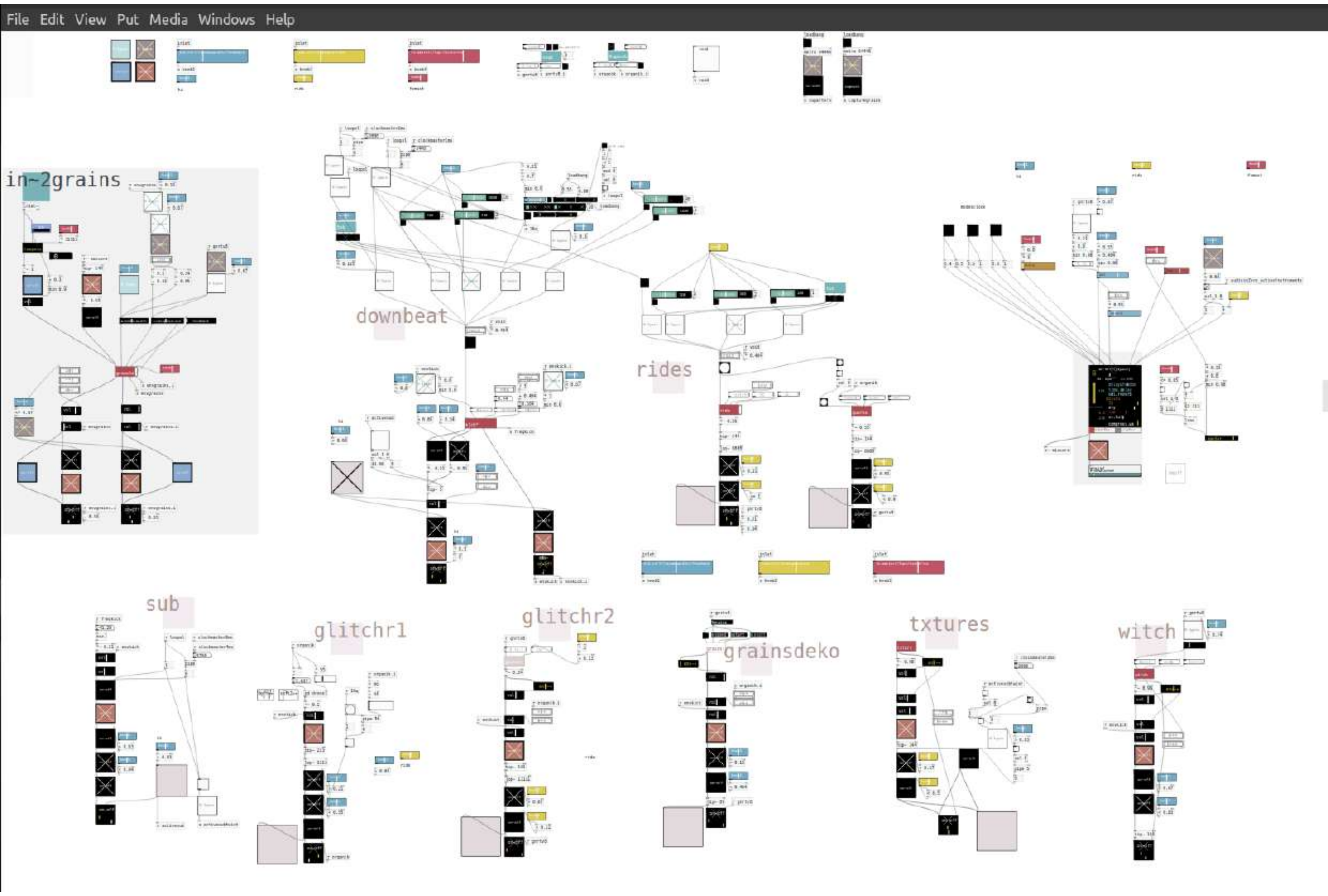
Instrument per a la performance sonora integrant el concepte pedal (post procés) i el concepte de sintetitzador algoritmic.

Check out the document ***WaveFunctionTUTORIAL_How2BuildaGnrtvAlgorythmicPedalFromScratch.pdf*** *in order to build one of thoswe instruments from scratch.*






5VSYS
GND
UART0 TX
UART0 RX
UART0 TX
UART0 RX
SWAP
RX <-> TX
3.3V(36)
GP28
AGND
GP27
GP26
GND
VSYS
GND
WIRING


File Edit View Put Media Windows Help
in~2grains
downbeat
rides
sub
glitchr1
glitchr2
grainsdeko
txtures
witch

Wave Function is the name of this unconventional instrument.
The type of algorythmic language used (pd/gnrtv.cells) allows
a high sonic complexity in a very tiny and compact computation platform,
therefore a nice **permacompùting / low consumption platform**.

It works as a **'pedal'** or a **post FX processor** unit that deconstructs and texturizes
the incoming audio signal with granular synthesis techniques.

In addition when no audio input is connected,
it behaves as a **generative pattern synthesizer**
creating minimal steps and patterns crossing over musical genres
like (minimal/deep/dub) techno, glitch and soundscaping.

The instrument approach drifts among generative states
which are emerging with several sonic ingredients.

Therefore as performers we do not control one parameter of a certain sonic event :
rather we **drift** and flow about *growing sonic structures* combining
texturization and nonconventional sounds and beats
from chill ***states*** to other more oversized hi intensity ***states***...

. ..具.. .

Espais Latents

LAB & TUTORIALS

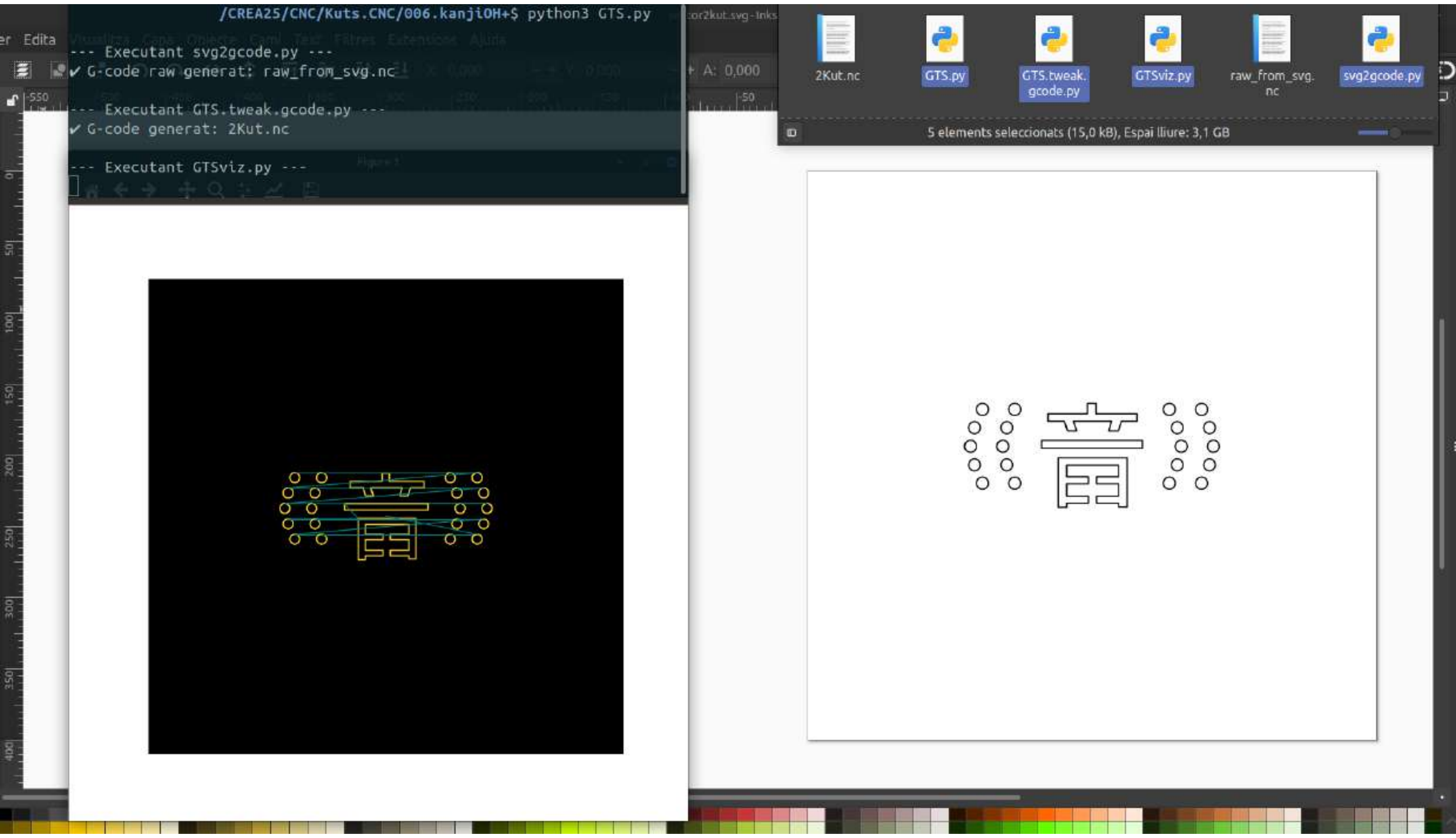
MakerSpace : CNC methods
How to convert an Svg vector to a gcode for CNC cutting

Aquest és un mètode molt compacte i minimalista per a convertir dissenys en format **.svg** (format de dibuix vectorial obert / w3c) a format **gcode (.nc)** que és el format que interpreten les màquines **CNC** de fabricació digital en tall laser ò fresat. Un cop generat el **gcode (.nc)** podem copiar-lo a la targeta SD que insertarem al controlador offline de la màquina (normalment disposen de controlador offline o directament la inserció de la SD a la estructura de la màquina.

Nota : Aquest mètode, per tant és molt simplificat i no requereix de programari intermediari però necessitarem que la màquina CNC a operar pugui tenir el control offline.

Es tracta d'una sèrie d'scripts amb el llenguatge python que realitzen els següents passos :
svg2gcode.py converteix un svg a un gcode en 'brut'.
GTS.tweak.gcode.py recull el gcode en brut i el reformata i el fa més compacte.
GTSviz.py Ens genera una visualització / simulador de com quedarà el tall, destacant en dos colors el tall amb color groc i el traçat del capçal 'a l'aire' en blau turquesa per a connectar diverses formes o siluetes aillades que contingui el disseny.

Per a fer-lo més usable i pràctic aquesta tríada d'scripts s'executa amb un script 'macro' *GTS.py* que els va executant progressivament i ens simplifica el procés amb una única comanda >
python3 GTS.py



Per a executar GTS.py necessitareu tenir instal·lat python3 que via terminal a raspbian o altres distribucions linux podeu instal·lar amb
sudo apt install python3

també instal.lar les llibreries matplotlib i svgpathtools que necessitarem :

python3 -m pip install --upgrade pip
i quan acabi instal·lar

pip3 install matplotlib
pip3 install svgpathtools

A continuació podeu copiar cadascun dels scripts i salvarlo com a .py amb un editor.

svg2gcode.py converteix un svg a un gcode en 'brut'.
codi :

```
#!/usr/bin/env python3
from svgpathtools import svg2paths2, Path, Line, CubicBezier, QuadraticBezier, Arc
import math

# -------------------- CONFIG --------------------
INPUT_SVG = "vector2kut.svg"
OUTPUT_NC = "raw_from_svg.nc"
Z_SAFE = 15
Z_CUT = -9
F_CUT = 200
STEP_MM = 0.5    # discretització de corbes
DECIMALS = 3     # decimals per coordenades
MIRROR_Y = True  # invertir eix Y
TABLE_WIDTH = 400
TABLE_HEIGHT = 400

# -------------------- AUX --------------------
def discretize_segment(seg, step=STEP_MM):
    """Retorna una llista de punts (x,y) aproximant un segment"""
    length = seg.length()
    num_points = max(int(math.ceil(length / step)), 1)
    points = [seg.point(t) for t in [i/num_points for i in range(num_points + 1)]]
    return [(round(p.real, DECIMALS), round(p.imag, DECIMALS)) for p in points]

def discretize_subpath(subpath, step=STEP_MM):
    """Discretitza tots els segments d'una subpath"""
    points = []
    for seg in subpath:
        points.extend(discretize_segment(seg, step))
    return points

# -------------------- LLEGIR SVG --------------------
paths, attributes, svg_att = svg2paths2(INPUT_SVG)

# centrat
x_offset = TABLE_WIDTH / 2
y_offset = TABLE_HEIGHT / 2

with open(OUTPUT_NC, "w") as f:
    block_count = 1

    for path in paths:
        # svgpathtools no separa automàticament subpaths,
        # així que creem subpaths quan hi ha un Move (M)
        subpaths = []
        current_sub = Path()
        for seg in path:
            # si segment és un Line/Cubic/etc i comença en un punt diferent del darrer, és Move
            if len(current_sub) > 0 and seg.start != current_sub[-1].end:
                subpaths.append(current_sub)
                current_sub = Path()
            current_sub.append(seg)
        if len(current_sub) > 0:
            subpaths.append(current_sub)

        # ara cada subpath és un bloc
        for sub in subpaths:
            points = discretize_subpath(sub)
            if not points:
                continue

            f.write(f"(Block-name: SVG_Block_{block_count})\n")
            f.write(f"G0 Z{Z_SAFE}\n")

            started = False
            for x, y in points:
                # mirall i centrat
                if MIRROR_Y:
```

```
                y = -(y - y_offset)
            else:
                y = y - y_offset
            x = x - x_offset

            if not started:
                f.write(f"G0 X{x} Y{y}\n")       # desplaçament primer punt
                f.write(f"G1 Z{Z_CUT} F{F_CUT}\n") # baixar fresa + primer F
                started = True
            else:
                f.write(f"G1 X{x} Y{y}\n")       # moviment següent sense F
        f.write(f"G0 Z{Z_SAFE}\n")  # aixecar al final del bloc
        block_count += 1

print(f"      G-code raw generat: {OUTPUT_NC}")
```

GTS.tweak.gcode.py recull el gcode en brut i el reformata i el fa més compacte.
codi :

```
#!/usr/bin/env python3

INPUT_FILE  = "raw_from_svg.nc"
OUTPUT_FILE = "2Kut.nc"

Z_SAFE   = 15
Z_CUT    = -9
F_CUT    = 200
SPINDLE  = 800

skipping_raw_header = False
skipping_raw_footer = False
header_written = False
awaiting_xy = False

last_line = None  # <-- Només eliminarem duplicats consecutius

with open(INPUT_FILE) as fin, open(OUTPUT_FILE, "w") as fout:
    for line in fin:
        line = line.rstrip()

        # --- Detectar Header brut ---
        if "(Block-name: Header)" in line:
            skipping_raw_header = True
            continue

        if skipping_raw_header:
            if "(Block-name:" in line:
                skipping_raw_header = False
            else:
                continue

        # --- Detectar Footer brut ---
        if "(Block-name: Footer)" in line:
            skipping_raw_footer = True
            continue

        if skipping_raw_footer:
            continue  # saltem tot el footer original

        # --- Bloc nou ---
        if "(Block-name:" in line:
            if not header_written:
                # Escriure header inicial
                for l in [
                    ";____________________",
                    "; CNC-_-Kut",
                    ";____________________",
                    "G21",
                    "G90",
                    f"G0 Z{Z_SAFE}",
                    f"M3 S{SPINDLE}",
                    "G4 P3"
                ]:
                    if l != last_line:
                        fout.write(l + "\n")
                        last_line = l
                header_written = True
            else:
                # Aixecar Z abans de començar el nou bloc
                z_safe_line = f"G0 Z{Z_SAFE}"
                if z_safe_line != last_line:
                    fout.write(z_safe_line + "\n")
                    last_line = z_safe_line

            # Escriure el nom del bloc
```

```
            if line != last_line:
                fout.write(line + "\n")
                last_line = line

            awaiting_xy = True
            continue

        # --- Quan veiem coordenades inicials G0 X... ---
        if awaiting_xy and line.startswith("G0 X"):
            if line != last_line:
                fout.write(line + "\n")
                last_line = line
            z_line = f"G1 Z{Z_CUT} F{F_CUT}"
            if z_line != last_line:
                fout.write(z_line + "\n")
                last_line = z_line
            awaiting_xy = False
            continue

        # --- Escriure la resta del G-code, eliminant només consecutius ---
        if line != last_line:
            fout.write(line + "\n")
            last_line = line

    # --- Footer definitiu ---
    for l in [f"G0 Z{Z_SAFE}", "M5", "G0 X0 Y0"]:
        if l != last_line:
            fout.write(l + "\n")
            last_line = l

print(f"   G-code generat: {OUTPUT_FILE}")
```

GTSviz.py
codi :

```
#!/usr/bin/env python3
import matplotlib.pyplot as plt

INPUT_FILE = "2Kut.nc"
Z_SAFE = 15
Z_CUT = -9

# Preparar llista de línies XY
x_coords = []
y_coords = []
colors = []

current_z = Z_SAFE
last_x = None
last_y = None

with open(INPUT_FILE) as f:
    for line in f:
        line = line.strip()
        if line.startswith("G0") or line.startswith("G1"):
            parts = line.split()
            x = y = None
            z = None
            for p in parts:
                if p.startswith("X"):
                    x = float(p[1:])
                elif p.startswith("Y"):
                    y = float(p[1:])
                elif p.startswith("Z"):
                    z = float(p[1:])
            if z is not None:
                current_z = z
            if x is not None or y is not None:
                # Si no hi ha valor, mantenim l'anterior
                if x is None: x = last_x
                if y is None: y = last_y
                if last_x is not None and last_y is not None:
                    x_coords.append([last_x, x])
                    y_coords.append([last_y, y])
                    # Assignar color segons Z
                    if current_z <= Z_CUT:
                        colors.append("#008080")  # tall
                    else:
                        colors.append("#FFD42A")  # Z safe
                last_x, last_y = x, y

#fig, ax = plt.subplots(figsize=(8,8), dpi=100)  # 800x800 px exactes
fig, ax = plt.subplots(figsize=(8,8), dpi=96)  # 800x800 px exactes
ax.set_facecolor("#000000")
for xs, ys, c in zip(x_coords, y_coords, colors):
    ax.plot(xs, ys, color=c, linewidth=1)

ax.set_xlim(-200, 200)
ax.set_ylim(-200, 200)
ax.set_aspect('equal')
ax.set_title("G-code Preview", color="white")
ax.tick_params(colors='white')
plt.show()
```

Finalment un cop copiats els scripts necessitarem l script 'master' que serà el que automatitzi tot amb una única comanda al terminal > python3 GTS.py

GTSviz.py
codi :

```
#!/usr/bin/env python3
# code by Xa Manzanares IG+Github @xamanza 2026 / code tweaks Transformer Assistants (GP&DS) // GPL v3
import subprocess
import os
import sys

# -------------------- CONFIG --------------------
SVG_FILE = "vector2kut.svg"   # fitxer SVG d'entrada
RAW_NC = "raw_from_svg.nc"    # generat per svg2gcode.py
GTS_OUTPUT = "2Kut.nc"        # generat per GTS.py

# -------------------- FUNCIONS --------------------
def run_script(script, args=[]):
    """Executa un script Python amb subprocess i mostra errors si n'hi ha"""
    cmd = [sys.executable, script] + args
    print(f"\n--- Executant {script} ---")
    result = subprocess.run(cmd)
    if result.returncode != 0:
        print(f"    Error en {script}")
        sys.exit(1)

# -------------------- PIPELINE --------------------
# 1 SVG → raw G-code
run_script("svg2gcode.py", [SVG_FILE])

# 2  raw G-code → GTS tunejat
run_script("GTS.tweak.gcode.py", [RAW_NC, GTS_OUTPUT])

# 3 Mostrar preview amb GTSviz.py
run_script("GTSviz.py", [GTS_OUTPUT])

print("\n    Pipeline complet executat amb èxit!")
```

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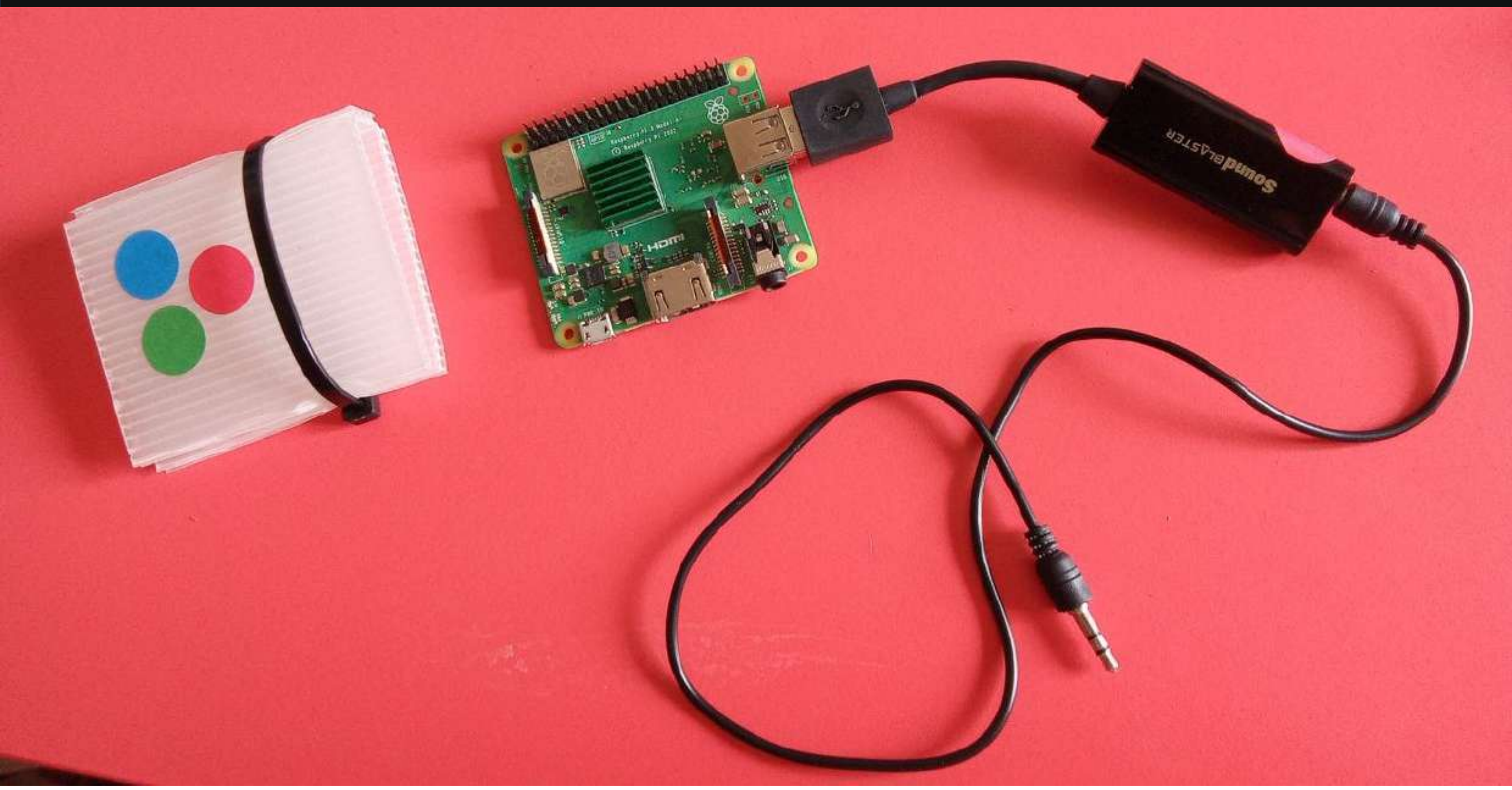
Espais Latents

LAB & TUTORIALS

OSC.Transmitter.Tutorial

How 2 Build an OSC Transmitter



How 2 Build an OSC Transmitter

by Xavi.Manzanares @xamanza // 2025

Physical audio analysis module and OSC message transmission.

Ingredients:

Raspberry Pi model 3A
Creative Sound Blaster USB SoundCard

0

Requirements >

Download and install ***Raspbian OS***
Raspbian OS > https://downloads.raspberrypi.com/raspios_arm64/images/raspios_arm64-2025-05-13/2025-05-13-raspios-bookworm-arm64.img.xz
(recommended Software Imager to copy .img to a 16gb or 32gb SD card
Imager >
from linux OS https://downloads.raspberrypi.org/imager/imager_latest_amd64.deb
from win OS https://downloads.raspberrypi.org/imager/imager_latest.exe
from mac OS https://downloads.raspberrypi.org/imager/imager_latest.dmg

1

Start the Rpi with a keyboard, mouse and screen Install Pd vanilla
Open terminal and execute

sudo apt-get install puredata

2

Copy the following code into a text file and save it as ***Audio2OSC-__-transmission.pd***
in /home/pi/Desktop (if you change the RPI number it will be /home/whatever/Desktop)

```
#N canvas 715 437 399 223 12;
#X obj 44 38 cnv 15 282 145 empty empty empty 20 12 0 14 #dcdcdc #404040
0;
#N canvas 1727 33 1564 1012.06 Audio·2·OSC-__-transmission 1;
#X obj 14 547 cnv 15 818 19 empty empty empty 20 12 0 14 #c4f01c #404040
0;
#X obj 3 576 cnv 15 860 123 empty empty empty 20 12 0 14 #f8f8f8 #404040
0;
#N canvas 429 98 1920 1022 (subpatch) 0;
#X obj 655 224 unpack;
#X floatatom 655 274 0 0 0 0 - - -;
#X floatatom 669 251 0 0 0 0 - - -;
#X obj 655 184 route 1 2 3 4;
#X obj 738 224 unpack;
#X floatatom 738 274 0 0 0 0 - - -;
#X floatatom 752 251 0 0 0 0 - - -;
#X obj 822 224 unpack;
#X floatatom 822 274 0 0 0 0 - - -;
#X floatatom 836 251 0 0 0 0 - - -;
#X obj 295 225 unpack;
#X floatatom 295 275 0 0 0 0 - - -;
#X floatatom 309 252 0 0 0 0 - - -;
#X floatatom 421 287 0 0 0 0 - - -;
#X obj 171 223 print attack;
#X obj 20 203 print pitch;
#X text 601 91 number of pitch outlets (1-3 \, default 1);
#X text 603 112 number of peaks to find (1-100 \, default 20);
#X text 603 134 number of peaks to output (default 0.);
#X obj 300 102 fiddle~ 1024 1 20 3;
#X text 662 298 individual sinusoidal components;
#X text 414 310 amplitude;
#X text 419 334 (dB);
#X text 282 301 raw pitch;
#X text 269 325 and amplitude;
#X text 257 342 (up to 3 outputs);
#X text 177 243 bang on;
#X text 177 265 attack;
#X text 15 223 cooked pitch;
#X text 32 240 output;
#X text 473 101 ------ arguments:;
#X text 601 69 window size (128-2048 \, default 1024);
#X obj 159 -67 inlet~;
#X obj 119 222 outlet;
#X obj 475 284 outlet;
#X obj 655 368 outlet;
#X text 518 230 tweak!;
#X obj 217 12 *~;
#X obj 297 -26 inlet;
#X obj 6 -21 cnv 15 55 20 empty empty empty 3 8 0 8 #000000 #404040
0;
#X obj 9 -9 hsl 44 8 0 1 0 1 empty empty empty 7 7 0 11 #000000 #50c0c0
#50c0c0 0 1;
#X obj 37 -12 cnv 11 15 9 empty empty in~ -10 -3 0 14 #08ccb4 #fcfcfc
0;
#X obj 475 229 / 100;
#X obj 823 367 outlet;
#X obj 823 329 * 4;
#X connect 0 0 1 0;
#X connect 0 1 2 0;
#X connect 1 0 35 0;
#X connect 3 0 0 0;
#X connect 3 1 4 0;
#X connect 3 2 7 0;
#X connect 4 0 5 0;
#X connect 4 1 6 0;
#X connect 7 0 8 0;
#X connect 7 1 9 0;
#X connect 8 0 44 0;
#X connect 10 0 11 0;
#X connect 10 1 12 0;
#X connect 19 1 33 0;
#X connect 19 2 10 0;
#X connect 19 3 13 0;
#X connect 19 3 42 0;
#X connect 19 4 3 0;
#X connect 32 0 37 0;
```

```
#X connect 37 0 19 0;
#X connect 38 0 37 1;
#X connect 42 0 34 0;
#X connect 42 0 40 0;
#X connect 44 0 43 0;
#X coords 0 1 100 -1 55 24 1 6 -23;
#X restore 187 138 graph;
#X obj 235 118 hsl 57 12 0 1 0 1 empty empty vol.in 19 5 0 11 #fcfcfc
#08ccb4 #b0b0b0 5600 1;
#X obj 1 -1 cnv 15 100 19 empty empty empty 20 12 0 14 #f0f0f0 #404040
0;
#N canvas 1600 28 1920 1027 (subpatch) 0;
#X obj -175 -157 cnv 15 100 60 empty empty empty 20 12 0 14 #e0e0e0
#404040 0;
#X obj 6 -23 cnv 15 55 55 empty empty empty 5 12 0 14 #fcfcfc #404040
0;
#X obj 6 -21 cnv 15 55 27 empty empty O.S.C. 3 5 0 14 #000000 #fcfcfc
0;
#X obj 6 3 cnv 15 55 27 empty empty send 4 19 0 14 #74a8e0 #fcfcfc
0;
#X obj -154 218 oscparse;
#X msg 609 212 disconnect;
#X obj 362 350 netsend -u -b;
#X obj 211 157 list prepend send;
#X obj 211 182 list trim;
#X text 702 211 don't send;
#X obj 112 -178 cnv 15 300 111 empty empty send 20 12 0 14 #6cfcd8
#404040 0;
#X obj 212 87 oscformat;
#X obj 212 38 inlet;
#X obj 479 74 inlet;
#X obj 6 3 tgl 16 0 empty empty print 17 7 0 10 #000000 #fcfcfc #fcfcfc
0 1;
#X obj -36 325 spigot;
#X obj 706 76 inlet;
#X obj -130 20 inlet;
#X obj -156 254 list split 1;
#X obj 28 358 print OSC.send;
#X obj -56 370 outlet;
#X connect 4 0 18 0;
#X connect 5 0 6 0;
#X connect 5 0 19 0;
#X connect 7 0 8 0;
#X connect 8 0 6 0;
#X connect 11 0 7 0;
#X connect 11 0 4 0;
#X connect 12 0 11 0;
#X connect 12 0 19 0;
#X connect 13 0 6 0;
#X connect 13 0 19 0;
#X connect 14 0 15 1;
#X connect 15 0 19 0;
#X connect 15 0 20 0;
#X connect 16 0 5 0;
#X connect 17 0 11 0;
#X connect 18 1 15 0;
#X coords 0 1 100 -1 55 55 1 6 -23;
#X restore 213 608 graph;
#X obj 277 608 bng 21 250 50 0 empty empty disconnect 1 -9 0 10 #000000
#fcfcfc #000000;
#X obj 229 523 loadbang;
#X obj 213 664 nbx 5 14 -1e+37 1e+37 0 0 empty empty empty 0 -8 0 9
#74a8e0 #000000 #000000 0 256 0;
#X obj 245 571 r connect;
#X obj 14 378 cnv 15 812 113 empty empty empty 20 12 0 14 #e8bcbc #404040
0;
#N canvas 1600 28 1920 1027 (subpatch) 0;
#X obj -175 -157 cnv 15 100 60 empty empty empty 20 12 0 14 #e0e0e0
#404040 0;
#X obj 6 -23 cnv 15 55 55 empty empty empty 5 12 0 14 #fcfcfc #404040
0;
#X obj 6 -21 cnv 15 55 27 empty empty O.S.C. 3 5 0 14 #000000 #fcfcfc
0;
#X obj 6 3 cnv 15 55 27 empty empty send 4 19 0 14 #74a8e0 #fcfcfc
0;
#X obj -154 218 oscparse;
#X msg 609 212 disconnect;
```

```
#X obj 362 350 netsend -u -b;
#X obj 211 157 list prepend send;
#X obj 211 182 list trim;
#X text 702 211 don't send;
#X obj 112 -178 cnv 15 300 111 empty empty send 20 12 0 14 #6cfcd8
#404040 0;
#X obj 212 87 oscformat;
#X obj 212 38 inlet;
#X obj 479 74 inlet;
#X obj 6 3 tgl 16 0 empty empty print 17 7 0 10 #000000 #fcfcfc #fcfcfc
0 1;
#X obj -36 325 spigot;
#X obj 706 76 inlet;
#X obj -130 20 inlet;
#X obj -156 254 list split 1;
#X obj 28 358 print OSC.send;
#X obj -56 370 outlet;
#X connect 4 0 18 0;
#X connect 5 0 6 0;
#X connect 5 0 19 0;
#X connect 7 0 8 0;
#X connect 8 0 6 0;
#X connect 11 0 7 0;
#X connect 11 0 4 0;
#X connect 12 0 11 0;
#X connect 12 0 19 0;
#X connect 13 0 6 0;
#X connect 13 0 19 0;
#X connect 14 0 15 1;
#X connect 15 0 19 0;
#X connect 15 0 20 0;
#X connect 16 0 5 0;
#X connect 17 0 11 0;
#X connect 18 1 15 0;
#X coords 0 1 100 -1 55 55 1 6 -23;
#X restore 376 608 graph;
#X obj 440 608 bng 21 250 50 0 empty empty disconnect 1 -9 0 10 #000000
#fcfcfc #000000;
#X obj 392 523 loadbang;
#X obj 376 664 nbx 5 14 -1e+37 1e+37 0 0 empty empty empty 0 -8 0 9
#74a8e0 #000000 #000000 0 256 0;
#X obj 408 571 r connect;
#N canvas 1600 28 1920 1027 (subpatch) 0;
#X obj -175 -157 cnv 15 100 60 empty empty empty 20 12 0 14 #e0e0e0
#404040 0;
#X obj 6 -23 cnv 15 55 55 empty empty empty 5 12 0 14 #fcfcfc #404040
0;
#X obj 6 -21 cnv 15 55 27 empty empty O.S.C. 3 5 0 14 #000000 #fcfcfc
0;
#X obj 6 3 cnv 15 55 27 empty empty send 4 19 0 14 #74a8e0 #fcfcfc
0;
#X obj -154 218 oscparse;
#X msg 609 212 disconnect;
#X obj 362 350 netsend -u -b;
#X obj 211 157 list prepend send;
#X obj 211 182 list trim;
#X text 702 211 don't send;
#X obj 112 -178 cnv 15 300 111 empty empty send 20 12 0 14 #6cfcd8
#404040 0;
#X obj 212 87 oscformat;
#X obj 212 38 inlet;
#X obj 479 74 inlet;
#X obj 6 3 tgl 16 0 empty empty print 17 7 0 10 #000000 #fcfcfc #fcfcfc
0 1;
#X obj -36 325 spigot;
#X obj 706 76 inlet;
#X obj -130 20 inlet;
#X obj -156 254 list split 1;
#X obj 28 358 print OSC.send;
#X obj -56 370 outlet;
#X connect 4 0 18 0;
#X connect 5 0 6 0;
#X connect 5 0 19 0;
#X connect 7 0 8 0;
#X connect 8 0 6 0;
#X connect 11 0 7 0;
#X connect 11 0 4 0;
```

```
#X connect 12 0 11 0;
#X connect 12 0 19 0;
#X connect 13 0 6 0;
#X connect 13 0 19 0;
#X connect 14 0 15 1;
#X connect 15 0 19 0;
#X connect 15 0 20 0;
#X connect 16 0 5 0;
#X connect 17 0 11 0;
#X connect 18 1 15 0;
#X coords 0 1 100 -1 55 55 1 6 -23;
#X restore 540 608 graph;
#X obj 603 608 bng 21 250 50 0 empty empty disconnect 1 -9 0 10 #000000
#fcfcfc #000000;
#X obj 556 523 loadbang;
#X obj 540 664 nbx 5 14 -1e+37 1e+37 0 0 empty empty empty 0 -8 0 9
#74a8e0 #000000 #000000 0 256 0;
#X obj 572 571 r connect;
#N canvas 1600 28 1920 1027 (subpatch) 0;
#X obj -175 -157 cnv 15 100 60 empty empty empty 20 12 0 14 #e0e0e0
#404040 0;
#X obj 6 -23 cnv 15 55 55 empty empty empty 5 12 0 14 #fcfcfc #404040
0;
#X obj 6 -21 cnv 15 55 27 empty empty O.S.C. 3 5 0 14 #000000 #fcfcfc
0;
#X obj 6 3 cnv 15 55 27 empty empty send 4 19 0 14 #74a8e0 #fcfcfc
0;
#X obj -154 218 oscparse;
#X msg 609 212 disconnect;
#X obj 362 350 netsend -u -b;
#X obj 211 157 list prepend send;
#X obj 211 182 list trim;
#X text 702 211 don't send;
#X obj 112 -178 cnv 15 300 111 empty empty send 20 12 0 14 #6cfcd8
#404040 0;
#X obj 212 87 oscformat;
#X obj 212 38 inlet;
#X obj 479 74 inlet;
#X obj 6 3 tgl 16 0 empty empty print 17 7 0 10 #000000 #fcfcfc #fcfcfc
0 1;
#X obj -36 325 spigot;
#X obj 706 76 inlet;
#X obj -130 20 inlet;
#X obj -156 254 list split 1;
#X obj 28 358 print OSC.send;
#X obj -56 370 outlet;
#X connect 4 0 18 0;
#X connect 5 0 6 0;
#X connect 5 0 19 0;
#X connect 7 0 8 0;
#X connect 8 0 6 0;
#X connect 11 0 7 0;
#X connect 11 0 4 0;
#X connect 12 0 11 0;
#X connect 12 0 19 0;
#X connect 13 0 6 0;
#X connect 13 0 19 0;
#X connect 14 0 15 1;
#X connect 15 0 19 0;
#X connect 15 0 20 0;
#X connect 16 0 5 0;
#X connect 17 0 11 0;
#X connect 18 1 15 0;
#X coords 0 1 100 -1 55 55 1 6 -23;
#X restore 703 608 graph;
#X obj 766 608 bng 21 250 50 0 empty empty disconnect 1 -9 0 10 #000000
#fcfcfc #000000;
#X obj 719 523 loadbang;
#X obj 703 664 nbx 5 14 -1e+37 1e+37 0 0 empty empty empty 0 -8 0 9
#74a8e0 #000000 #000000 0 256 0;
#X obj 735 571 r connect;
#X obj 874 588 r disconnect;
#X text 715 498 continuum;
#X obj 703 305 hsl 50 12 0 1 0 1 empty empty env~ 7 7 0 11 #000000
#50c0c0 #50c0c0 0 1;
#X text 712 346 envelope;
#N canvas 1600 28 1920 1027 (subpatch) 0;
```

```
#X obj -175 -157 cnv 15 100 60 empty empty empty 20 12 0 14 #e0e0e0
#404040 0;
#X obj 6 -23 cnv 15 55 55 empty empty empty 5 12 0 14 #fcfcfc #404040
0;
#X obj 6 -21 cnv 15 55 27 empty empty O.S.C. 3 5 0 14 #000000 #fcfcfc
0;
#X obj 6 3 cnv 15 55 27 empty empty send 4 19 0 14 #74a8e0 #fcfcfc
0;
#X obj -154 218 oscparse;
#X msg 609 212 disconnect;
#X obj 362 350 netsend -u -b;
#X obj 211 157 list prepend send;
#X obj 211 182 list trim;
#X text 702 211 don't send;
#X obj 112 -178 cnv 15 300 111 empty empty send 20 12 0 14 #6cfcd8
#404040 0;
#X obj 212 87 oscformat;
#X obj 212 38 inlet;
#X obj 479 74 inlet;
#X obj 6 3 tgl 16 0 empty empty print 17 7 0 10 #000000 #fcfcfc #fcfcfc
0 1;
#X obj -36 325 spigot;
#X obj 706 76 inlet;
#X obj -130 20 inlet;
#X obj -156 254 list split 1;
#X obj 28 358 print OSC.send;
#X obj -56 370 outlet;
#X connect 4 0 18 0;
#X connect 5 0 6 0;
#X connect 5 0 19 0;
#X connect 7 0 8 0;
#X connect 8 0 6 0;
#X connect 11 0 7 0;
#X connect 11 0 4 0;
#X connect 12 0 11 0;
#X connect 12 0 19 0;
#X connect 13 0 6 0;
#X connect 13 0 19 0;
#X connect 14 0 15 1;
#X connect 15 0 19 0;
#X connect 15 0 20 0;
#X connect 16 0 5 0;
#X connect 17 0 11 0;
#X connect 18 1 15 0;
#X coords 0 1 100 -1 55 55 1 6 -23;
#X restore 47 608 graph;
#X obj 114 608 bng 21 250 50 0 empty empty disconnect 1 -9 0 10 #000000
#fcfcfc #000000;
#X obj 63 523 loadbang;
#X obj 47 664 nbx 5 14 -1e+37 1e+37 0 0 empty empty empty 0 -8 0 9
#74a8e0 #000000 #000000 0 256 0;
#X msg 63 430 0;
#X msg 47 380 1;
#X obj 47 354 bng 21 50 10 0 empty empty empty 17 7 0 10 #000000 #fcfcfc
#000000;
#X obj 79 571 r connect;
#X text 65 498 trigger;
#X msg 392 547 set mid;
#X msg 557 547 set hi;
#X msg 719 547 set env;
#X msg 63 547 set tri;
#X obj 115 68 adc~ 2;
#X obj 339 78 loadbang;
#X msg 308 98 1;
#X obj 44 68 adc~ 1;
#X text 136 413 111;
#X text 131 389 1111;
#X text 392 498 continuum;
#X text 240 498 continuum;
#X text 555 498 continuum;
#X text 85 322 peaks;
#X obj 214 469 hsl 74 21 0 1 0 1 empty empty lo 7 7 0 11 #000000 #50c0c0
#50c0c0 949 1;
#X text 238 346 baixos;
#X text 402 346 mitjos;
#X text 562 346 aguts;
#X obj 376 469 hsl 74 21 0 1 0 1 empty empty mid 7 7 0 11 #000000 #50c0c0
```

#50c0c0 1533 1;
#X obj 539 469 hsl 74 21 0 1 0 1 empty empty hi 7 7 0 11 #000000 #50c0c0
#50c0c0 2482 1;
#X obj 465 12 cnv 15 585 263 empty empty empty 20 12 0 14 #f2f2f2 #404040
0;
#X obj 215 305 hsl 50 12 0 1 0 1 empty empty f 7 7 0 11 #000000 #50c0c0
#50c0c0 0 1;
#X obj 376 305 hsl 50 12 0 1 0 1 empty empty f 7 7 0 11 #000000 #50c0c0
#50c0c0 0 1;
#X obj 539 305 hsl 50 12 0 1 0 1 empty empty f 7 7 0 11 #000000 #50c0c0
#50c0c0 0 1;
#X obj 703 469 hsl 74 21 0 1 0 1 empty empty env 7 7 0 11 #000000 #50c0c0
#50c0c0 0 1;
#X obj 47 469 hsl 74 21 0 1 0 1 empty empty tri\ \ gger\ peak 7 7 0
11 #000000 #50c0c0 #50c0c0 0 1;
#X obj 64 406 pipe 256;
#X obj 703 397 * 1;
#X obj 759 427 + 0;
#X text 78 346 triggers detected;
#N canvas 1600 33 1920 1022 (subpatch) 0;
#X obj -174 -67 loadbang, f 13;
#X obj 1 1 cnv 15 57 18 empty empty OSC 10 6 0 14 #e0e0e0 #404040 0
;
#X obj 135 18 sel 1 0;
#X obj 63 -48 inlet;
#X obj 123 186 outlet;
#X obj 192 186 outlet;
#X obj 104 93 bng 15 250 50 0 empty empty empty 17 7 0 10 #fcfcfc #000000
#000000;
#X obj 148 93 bng 15 250 50 0 empty empty empty 17 7 0 10 #fcfcfc #000000
#000000;
#X connect 2 0 6 0;
#X connect 2 1 7 0;
#X connect 3 0 2 0;
#X connect 6 0 4 0;
#X connect 7 0 5 0;
#X coords 0 1 100 -1 59 20 1 0 0;
#X restore 530 172 graph;
#X obj 530 152 hradio 15 0 1 2 empty empty disconnect_____connect
-63 6 0 10 #000000 #74a8e0 #383838 1;
#X obj 630 169 s disconnect;
#X obj 530 222 s connect;
#X msg 771 175 connect 192.168.0.100 11111;
#X text 968 121 mariana;
#X msg 771 242 connect localhost 11111;
#X msg 771 122 connect 192.168.0.104 11111;
#X obj 529 18 loadbang;
#X obj 530 68 bng 15 250 50 0 empty empty empty 17 7 0 10 #fcfcfc #000000
#000000;
#X msg 530 107 0;
#X msg 568 107 1;
#X obj 568 83 pipe 8000;
#X obj 529 43 pipe 3000;
#X msg 771 199 connect 192.168.0.101 11111;
#X msg 771 79 connect 192.168.0.105 11111;
#X text 976 77 xa;
#X text 862 155 ips wearables;
#X msg 771 40 connect 192.168.0.255 11111;
#X msg 530 196 connect 192.168.0.255 11111;
#X obj 872 345 cnv 15 181 184 empty empty empty 20 12 0 14 #78bc98
#404040 0;
#X text 797 156 esp32;
#X obj 703 427 - 0.13;
#X msg 229 547 set test;
#X obj 539 397 * 0.89;
#X obj 376 397 * 0.89;
#X obj 214 397 * 0.89;
#X obj 539 427 + 0.34;
#X obj 376 427 + 0.21;
#X obj 214 427 + 0.13;
#X text 772 17 all.devices.broadcast;
#X text 946 241 osc localhost;
#X obj 47 305 bng 21 50 10 0 empty empty empty 17 7 0 10 #000000 #fcfcfc
#000000;
#X text 974 185 etc...;
#X obj 908 359 loadbang;
#X obj 909 410 bng 15 250 50 0 empty empty empty 17 7 0 10 #fcfcfc

```
#000000 #000000;
#X msg 909 449 0;
#X msg 947 449 1;
#X msg 909 480 \; pd dsp \$1;
#X obj 947 425 pipe 5000;
#X obj 909 383 pipe 2000;
#X text 907 316 dsp auto.launch;
#N canvas 0 208 1600 847.031 Audio^______^Analisis::::::: 0;
#X obj -17 56 inlet~;
#X obj 161 56 inlet;
#X obj 161 118 bng 21 50 10 0 empty empty empty 17 7 0 10 #000000 #50c0c0
#000000;
#X obj 296 56 inlet;
#X obj 405 56 inlet;
#X obj 543 56 inlet;
#X text -27 21 adc income;
#X text 161 21 peaks;
#X text 291 21 env;
#X text 399 21 freqdetected.lo;
#X text 542 21 freqdetected.midhi;
#X obj 537 682 outlet;
#X obj 419 682 outlet;
#X obj 301 682 outlet;
#X obj 183 682 outlet;
#X obj 655 682 outlet;
#X obj 77 519 bng 21 50 10 0 empty empty empty 17 7 0 10 #000000 #50c0c0
#000000;
#X obj 732 615 hsl 50 12 0 1 0 1 empty empty env~ 7 7 0 11 #000000
#50c0c0 #50c0c0 0 1;
#X text 88 606 peaks;
#X obj 732 238 + 0;
#X obj 229 238 + 0;
#X obj 199 486 hsl 50 12 0 1 0 1 empty empty f 7 7 0 11 #000000 #50c0c0
#50c0c0 0 1;
#X obj 225 337 tgl 15 0 empty empty empty 17 7 0 10 #fcfcfc #000000
#000000 1 1;
#X obj 199 362 spigot;
#X obj 254 337 sel 0;
#X msg 254 362 0;
#X obj 391 238 + 0;
#X obj 361 486 hsl 50 12 0 1 0 1 empty empty f 7 7 0 11 #000000 #50c0c0
#50c0c0 0 1;
#X obj 384 337 tgl 15 0 empty empty empty 17 7 0 10 #fcfcfc #000000
#000000 1 1;
#X obj 361 362 spigot;
#X obj 416 337 sel 0;
#X msg 416 362 0;
#X obj 554 238 + 0;
#X obj 523 486 hsl 50 12 0 1 0 1 empty empty f 7 7 0 11 #000000 #50c0c0
#50c0c0 0 1;
#X obj 550 337 tgl 15 0 empty empty empty 17 7 0 10 #fcfcfc #000000
#000000 1 1;
#X obj 524 362 spigot;
#X obj 579 337 sel 0;
#X msg 579 362 0;
#X obj 553 265 / 16000;
#X obj 276 291 nbx 5 14 -1e+37 1e+37 0 0 empty empty empty 0 -8 0 10
#fcfcfc #000000 #000000 0 256 0;
#N canvas 948 443 450 300 analyz 0;
#X obj 191 58 bng 15 250 50 0 empty empty empty 17 7 0 10 #fcfcfc #000000
#000000;
#X obj 174 95 spigot;
#X obj 228 65 tgl 15 0 empty empty empty 17 7 0 10 #fcfcfc #000000
#000000 1 1;
#X obj 184 122 nbx 5 14 -1e+37 1e+37 0 0 empty empty empty 0 -8 0 10
#fcfcfc #000000 #000000 0 256 0;
#X obj 56 7 inlet;
#X msg 131 22 1;
#X obj 262 200 outlet;
#X obj 40 74 loadbang;
#X obj 191 33 metro 233;
#X obj 279 153 spigot;
#X obj 333 123 tgl 15 0 empty empty empty 17 7 0 10 #fcfcfc #000000
#000000 0 1;
#X obj 354 60 inlet;
#X connect 0 0 2 0;
#X connect 1 0 3 0;
```

```
#X connect 2 0 1 1;
#X connect 3 0 9 0;
#X connect 4 0 1 0;
#X connect 5 0 8 0;
#X connect 7 0 5 0;
#X connect 8 0 0 0;
#X connect 9 0 6 0;
#X connect 10 0 9 1;
#X connect 11 0 10 0;
#X restore 275 265 pd analyz;
#X obj 238 310 < 1;
#X obj 399 310 < 1;
#X obj 560 310 < 1;
#X obj 523 419 * 1;
#X obj 199 419 * 1;
#X obj 227 265 / 300;
#X obj 266 614 hsl 50 12 0 1 0 1 empty empty f 7 7 0 11 #000000 #50c0c0
#50c0c0 0 1;
#X obj 430 614 hsl 50 12 0 1 0 1 empty empty f 7 7 0 11 #000000 #50c0c0
#50c0c0 0 1;
#X obj 593 615 hsl 50 12 0 1 0 1 empty empty f 7 7 0 11 #000000 #50c0c0
#50c0c0 0 1;
#X obj 361 419 * 1;
#X obj 218 452 + 0;
#X obj 380 452 + 0;
#X obj 543 452 + 0;
#X obj 388 265 / 5000;
#X obj -18 191 env~;
#X obj -18 214 nbx 5 14 -1e+37 1e+37 0 0 empty empty empty 0 -8 0 10
#fcfcfc #000000 #000000 0 256 0;
#X obj -18 265 tgl 32 0 empty empty empty 17 7 0 10 #fcfcfc #000000
#000000 1 1;
#X obj -18 300 sel 1;
#X msg -18 328 0;
#X msg 14 328 1;
#X obj 116 494 tgl 36 0 empty empty empty 17 7 0 10 #fcfcfc #000000
#000000 0 1;
#X obj 266 572 spigot;
#X obj -18 242 < 48;
#X obj 167 564 sel 0;
#X msg 167 593 0;
#X obj 430 572 spigot;
#X obj 593 572 spigot;
#X obj 732 572 spigot;
#X obj 77 572 spigot;
#X obj 55 188 env~;
#X obj 55 215 nbx 5 14 -1e+37 1e+37 0 0 empty empty empty 0 -8 0 10
#fcfcfc #000000 #000000 0 256 0;
#X obj 322 241 tgl 21 0 empty empty empty 17 7 0 10 #fcfcfc #000000
#000000 0 1;
#X obj 444 291 nbx 5 14 -1e+37 1e+37 0 0 empty empty empty 0 -8 0 10
#fcfcfc #000000 #000000 0 256 0;
#N canvas 948 443 450 300 analyz 0;
#X obj 191 58 bng 15 250 50 0 empty empty empty 17 7 0 10 #fcfcfc #000000
#000000;
#X obj 174 95 spigot;
#X obj 228 65 tgl 15 0 empty empty empty 17 7 0 10 #fcfcfc #000000
#000000 1 1;
#X obj 184 122 nbx 5 14 -1e+37 1e+37 0 0 empty empty empty 0 -8 0 10
#fcfcfc #000000 #000000 0 256 0;
#X obj 56 7 inlet;
#X msg 131 22 1;
#X obj 262 200 outlet;
#X obj 40 74 loadbang;
#X obj 191 33 metro 233;
#X obj 279 153 spigot;
#X obj 333 123 tgl 15 0 empty empty empty 17 7 0 10 #fcfcfc #000000
#000000 0 1;
#X obj 354 60 inlet;
#X connect 0 0 2 0;
#X connect 1 0 3 0;
#X connect 2 0 1 1;
#X connect 3 0 9 0;
#X connect 4 0 1 0;
#X connect 5 0 8 0;
#X connect 7 0 5 0;
#X connect 8 0 0 0;
```

```
#X connect 9 0 6 0;
#X connect 10 0 9 1;
#X connect 11 0 10 0;
#X restore 443 265 pd analyz;
#X obj 490 241 tgl 21 0 empty empty empty 17 7 0 10 #fcfcfc #000000
#000000 0 1;
#X obj 617 291 nbx 5 14 -1e+37 1e+37 0 0 empty empty empty 0 -8 0 10
#fcfcfc #000000 #000000 0 256 0;
#N canvas 948 443 450 300 analyz 0;
#X obj 191 58 bng 15 250 50 0 empty empty empty 17 7 0 10 #fcfcfc #000000
#000000;
#X obj 174 95 spigot;
#X obj 228 65 tgl 15 0 empty empty empty 17 7 0 10 #fcfcfc #000000
#000000 1 1;
#X obj 184 122 nbx 5 14 -1e+37 1e+37 0 0 empty empty empty 0 -8 0 10
#fcfcfc #000000 #000000 0 256 0;
#X obj 56 7 inlet;
#X msg 131 22 1;
#X obj 262 200 outlet;
#X obj 40 74 loadbang;
#X obj 191 33 metro 233;
#X obj 279 153 spigot;
#X obj 333 123 tgl 15 0 empty empty empty 17 7 0 10 #fcfcfc #000000
#000000 0 1;
#X obj 354 60 inlet;
#X connect 0 0 2 0;
#X connect 1 0 3 0;
#X connect 2 0 1 1;
#X connect 3 0 9 0;
#X connect 4 0 1 0;
#X connect 5 0 8 0;
#X connect 7 0 5 0;
#X connect 8 0 0 0;
#X connect 9 0 6 0;
#X connect 10 0 9 1;
#X connect 11 0 10 0;
#X restore 616 265 pd analyz;
#X obj 663 241 tgl 21 0 empty empty empty 17 7 0 10 #fcfcfc #000000
#000000 0 1;
#X obj 593 452 * 1;
#X obj 593 486 + 0;
#X obj 430 486 + 0;
#X obj 266 486 + 0;
#X obj 430 452 * 1;
#X obj 266 452 * 1;
#X obj 66 756 cnv 15 734 5 empty empty empty 20 12 0 14 #000000 #404040
0;
#X obj 55 57 inlet~;
#X obj 55 266 tgl 32 0 empty empty empty 17 7 0 10 #fcfcfc #000000
#000000 0 1;
#X obj 55 301 sel 1;
#X msg 55 329 0;
#X msg 87 329 1;
#X obj 55 243 < 48;
#X obj -89 -6 cnv 15 89 17 empty empty empty 20 12 0 14 #e0e0e0 #404040
0;
#X connect 0 0 55 0;
#X connect 1 0 2 0;
#X connect 2 0 16 0;
#X connect 3 0 19 0;
#X connect 4 0 20 0;
#X connect 5 0 26 0;
#X connect 5 0 32 0;
#X connect 16 0 69 0;
#X connect 17 0 15 0;
#X connect 19 0 68 0;
#X connect 20 0 46 0;
#X connect 22 0 23 1;
#X connect 22 0 24 0;
#X connect 23 0 45 0;
#X connect 23 0 84 0;
#X connect 24 0 25 0;
#X connect 25 0 45 0;
#X connect 25 0 84 0;
#X connect 26 0 54 0;
#X connect 28 0 29 1;
#X connect 28 0 30 0;
```

```
#X connect 29 0 50 0;
#X connect 29 0 83 0;
#X connect 30 0 31 0;
#X connect 31 0 50 0;
#X connect 31 0 83 0;
#X connect 32 0 38 0;
#X connect 34 0 35 1;
#X connect 34 0 36 0;
#X connect 35 0 44 0;
#X connect 35 0 79 0;
#X connect 36 0 37 0;
#X connect 37 0 44 0;
#X connect 37 0 79 0;
#X connect 38 0 35 0;
#X connect 38 0 43 0;
#X connect 38 0 77 0;
#X connect 40 0 39 0;
#X connect 41 0 22 0;
#X connect 42 0 28 0;
#X connect 43 0 34 0;
#X connect 44 0 53 0;
#X connect 45 0 51 0;
#X connect 46 0 23 0;
#X connect 46 0 40 0;
#X connect 46 0 41 0;
#X connect 47 0 13 0;
#X connect 48 0 12 0;
#X connect 49 0 11 0;
#X connect 50 0 52 0;
#X connect 51 0 21 0;
#X connect 52 0 27 0;
#X connect 53 0 33 0;
#X connect 54 0 29 0;
#X connect 54 0 42 0;
#X connect 54 0 74 0;
#X connect 55 0 56 0;
#X connect 56 0 63 0;
#X connect 57 0 58 0;
#X connect 58 0 59 0;
#X connect 58 1 60 0;
#X connect 59 0 61 0;
#X connect 60 0 61 0;
#X connect 61 0 62 1;
#X connect 61 0 64 0;
#X connect 61 0 66 1;
#X connect 61 0 67 1;
#X connect 61 0 68 1;
#X connect 61 0 69 1;
#X connect 62 0 47 0;
#X connect 63 0 57 0;
#X connect 64 0 65 0;
#X connect 65 0 47 0;
#X connect 65 0 48 0;
#X connect 65 0 49 0;
#X connect 65 0 17 0;
#X connect 66 0 48 0;
#X connect 67 0 49 0;
#X connect 68 0 17 0;
#X connect 69 0 14 0;
#X connect 70 0 71 0;
#X connect 72 0 40 1;
#X connect 74 0 73 0;
#X connect 75 0 74 1;
#X connect 77 0 76 0;
#X connect 78 0 77 1;
#X connect 79 0 80 0;
#X connect 80 0 67 0;
#X connect 81 0 66 0;
#X connect 82 0 62 0;
#X connect 83 0 81 0;
#X connect 84 0 82 0;
#X connect 86 0 70 0;
#X connect 87 0 88 0;
#X connect 88 0 89 0;
#X connect 88 1 90 0;
#X connect 89 0 61 0;
#X connect 90 0 61 0;
```

```
#X connect 91 0 87 0;
#X restore 130 220 pd Audio^_____^Analisis::::::: ~~~~~;
#X connect 2 0 111 2;
#X connect 2 1 111 3;
#X connect 2 2 111 4;
#X connect 2 3 111 5;
#X connect 3 0 2 1;
#X connect 5 0 8 0;
#X connect 6 0 5 3;
#X connect 7 0 92 0;
#X connect 9 0 5 2;
#X connect 11 0 14 0;
#X connect 12 0 11 3;
#X connect 13 0 39 0;
#X connect 15 0 11 2;
#X connect 16 0 19 0;
#X connect 17 0 16 3;
#X connect 18 0 40 0;
#X connect 20 0 16 2;
#X connect 21 0 24 0;
#X connect 22 0 21 3;
#X connect 23 0 41 0;
#X connect 25 0 21 2;
#X connect 26 0 22 0;
#X connect 26 0 17 0;
#X connect 26 0 12 0;
#X connect 26 0 6 0;
#X connect 26 0 31 0;
#X connect 28 0 66 0;
#X connect 30 0 33 0;
#X connect 31 0 30 3;
#X connect 32 0 42 0;
#X connect 34 0 64 0;
#X connect 35 0 64 0;
#X connect 35 0 65 0;
#X connect 36 0 35 0;
#X connect 37 0 30 2;
#X connect 39 0 11 1;
#X connect 40 0 16 1;
#X connect 41 0 21 1;
#X connect 42 0 30 1;
#X connect 43 0 2 0;
#X connect 44 0 45 0;
#X connect 45 0 3 0;
#X connect 46 0 2 0;
#X connect 46 0 111 0;
#X connect 53 0 5 0;
#X connect 57 0 11 0;
#X connect 58 0 16 0;
#X connect 60 0 95 0;
#X connect 61 0 94 0;
#X connect 62 0 93 0;
#X connect 63 0 21 0;
#X connect 64 0 30 0;
#X connect 65 0 34 0;
#X connect 66 0 91 0;
#X connect 69 0 88 0;
#X connect 69 1 71 0;
#X connect 70 0 69 0;
#X connect 77 0 82 0;
#X connect 78 0 79 0;
#X connect 78 0 81 0;
#X connect 79 0 70 0;
#X connect 80 0 70 0;
#X connect 81 0 80 0;
#X connect 82 0 78 0;
#X connect 88 0 72 0;
#X connect 91 0 63 0;
#X connect 92 0 5 1;
#X connect 93 0 96 0;
#X connect 94 0 97 0;
#X connect 95 0 98 0;
#X connect 96 0 58 0;
#X connect 97 0 57 0;
#X connect 98 0 53 0;
#X connect 101 0 36 0;
#X connect 103 0 109 0;
```

```
#X connect 104 0 105 0;
#X connect 104 0 108 0;
#X connect 105 0 107 0;
#X connect 106 0 107 0;
#X connect 108 0 106 0;
#X connect 109 0 104 0;
#X connect 111 0 101 0;
#X connect 111 1 60 0;
#X connect 111 2 61 0;
#X connect 111 3 62 0;
#X connect 111 4 28 0;
#X restore 78 97 pd Audio·2·OSC-__-transmission;
```

2.1

By default the **Audio2OSC-__-transmission.pd** *will perform an Audio analysis detecting the following ingredients:*

__percussive volume peaks / triggers (osc ID tri)
__detection of low frequencies (osc ID lo)
__detection of medium frequencies (osc ID mid)
__detection of high frequencies (osc ID hi)
__Render the envelope / similar to the current volumes (osc ID env)
OSC is sent directly to the entire network via port 11111, with the message
[connect 192.168.0.255 11111[
The Audio will be set automatically <u>although before we must open the patch from the same RPi3 with a screen or Projector to set the SoundCard (SoundBlaster input 1 channel + SoundBlaster output 2 channels) and that it is always configured.</u>

3

Create autostart **pd.desktop** *file in /home/pi/.config/autostart*
note **.config** *is a hidden dir, you can view it with Control+H (hide)*
By default the autostart folder will be created, otherwise you can create it.

Open a text editor (you will find one in the Raspbian system menu) and copy the following content:

```
(Desktop Entry)
Type=Application
Name=Pd.App
Exec=/usr/bin/puredata /home/pi/Desktop/Audio2OSC-__-transmission.pd
Terminal=true
```

Save it in /home/pi/.config/autostart
note: The file you wanted to autorun will be the one that appears in the Exec line, in this case /usr/bin/puredata starts Pd and the specific patch with its path, in this case /home/pi/Desktop/Audio2OSC-__-transmission.pd

4

Be sure all previous files has execution permissions. In a terminal execute :

```
chmod +x /home/pi/.config/autostart/pd.autostart
chmod +x /home/pi/Desktop/Audio2OSC-__-transmission.pd
```

5

***Reboot** the system and we will have the patch working.*
If it doesn't work, verify that all files have execution permissions, and that soundCard is already set up.

. ..具.. .
Espais Latents

LAB & TUTORIALS

How 2 Build an Automat with RPI

How 2 Build an Automat with RPI

by Xavi.Manzanares @xamanza // 2025

Ingredients :
__Raspberry Pi model 3A (XS projects)
ò
__Raspberry Pi model 4 2Gb (S projects)
ò
__Raspberry Pi model 4 4Gb (M projects)
ò
__Raspberry Pi model 5 4Gb (L projects)

__Creative Sound Blaster Play!3 USB SoundCard
__Targeta microSD de 16gb o 32gb (best)

Download Raspbian OS

Raspbian OS > https://downloads.raspberrypi.com/raspios_arm64/images/raspios_arm64-2025-
(recomenat Software Imager per a copiar .img en una targeta micro SD de 16gb o 32gb
Imager >
desde linux OS https://downloads.raspberrypi.org/imager/imager_latest_amd64.deb
desde win OS https://downloads.raspberrypi.org/imager/imager_latest.exe
desde mac OS https://downloads.raspberrypi.org/imager/imager_latest.dmg

1

Arrencar la Rpi amb un teclat, mouse i pantalla.
Instal·lar **Pd vanilla**
Obrir terminal i executar

sudo apt-get install puredata

2

copiar el teu projecte.pd (patx) (per exemple via pendrive) a una ubicació concreta de la Rpi, per exemple el desktop. Si no canvies el nom de la maquina per defecte es diu pi

/home/pi/Desktop/elmeupatch.pd

3

Crear l'Arxiu d'Autoarrancada **pd.desktop** a /home/pi/.config/autostart
nota .config és un dir ocult, podeu visualitzar-lo amb Control+H (amagar)
Per defecte la carpeta autostart estarà creada, en cas contrari la pots crear.
Obrir un editor de text (en el menú del sistema Raspbian la trobareu per exemple 'leafpad' o altres editors) i copiar el següent contingut:

```
(Desktop Entry)
Type=Application
Name=Pd.App
Exec=/usr/bin/puredata /home/pi/Desktop/elmeupatch.pd
Terminal=true
```

Salvar-lo a **/home/pi/.config/autostart**
nota: L'arxiu que volgueu autoarrencar serà el que apareixerà a la línia Exec, en aquest cas /usr/bin/puredata arrenca Pd i l'aplicació custom amb la seva ruta /home/pi/Desktop/elmeupatch.pd

4

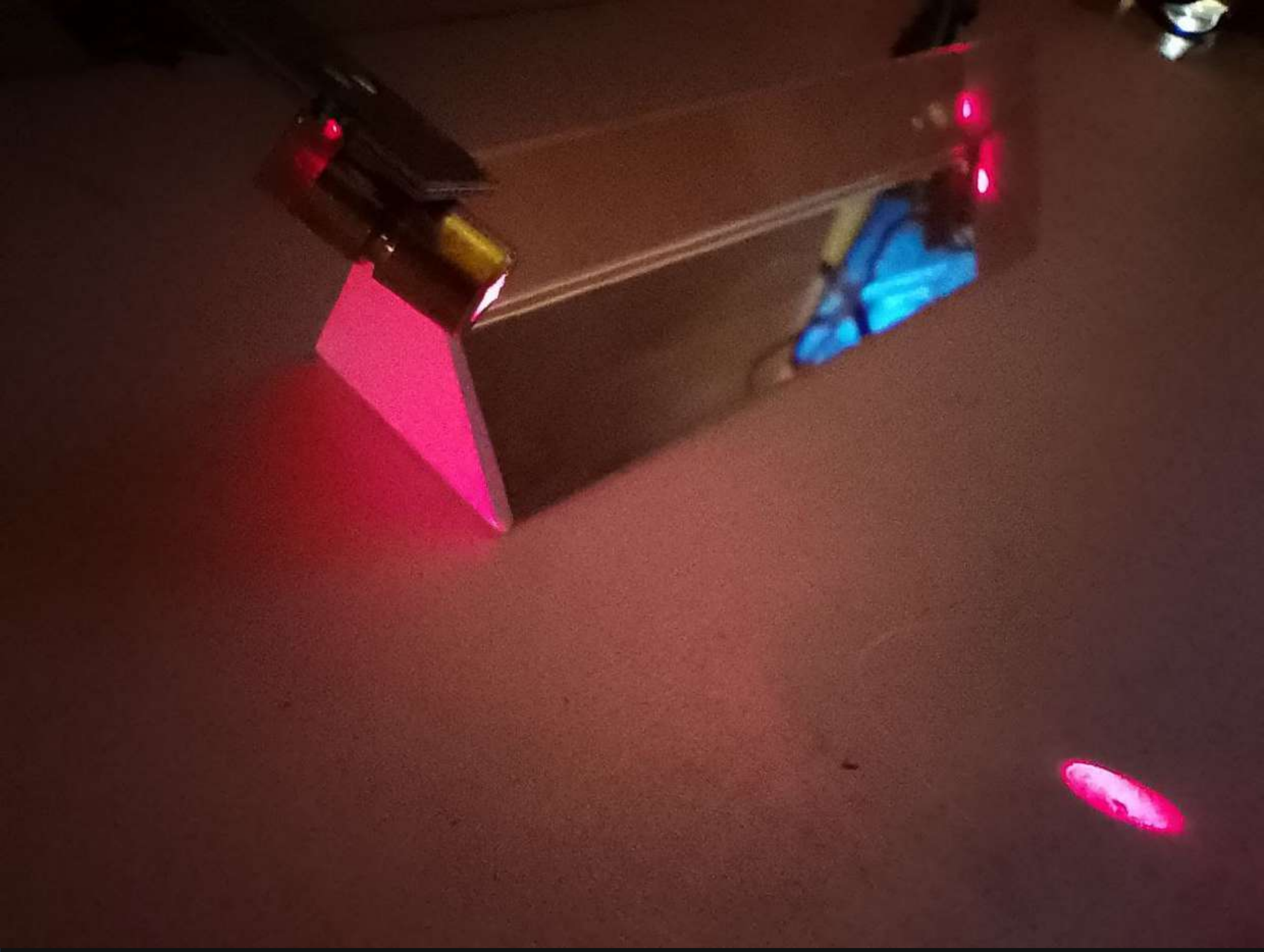
Assegureu-vos que tots els fitxers anteriors tinguin permisos d'execució. En un terminal executeu:

```
chmod +x /home/pi/.config/autostart/pd.autostart
chmod +x /home/pi/Desktop/Audio2OSC-__-transmission.pd
```

5

Obrir el patch amb la targeta de so connectada i fer el setup de la mateixa a Pd.
La entrada de la soundblaster te un canal (mono) i la sortida té dos. (stereo).
Un cop configurar, Salvem diverses vegades (save + apply changes) per a assegurar que sempre trobi la targeta quan arranqui automàticament.
Comprovar el pas 5 al següent reboot del sistema.

...I ja tindrem l'autòmata funcionant ^_^



. ..具.. .

Espais Latents

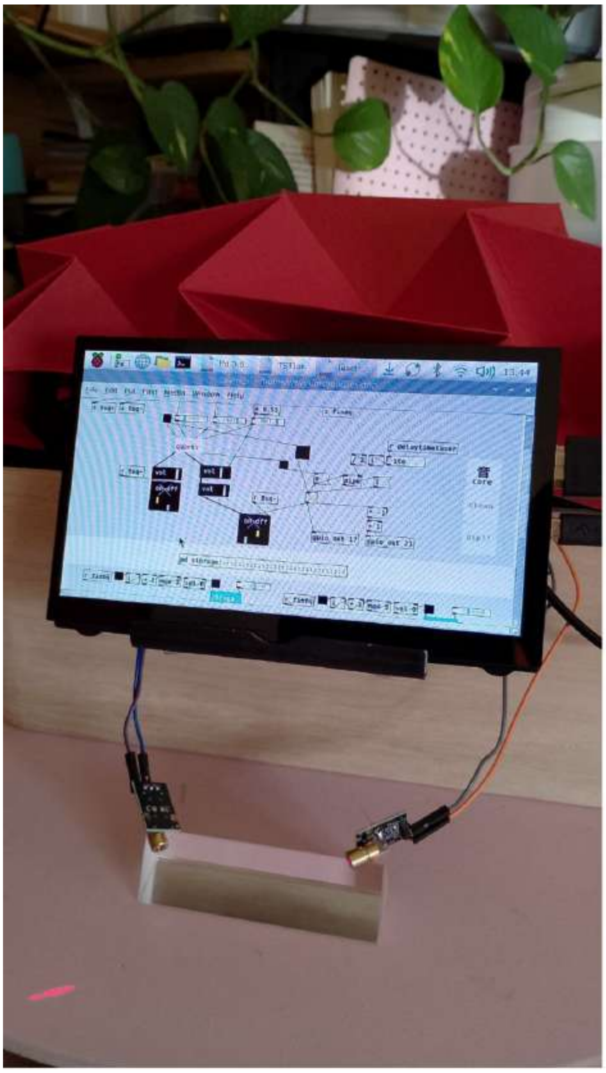
LAB

Reflexions Laser Xperiment

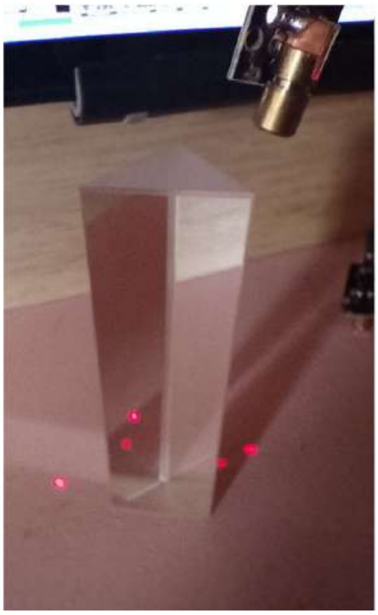
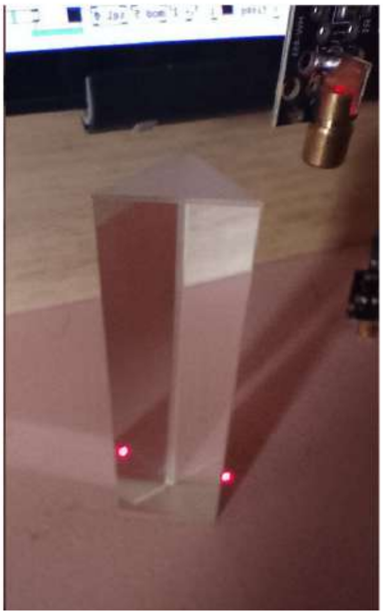
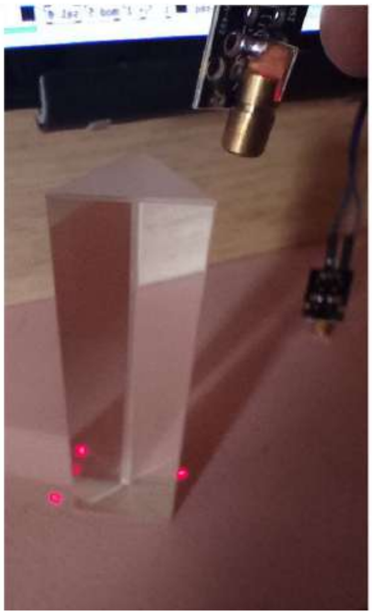
experiments & prototips

: // Laser·Reflexions

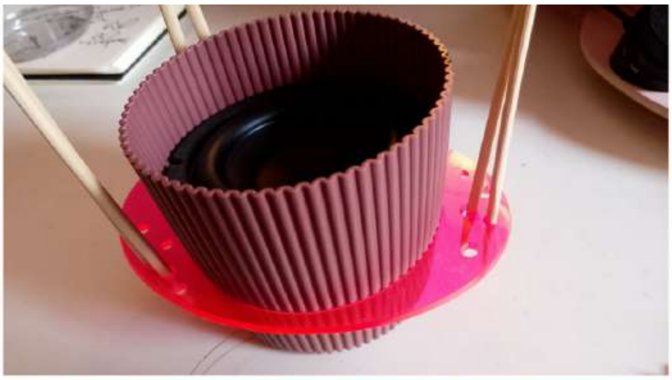
Hiwgfbuiofyhx ew eowhf cohfp ewrhe gfuioehugf rekghiuerhgiu erhgeurhgiohergerkgbhijrehgh rg eg reg hriehgfrhg ehgp euihgiuhj gihre iohwgjrehg irfewghurheg iujrfeghuperhgi rehg rpeoifg eohfgj roeghjro engjrbegew

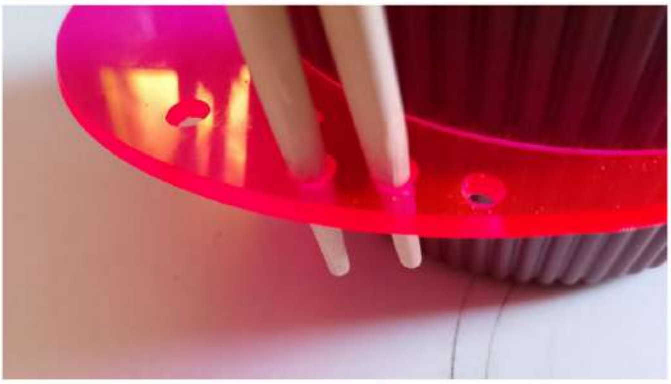


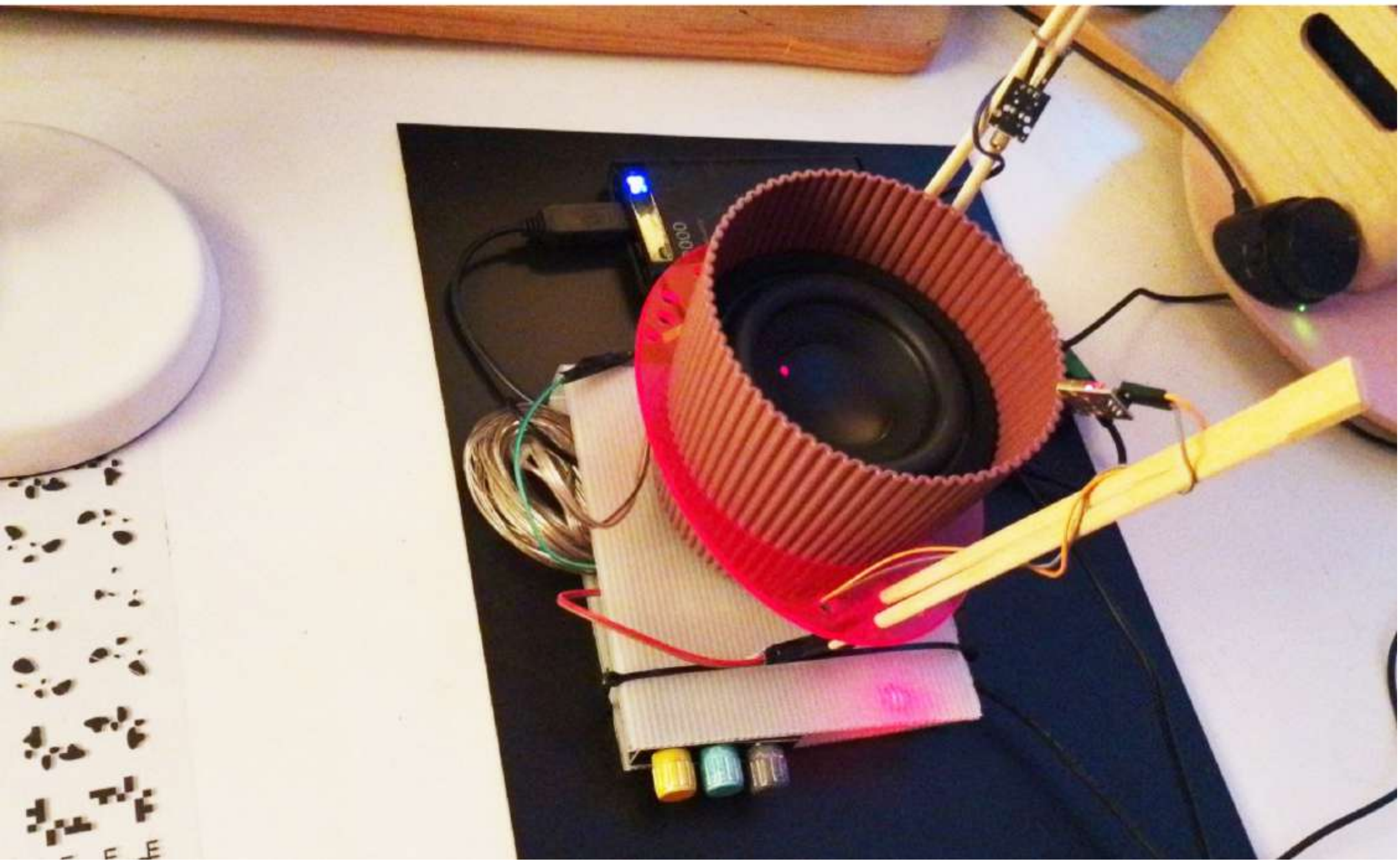



KRK

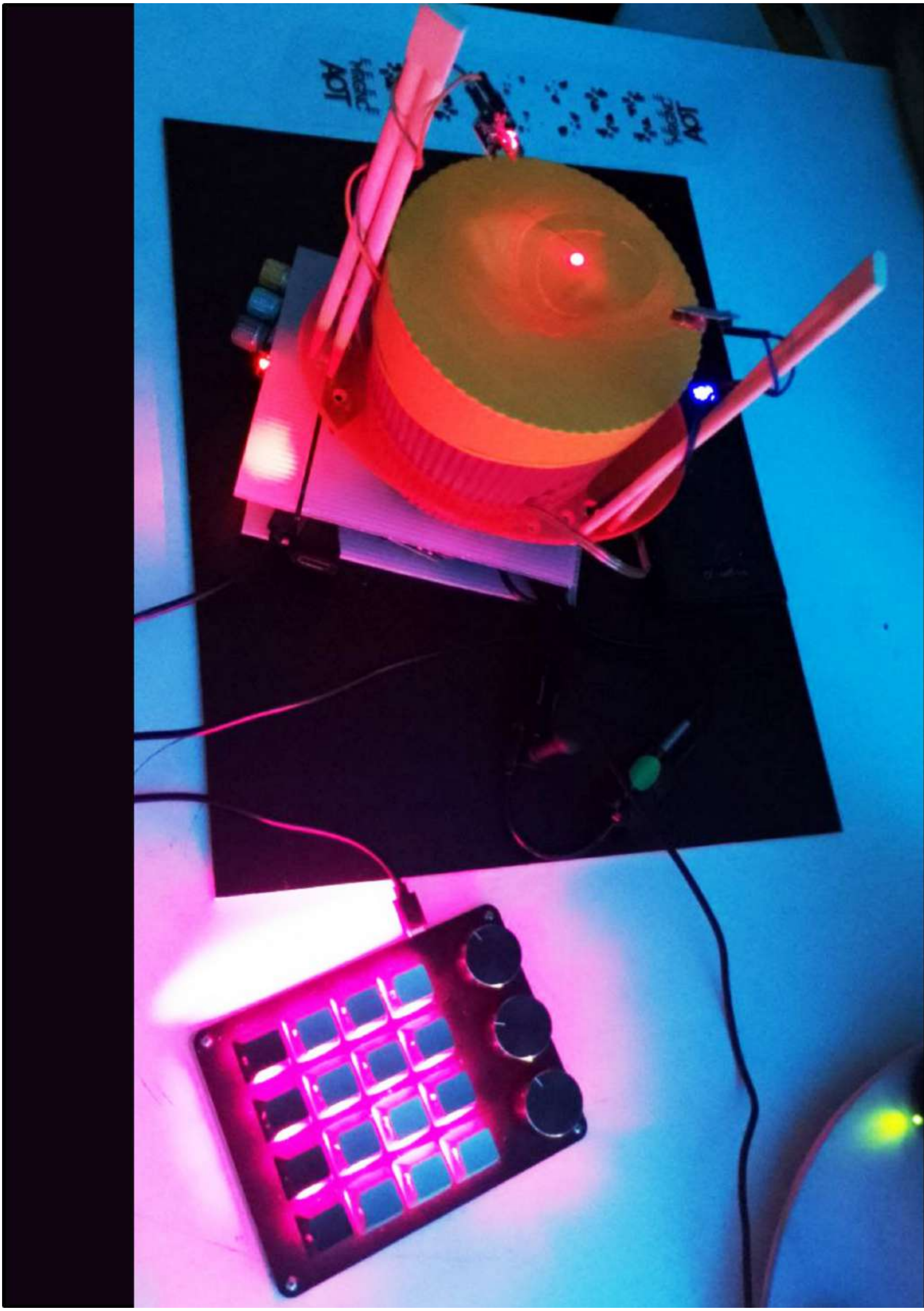


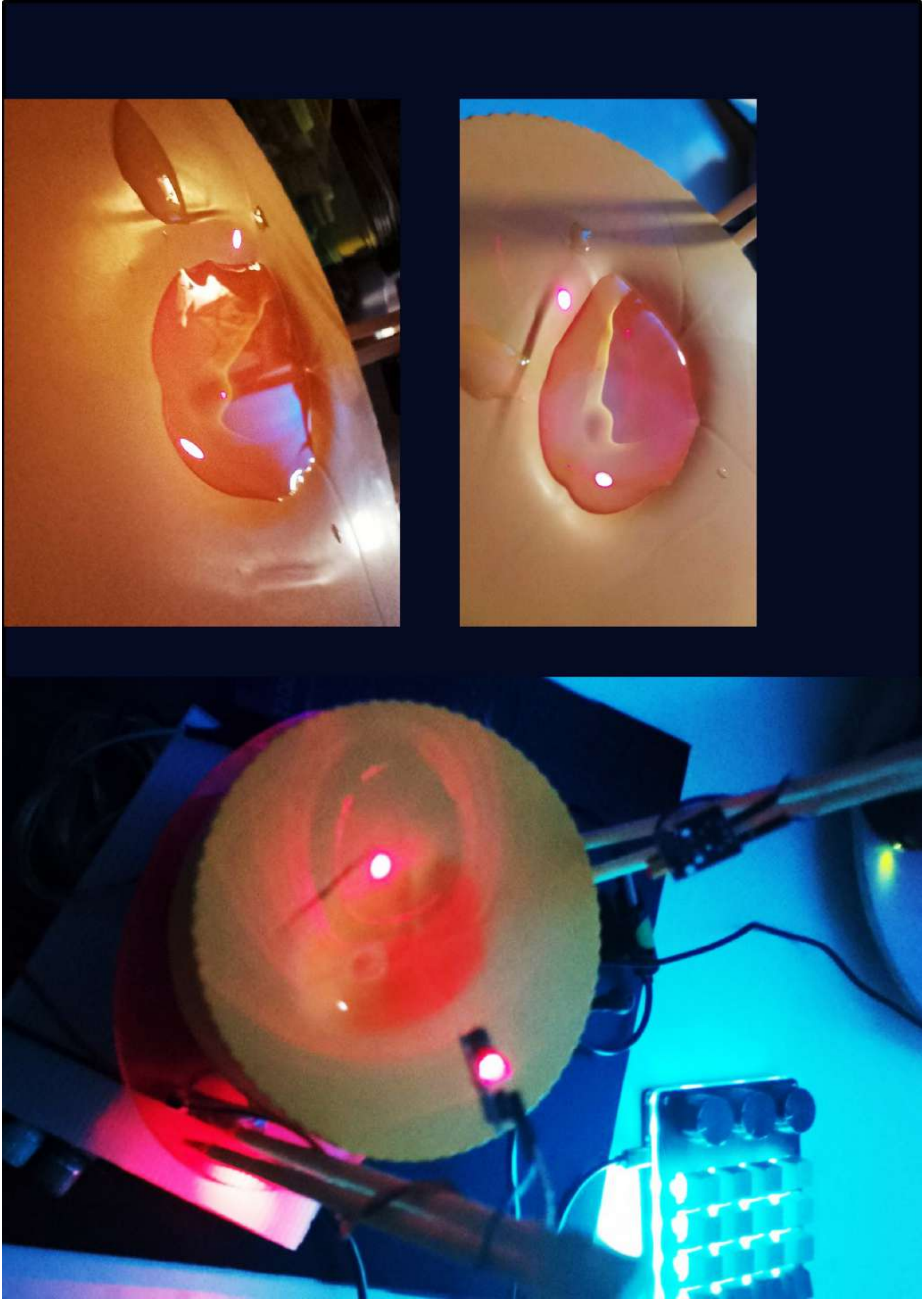










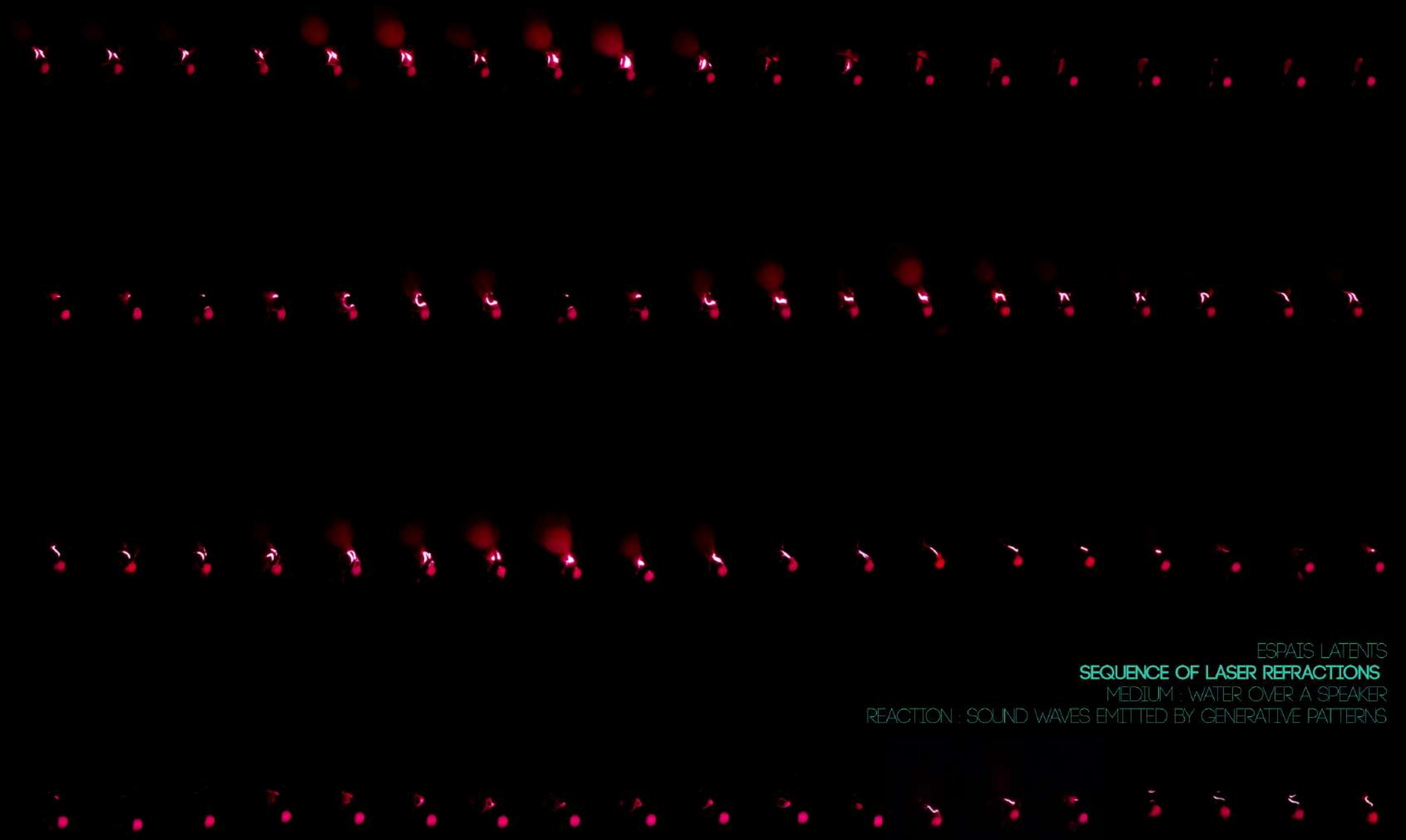



ESPAIS LATENTS

SEQUENCE OF LASER REFRACTIONS

MEDIUM : WATER OVER A SPEAKER

REACTION : SOUND WAVES EMITTED BY GENERATIVE PATTERNS


ESPAIS LATENTS
SEQUENCE OF LASER REFRACTIONS
MEDIUM : WATER OVER A SPEAKER
REACTION : SOUND WAVES EMITTED BY GENERATIVE PATTERNS

ESPAIS LATENTS

SEQUENCE OF LASER REFRACTIONS

MEDIUM : WATER OVER A SPEAKER

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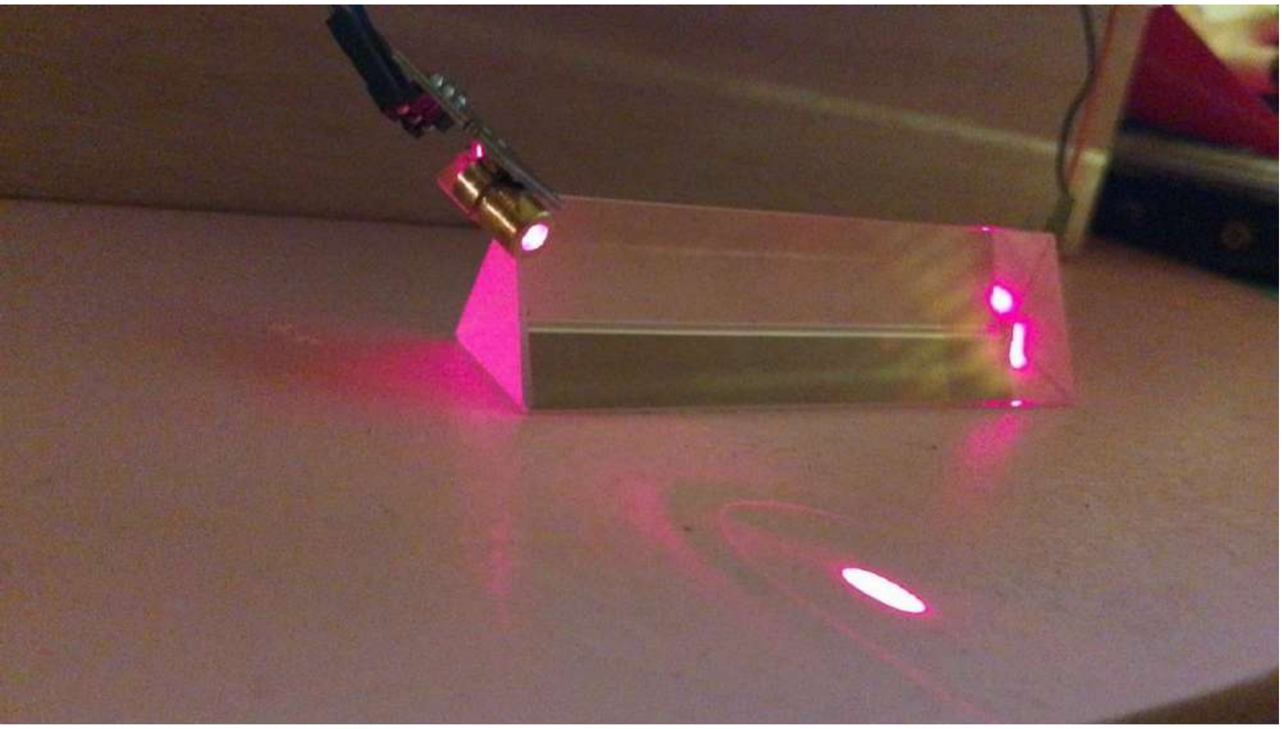
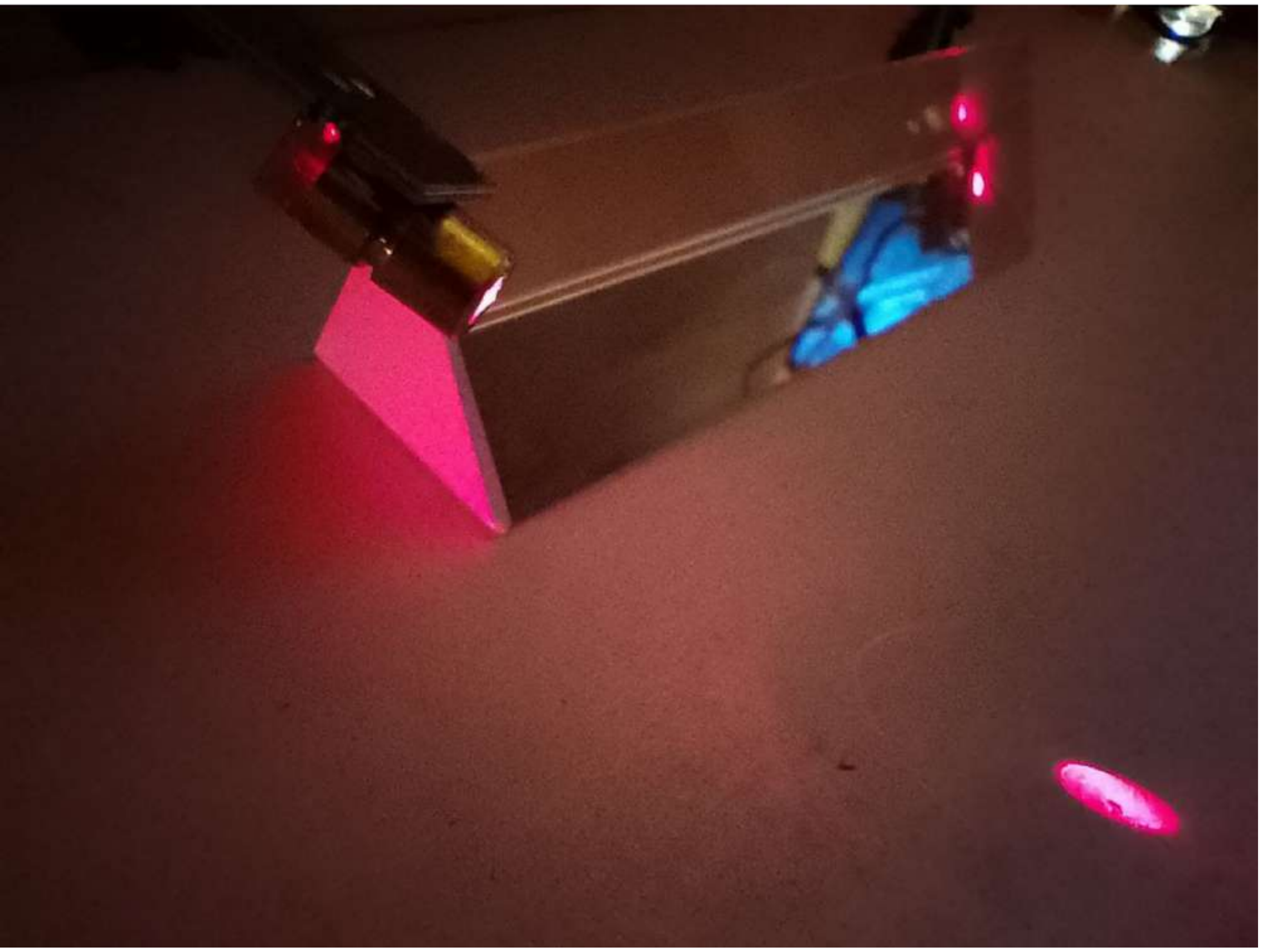
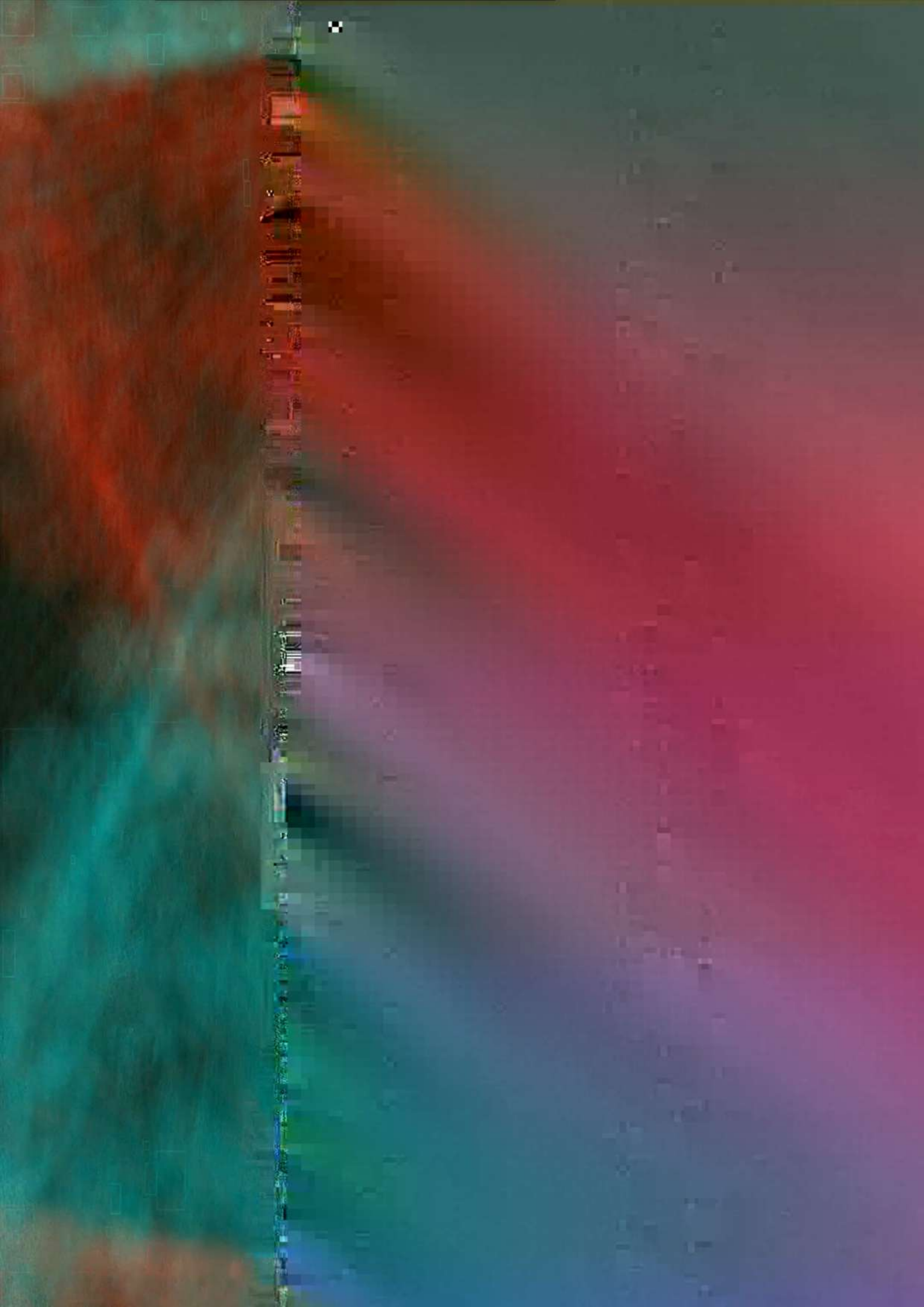
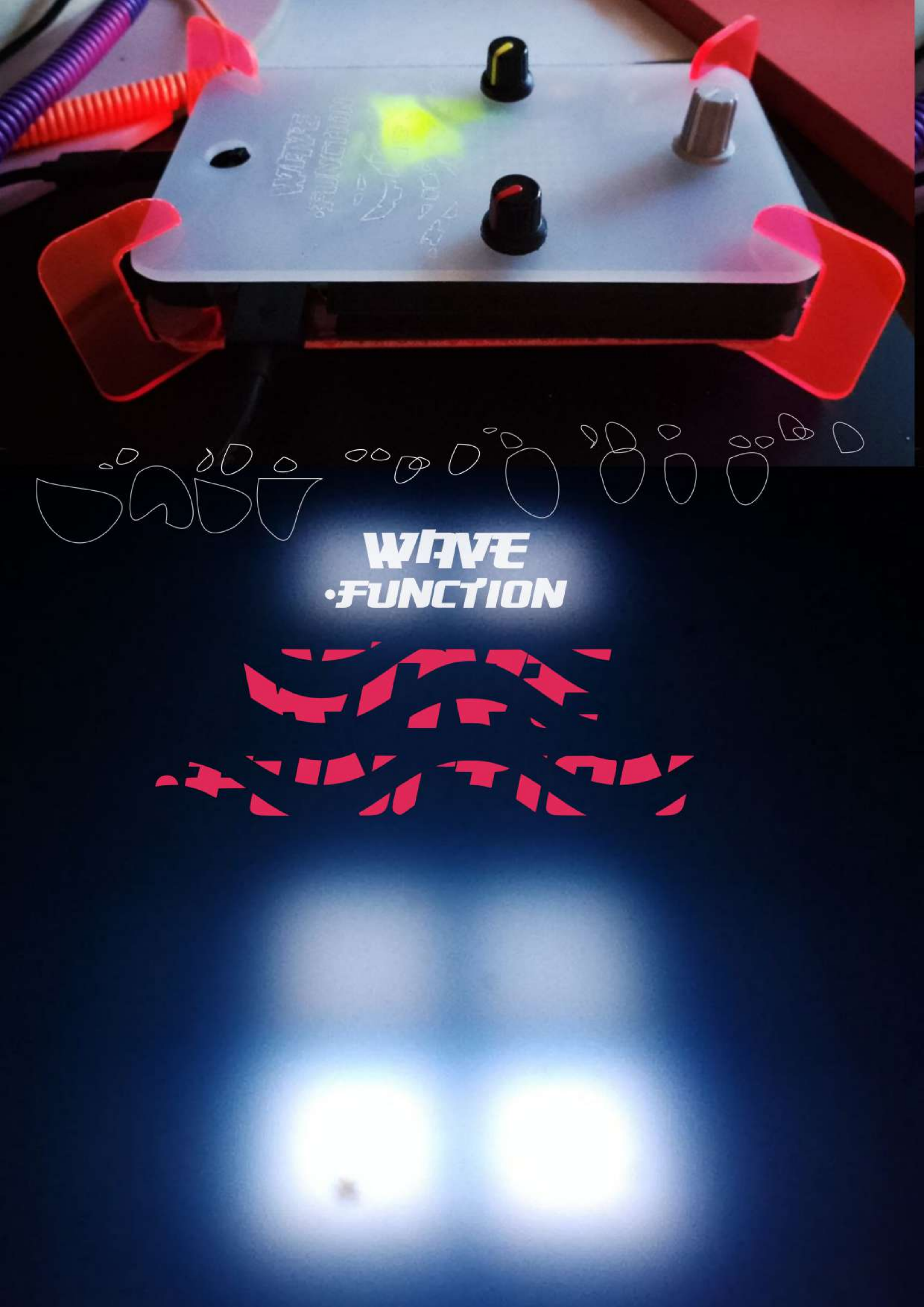
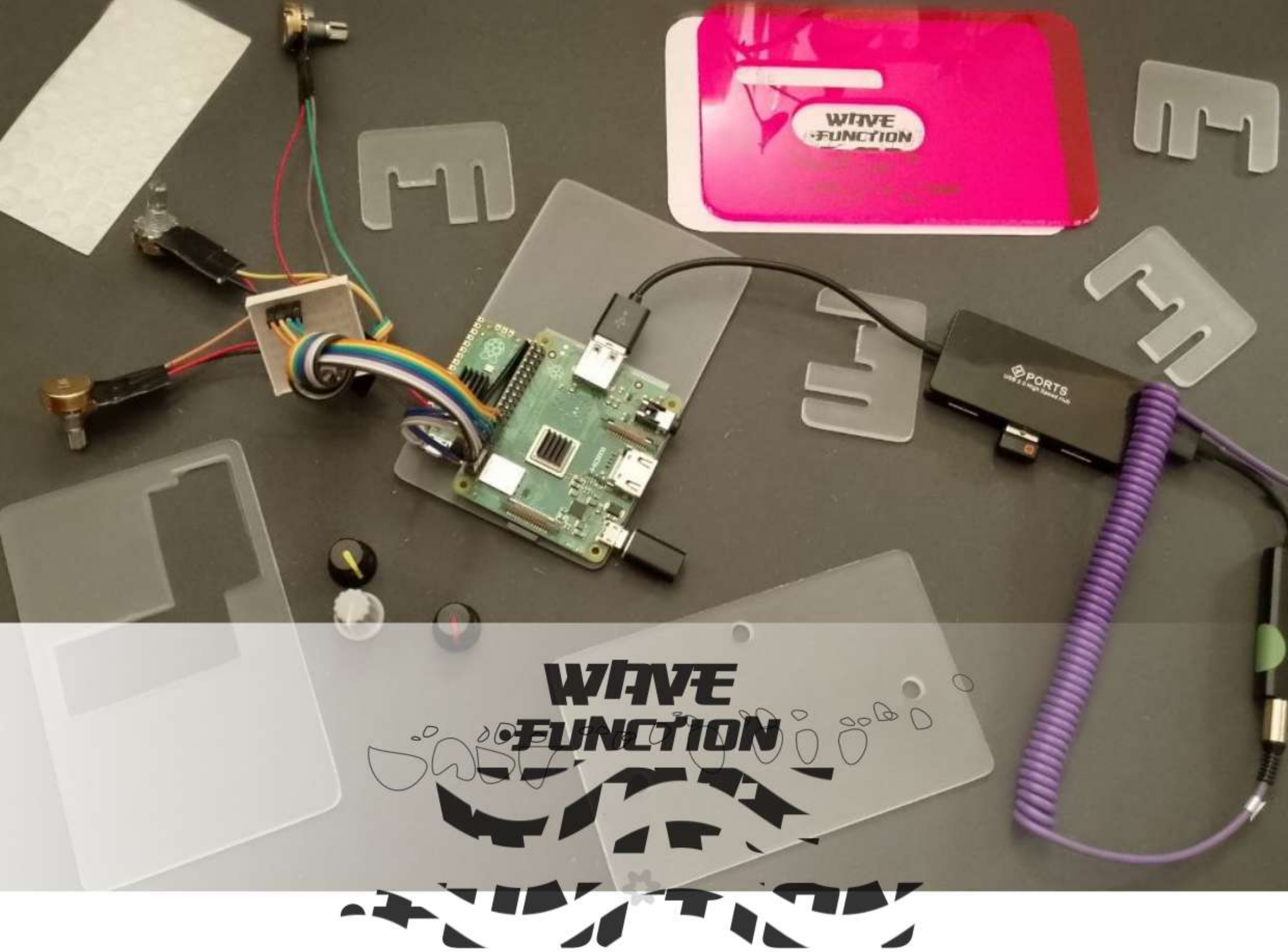


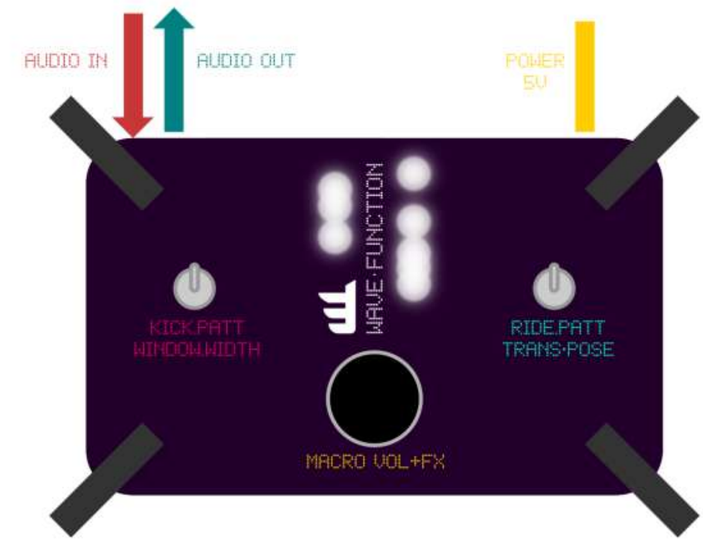


```
#include <stdio.h>
int main() { printf("Aloha!\n"); return 0; }
```




WAVE
•FUNCTION


WAVE
FUNCTION


AUDIO IN
AUDIO OUT
POWER
5V
WAVE·FUNCTION
KICK.PATT
WINDOW.WIDTH
RIDE.PATT
TRANS·POSE
MACRO VOL+FX


WAVE
FUNCTION
VOL+FX

LAB & TUTORIALS

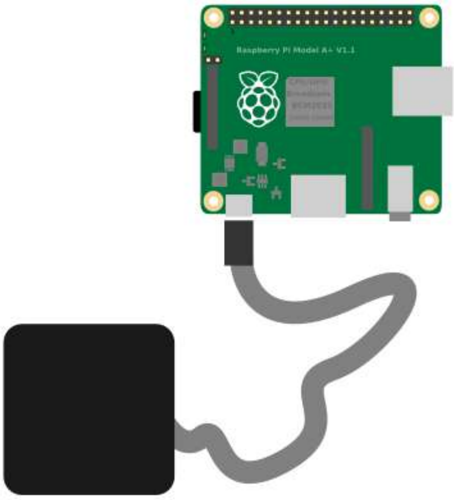
HOW 2 BUILD A POST FX GNRTV ALGORYTHMIC PEDAL FROM SCRATCH

note :
advanced electronic arts / creative coding skills required

. ..具.. .

Espais Latents
Xa.Manzanares
2026
@xamanza

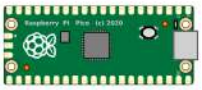
INGREDIENTS



RASPBERRY PI 3A+

POWER 5V 3A MICRO USB

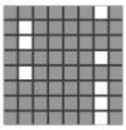
MINISD CARD 16GB



RASPBERRY PICO

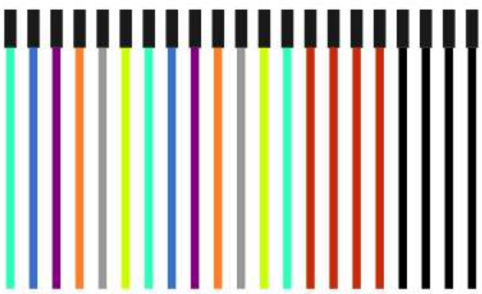
3KNOBS 10K

1 DIGITAL BUTTON 4 LEGS



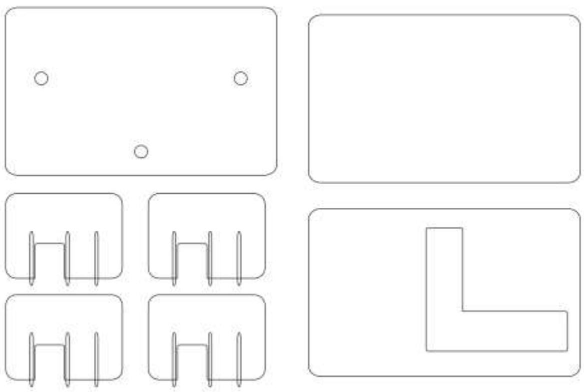
1 MATRIX LED KEM 12288AW

USB SOUND CARD SOUNDBLASTER PLAY3

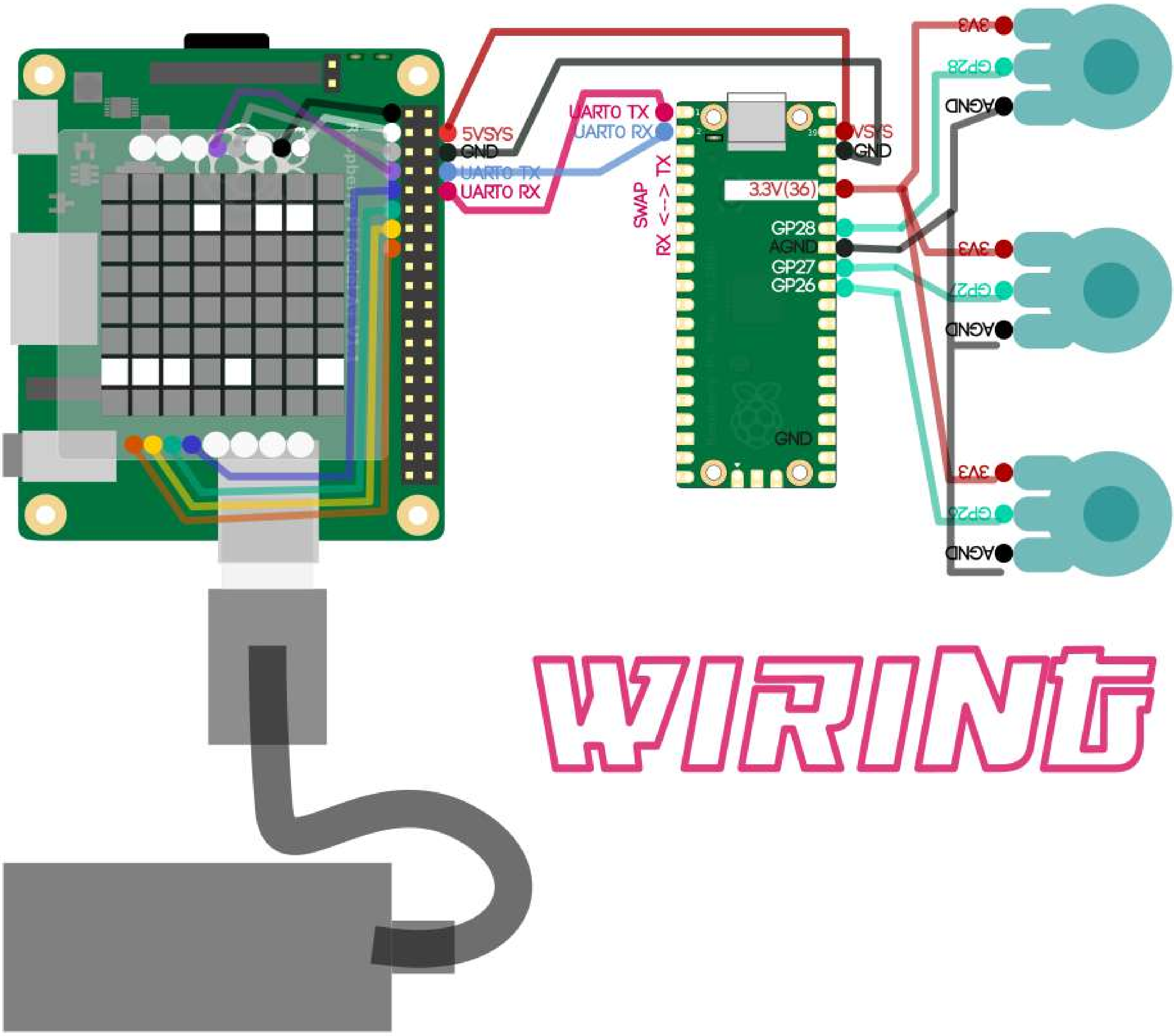


WIRING JUMPERS ABOUT 23

OPTIONAL CASE
VECTOR TO CNC CUT IN THIS DOCUMENT

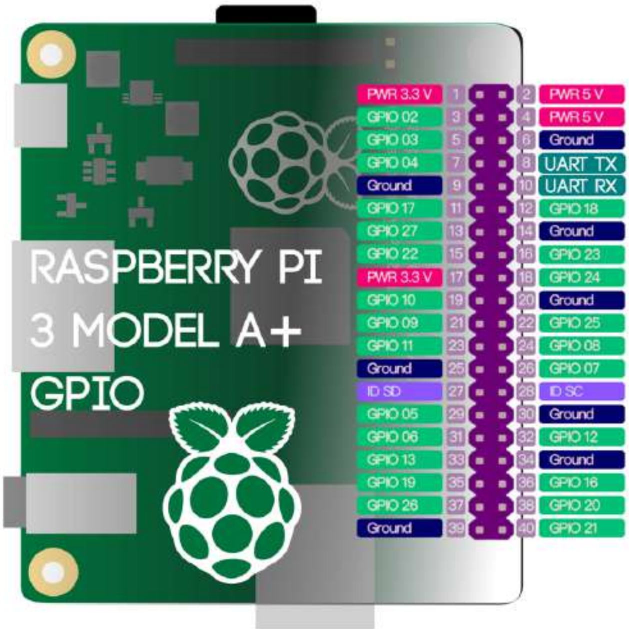


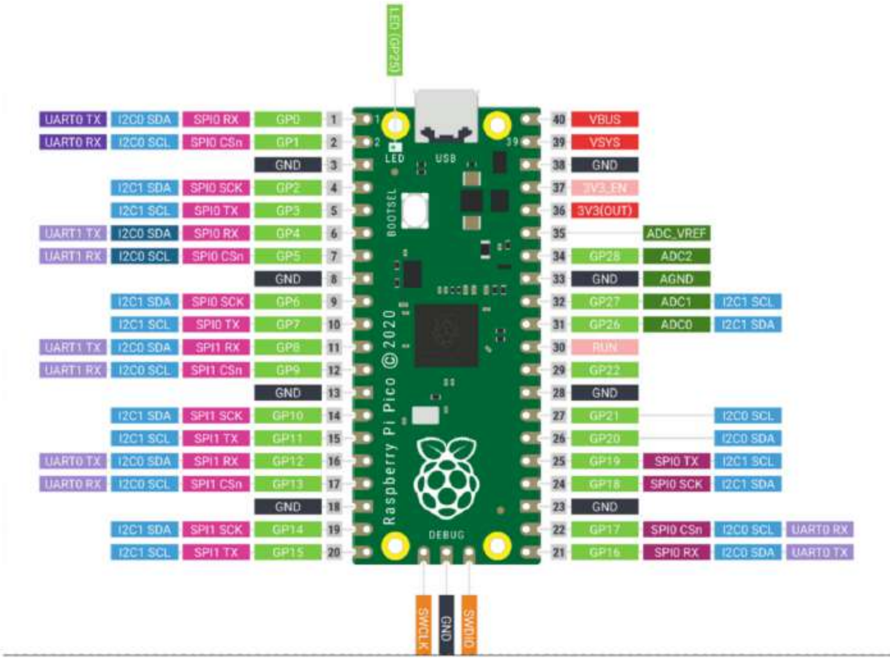
Before Coding these are the hardware connections and solderings :



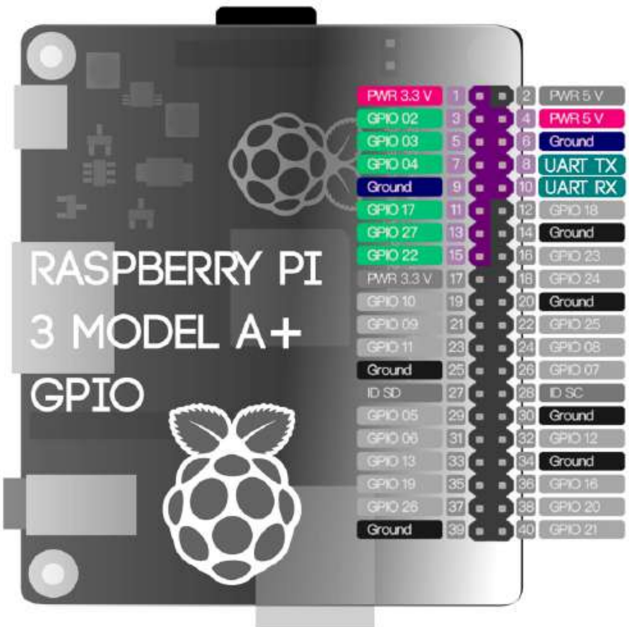
Notes:
All ports in the RPi3A can be solved without soldering using wiring jumpers .
Instead, all ports in the pico should be soldered unless you purchased a pico with soldered headers and then use wiring jumpers between both pcbs.

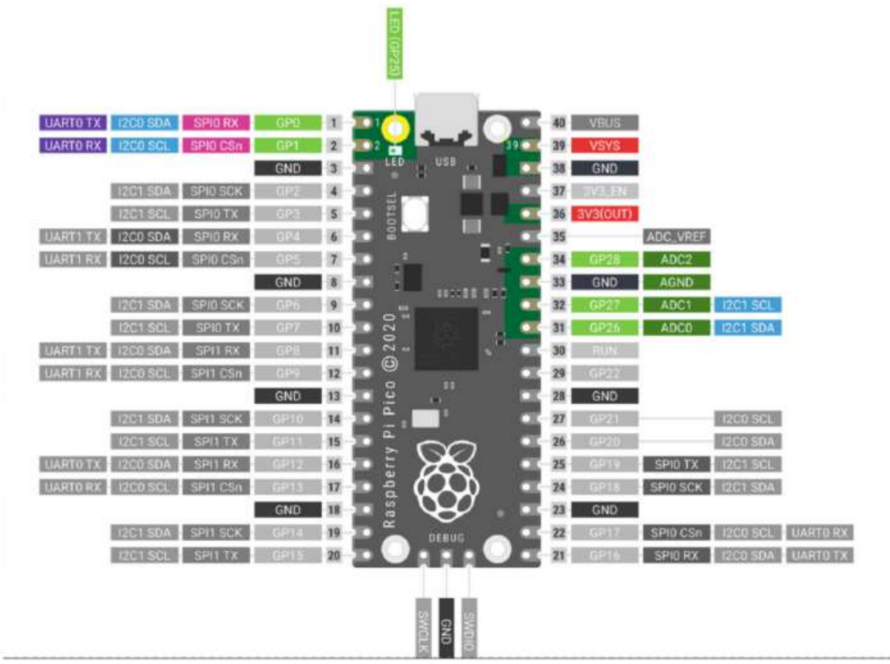
Full GPIO ports IDs





GPIO ports used for this project





Development Environment Setup

This project combines three main software layers:

1. Raspberry Pi Pico firmware (C / Pico SDK)

2. Pure Data externals C @ RaspberryPi 3A+

3. Pure Data / gnrtv.cells algorythmic programming

1. Raspberry Pi Pico firmware (C / Pico SDK)

Layer 1 development environment was tested primarily on linux mint (Debian / Ubuntu based distributions), although most firmware compilation steps can also be reproduced on Windows. (+ info page 15)

Required packages:
desktop environment dev, install by terminal the following packages
sudo apt update
sudo apt install build-essential
sudo apt install cmake
sudo apt install git
sudo apt install gcc
sudo apt install g++
sudo apt install make

2. Pure Data externals C @ RaspberryPi 3A+

Layer 2 development environment was tested primarily on
Raspbian GNU/Linux 13 (Trixie) 32 bits
You will need a monitor (via HDMI) mouse and keyboard for the RPI3A. Since RPi3A have only one usb port youll need a usb hub if you want to connect several hardware elements at once. (mouse keyboard sound card etc)
raspberry environment dev, install by terminal the following packages
sudo apt install update
sudo apt install gcc
sudo apt install g++
sudo apt install make
sudo apt install libpigpio-dev
sudo apt install pigpio
sudo apt install puredata
sudo apt install puredata-dev

These packages provide:
* GCC compiler
* Make
* CMake
* pigpio libraries
* Pure Data headers for compiling externals

/// Workflow Tips : Remote Control between your Laptop and Pi3A

In order to work with those environments and computers is very useful to remotely control your RaspPi 3 from your laptop. For this purpose, you have to be activated the ssh connection in the RPi3 :

Terminal @RPI3
sudo raspi-config
(go to 3 interface connections and then > I1 enable remote command line)

Exit or close the terminal with the mouse.
Then check your IP in the raspberry pi 3A in a terminal
ifconfig
it must appear in the something like (among a lot of more info)
wlan0 192.168.1.111

This will be the ip that once did this process yopu can easily connect your laptop and your pi.

In the laptop try :
Terminal

ssh pi@192.168.1.111
(or whatever ip was detected)
note : pi or the name you called your raspberry pi3A
after login with the password you'll have remote access via terminal and you can operate many things from your laptop remotely.

In addition you can use nautilus or Nemo or any other filemanager instead of the terminal.
In the location path bar you ll have to search or type >
sftp://pi@192.168.1.111/

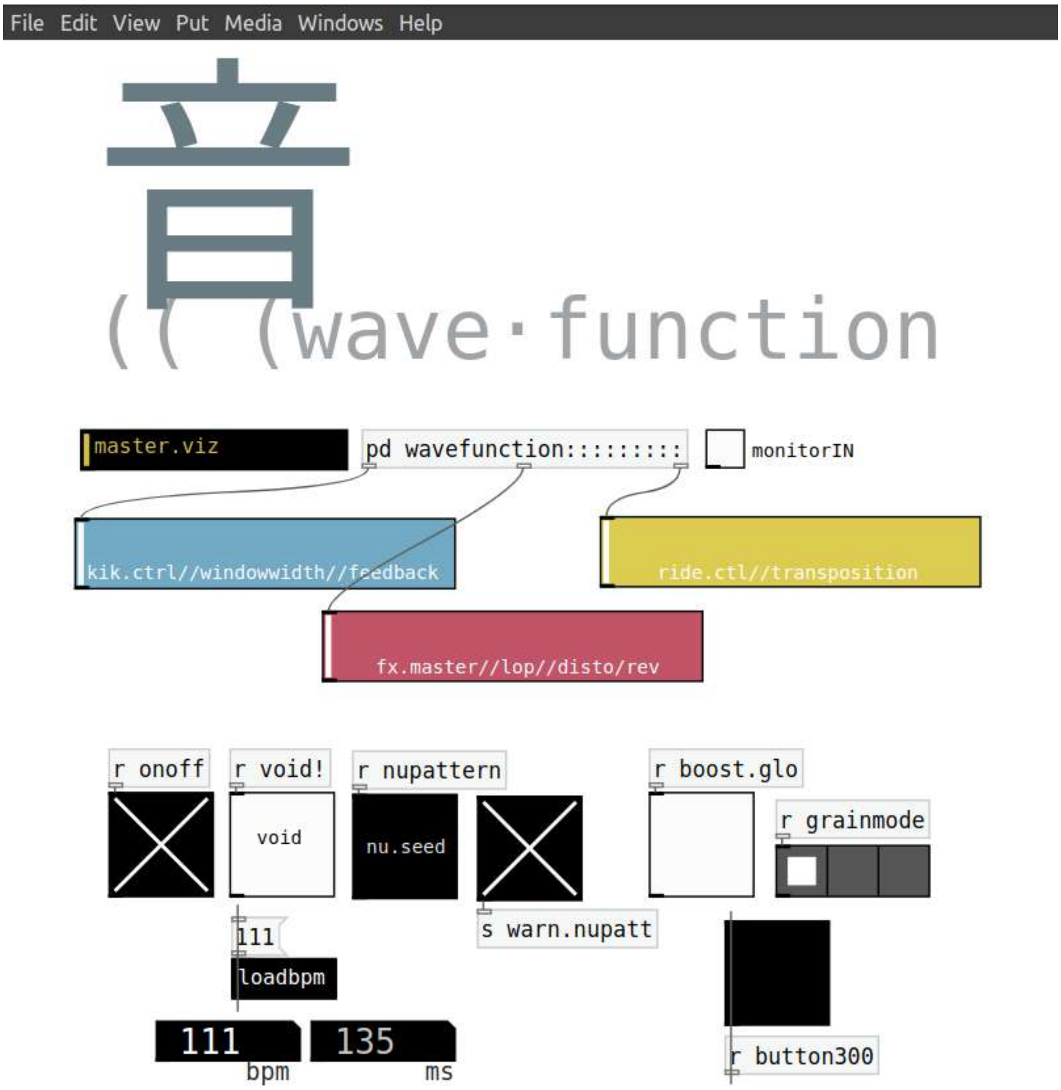
Those both methods (terminal or file manager) is very useful to develop and debug back and forth since well need many times operate between machines.

3. Pure Data / gnrtv.cells algorythmic programming

You can build your algorythm in your laptop and then copy it in any location /home/pi/Desktop/ for example.

The one which drives *wavefunction* is a pretty complex algorythm :

It works as a 'pedal' or a post FX processor unit that deconstructs and texturizes the incoming audio signal with granular synthesis techniques. When no audio input is processed, it behaves as a generative pattern synth, creating minimal beats crossing over musical genres like (minimal/deep/dub)techno, glitch and soundscaping.

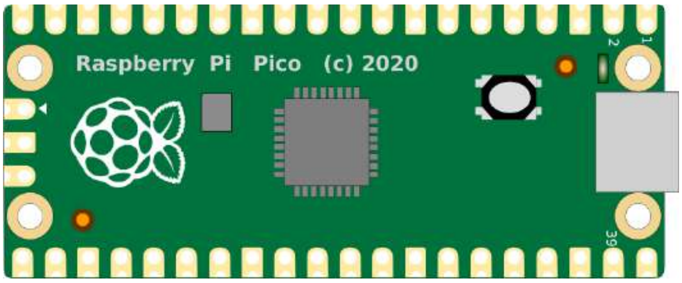


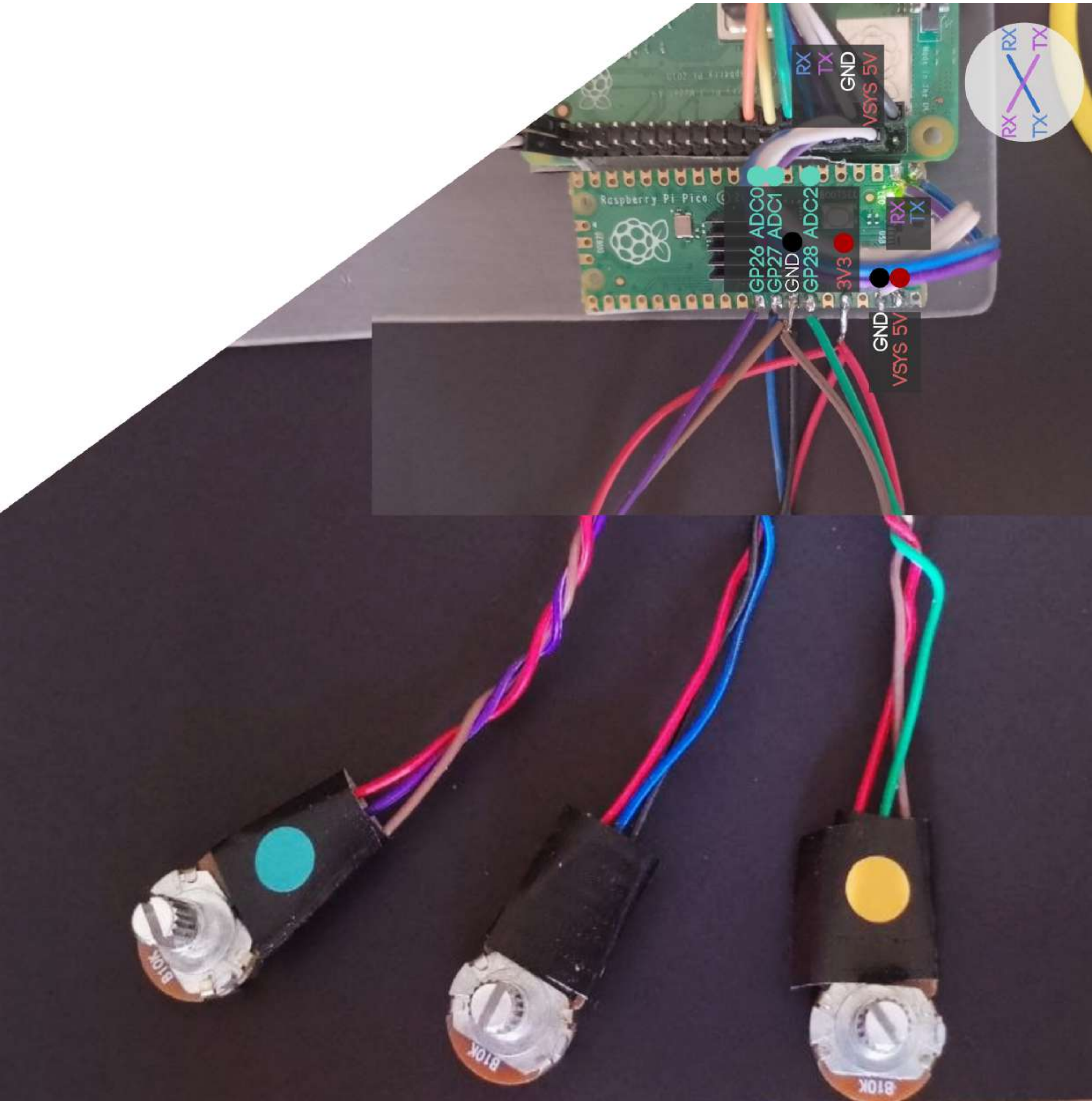
Once you have the app you want to load, check in ATOMATION BOOT section in order to make your app automatically loaded.

1

RASPBERRY PI PICO FIRMWARE (C / PICO SDK)

1. Raspberry Pi Pico firmware (C / Pico SDK)





1. Raspberry Pi Pico firmware (C / Pico SDK)

BEFORE START
Required packages desktop environment dev
Tested primarily on linux mint laptop (Debian / Ubuntu based)
laptop terminal

```
sudo apt update
sudo apt install build-essential
sudo apt install cmake
sudo apt install git
sudo apt install gcc
sudo apt install g++
sudo apt install make
```

SETUP @laptop
Clone the Pico SDK:
laptop terminal

```
git clone https://github.com/raspberrypi/pico-sdk.git
```

Set the SDK path:
laptop terminal

```
export PICO_SDK_PATH=/path/to/pico-sdk
```

note
/path/to/pico-sdk for example */home/user/*pico-sdk or whatever location you downloaded the git

(Optional)
Add the variable permanently:
laptop terminal

```
echo 'export PICO_SDK_PATH=/path/to/pico-sdk' >> ~/.bashrc
source ~/.bashrc
```

COMPILING
make a project folder > ex. Pico.Firmware
copy the C scripts (next pages) as tst.c and CmakeLists.txt inside

[PicoFirmware folder project]
tst.c
CmakeLists.txt

After this, create the **build** folder
laptop terminal
mkdir build

[PicoFirmware folder project]
tst.c
CmakeLists.txt
[build]

Once build is created change directory to 'build' folder
laptop terminal
cd build

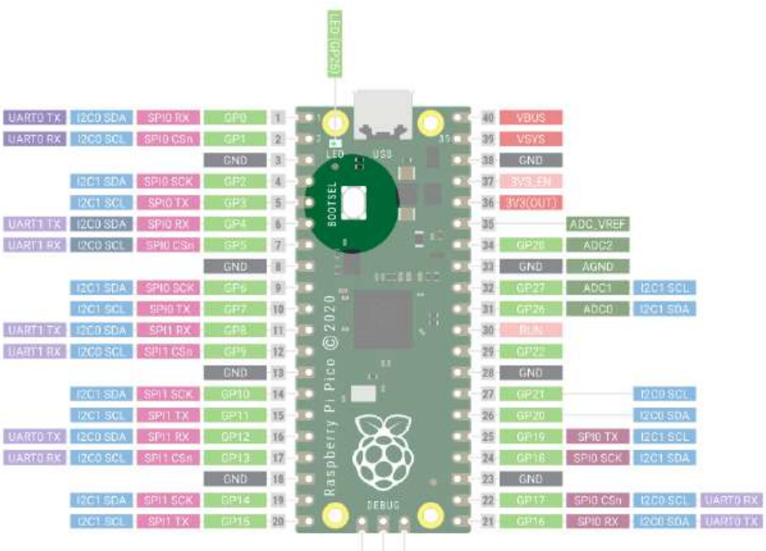
and then we compile >
laptop terminal
cmake
make

After compiling, search for .uf2 result in build folder among several files >
tst.uf2

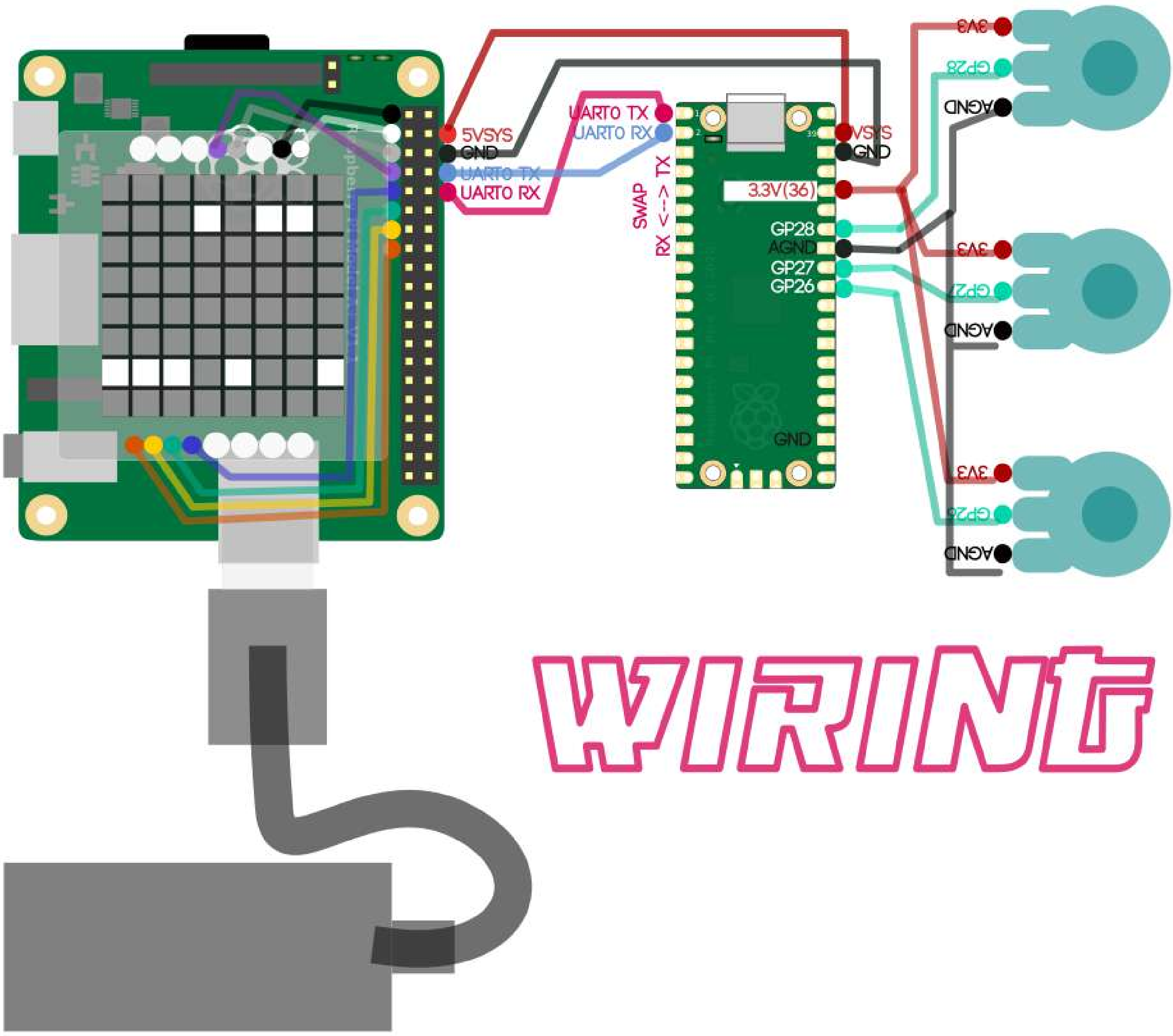
tst.uf2 is the firmware for Pico Board

UPLOAD FIRMWARE
Upload firmware to pico by:

1. Holding BOOTSEL on the Pico
2. Connecting USB to your laptop
3. Copying the generated UF2 file to the mounted Pico drive



The following C code (tst.c + makefileis the one in charge of managing the analog Input ports of the RaspberryPico in order to connect 3 Knobs (potentiometers) and sending back to the main computer (RaspberryPi 3A+) via serial communication throught the RX TX GPIO ports.
Once compiled it will be the firmware of the RaspberryPico.
*Pico will be powered trough **GND** and 5V SYS GPIO ports from the main computer.*



CODE

Pico Firmware C
Requires
tst.c code file
CMakeLists.txt

tst.c code file

```
#include <stdio.h>
#include "pico/stdlib.h"
#include "hardware/uart.h"
#include "hardware/adc.h"
#include "hardware/gpio.h"

#define UART_ID uart0
#define BAUD_RATE 115200

#define UART_TX_PIN 0
#define UART_RX_PIN 1

#define LED_PIN 25

// LED states
#define LED_FAST 200
#define LED_SLOW 2000

static absolute_time_t last_led_toggle;
static bool led_state = false;

static int mode = 0; // 0 = fast, 1 = slow, 2 = on, 3 = off

void set_led_mode(int m) {
  mode = m;
}

void led_task() {

  int interval = LED_SLOW;

  if (mode == 0) interval = LED_FAST;
  else if (mode == 1) interval = LED_SLOW;
  else if (mode == 2) {
    gpio_put(LED_PIN, 1);
    return;
  }
  else if (mode == 3) {
    gpio_put(LED_PIN, 0);
    return;
  }

  if (absolute_time_diff_us(last_led_toggle, get_absolute_time()) > interval * 1000) {
    led_state = !led_state;
    gpio_put(LED_PIN, led_state);
    last_led_toggle = get_absolute_time();
  }
}

int read_adc(int channel) {
  adc_select_input(channel);
  return adc_read();
}
```

```
void send_knobs() {

    int k1 = read_adc(0); // GP26
    int k2 = read_adc(1); // GP27
    int k3 = read_adc(2); // GP28

    char buf[64];

    sprintf(buf, "1:%d\n", k1);
    uart_puts(UART_ID, buf);

    sprintf(buf, "2:%d\n", k2);
    uart_puts(UART_ID, buf);

    sprintf(buf, "3:%d\n", k3);
    uart_puts(UART_ID, buf);

    set_led_mode(0); // streaming fast
}

int main() {

    uart_init(UART_ID, BAUD_RATE);

    gpio_set_function(UART_TX_PIN, GPIO_FUNC_UART);
    gpio_set_function(UART_RX_PIN, GPIO_FUNC_UART);

    uart_set_format(UART_ID, 8, 1, UART_PARITY_NONE);
    uart_set_fifo_enabled(UART_ID, false);

    gpio_init(LED_PIN);
    gpio_set_dir(LED_PIN, GPIO_OUT);

    adc_init();
    adc_gpio_init(26);
    adc_gpio_init(27);
    adc_gpio_init(28);

    sleep_ms(1000);

    last_led_toggle = get_absolute_time();

    while (1) {

        send_knobs();
        led_task();

        sleep_ms(50);
    }
}
```

CODE

CmakeLists.txt file

```
cmake_minimum_required(VERSION 3.13)

include(pico_sdk_import.cmake)

project(tst C CXX ASM)

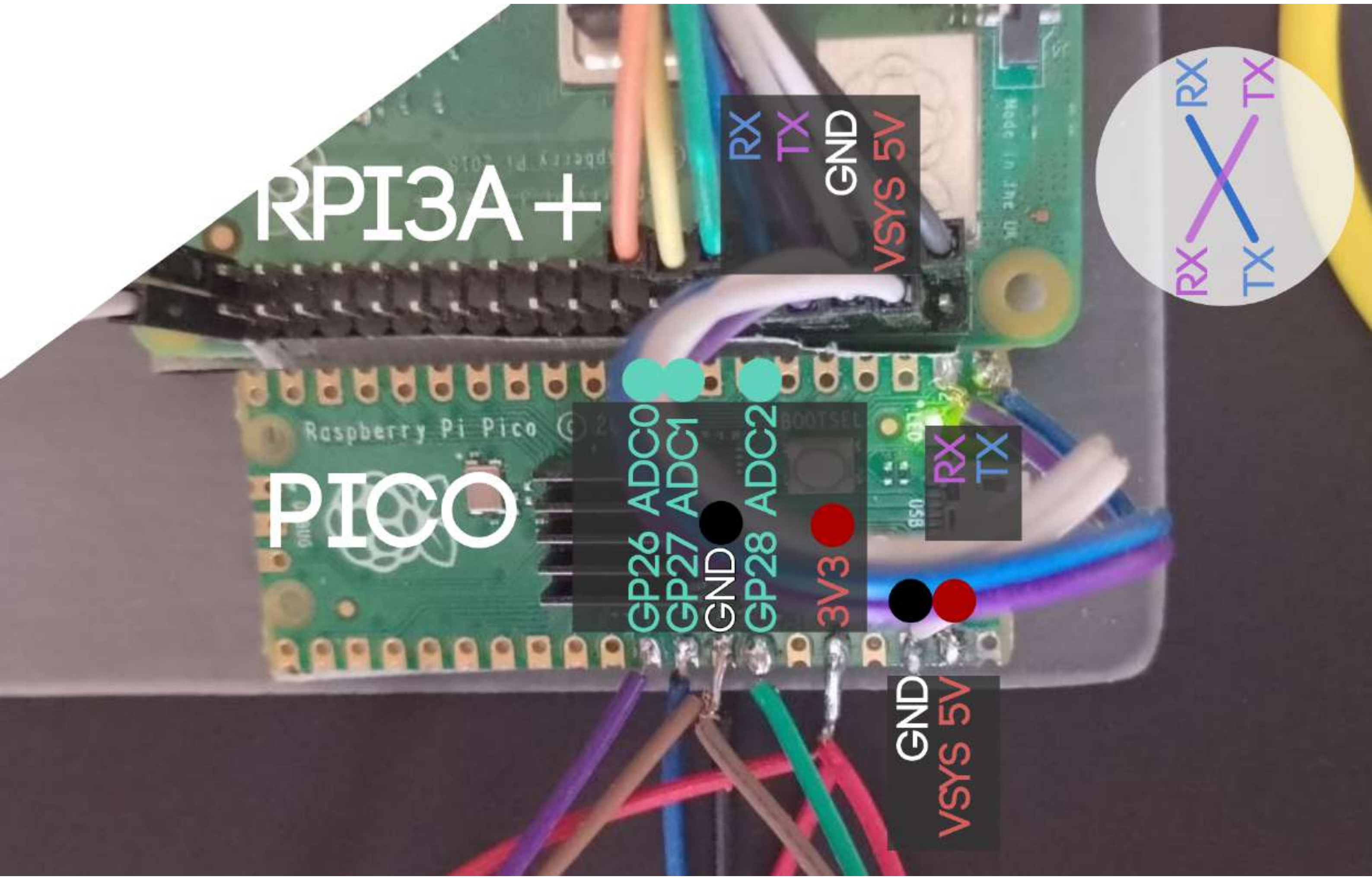
pico_sdk_init()

add_executable(tst
    tst.c
)

target_link_libraries(tst
    pico_stdlib
    hardware_uart
    hardware_adc
)

pico_enable_stdio_usb(tst 0)
pico_enable_stdio_uart(tst 1)

pico_add_extra_outputs(tst)
```



// Windows Environment

Windows support is mainly intended for Pico firmware development.

Recommended tools:

** Visual Studio Code*
** Pico SDK*
** CMake*
** ARM GCC Toolchain*
** Git*

Install:

** Raspberry Pi Pico Extension for VS Code*
** ARM GNU Toolchain*
** CMake*

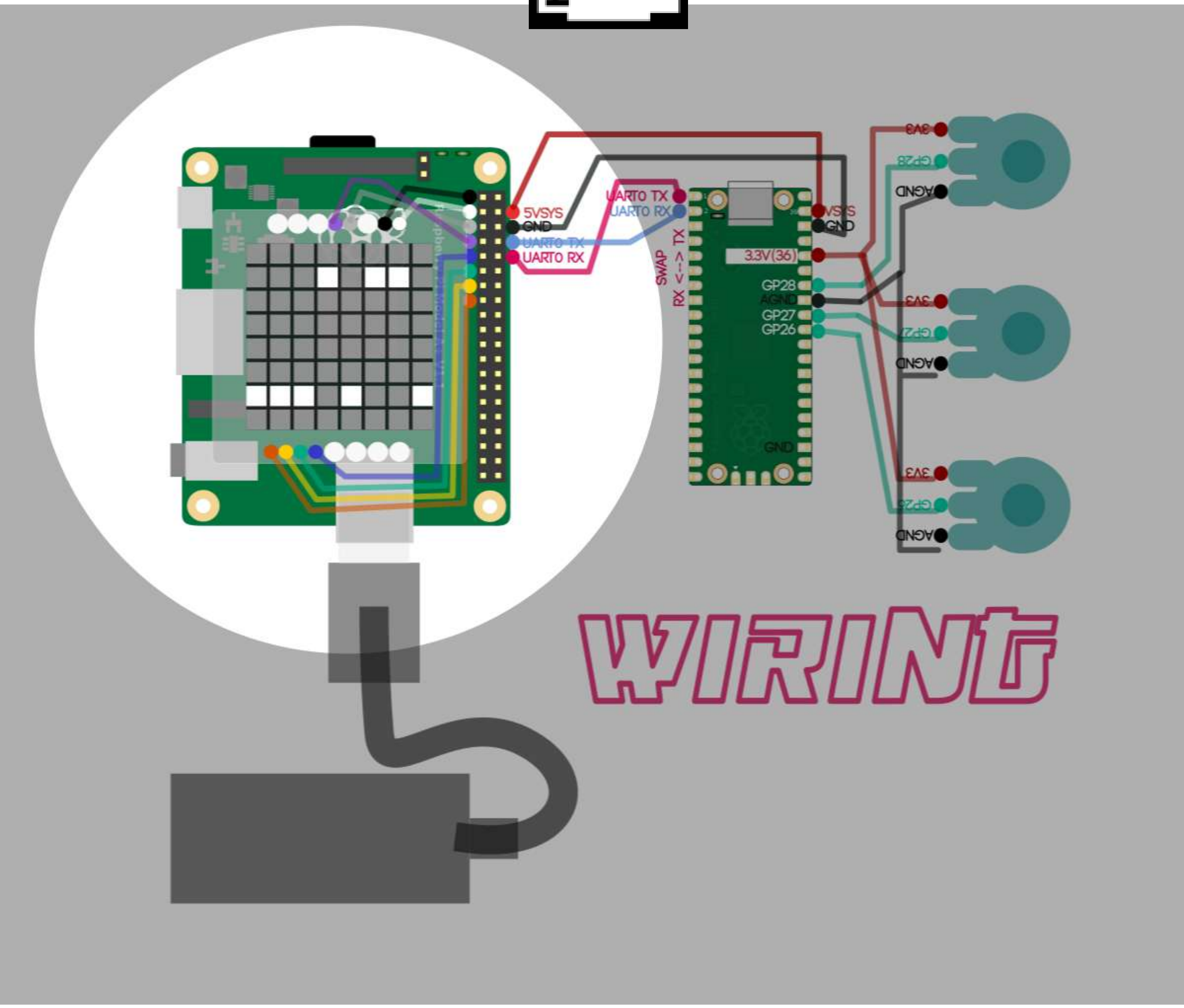
Firmware can be compiled directly from VS Code.

2

PURE DATA EXTERNALS C @ RASPBERRYPI 3A+

2. PURE DATA EXTERNALS C @ RASPBERRYPI 3A+

pd~



An external is a customized function (object) in pd language, programmed in C and after being compiled you can use as a regular object within pd.
In this case we are building 3 functions that will be in charge to interconnect the physical world with the intangible domain of code through the GPIO ports.

gpio_in *> external in charge to manage input GPIO ports and therefore integrate incoming physical buttons to pd (digital input).*

gpio_out *> external in charge to manage output GPIO ports and therefore integrate outcoming digital signals (for example blinking a led On/OFF). In this case we are controlling a matrix led which reads the envelope of the audio in live and creates a 64pixels low-res 'data visualization'.*

serialadc *> external in charge to interconnect the two PCBs (Pico & RPI·3A+) and reading the serial communication data through RX TX GPIO ports. In other words the analog input of the 3 Knobs will be read like it was a midi controller or another analog / variable incoming sensor.*

RASPBIAN OS

Development and testing were performed on:
Operating System:
Raspbian GNU/Linux 13 (Trixie) /// 13,4 /// 32 bits

Base Distribution:
Debian GNU/Linux 13.4

Target Hardware:
Raspberry Pi 3A+

The system image was written to the microSD card using Raspberry Pi Imager.
Raspberry Pi Imager is available for:

Linux
macOS
Windows

Download:
https://www.raspberrypi.com/software/

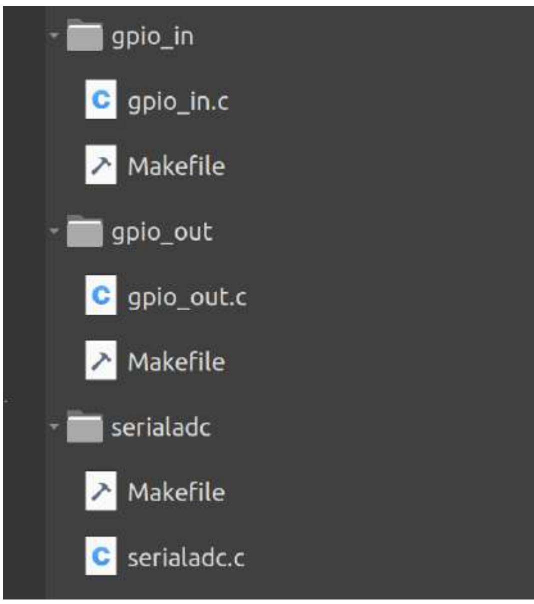
Requirements **miniSD 16gb**

Once created the image in your miniSD card, put it in your Rasperry Pi3A+
Boot the Pi3A+ and then install the following packages :

terminal @ RPi3

```
sudo apt install update
sudo apt install gcc
sudo apt install g++
sudo apt install make
sudo apt install libpigpio-dev
sudo apt install pigpio
sudo apt install puredata
sudo apt install puredata-dev
```

Then copy the following codes in the next pages
in a main folder called **gpio** for example in the Desktop with each
external to compile in its own folder (gpio_in, gpio_out and serialadc)



/home/pi/Desktop/gpio/
gpio_in
gpio_out
serialadc

COMPILING EXTERNALS

Once copied the files and installed the development packages of the previous page, we can compile :

terminal @ RPi3
cd /home/pi/Desktop/gpio/gpio_in
build process:
make clean
make
result:
gpio_in.pd_linux

terminal @ RPi3
cd /home/pi/Desktop/gpio/gpio_out
build process:
make clean
make
result:
gpio_out.pd_linux

terminal @ RPi3
cd /home/pi/Desktop/gpio/serialadc
build process:
make clean
make
result:
serialadc.pd_linux

These externals can then be loaded directly in Pure Data for managing the Pi's GPIO ports
How? loading the external paths/startup at Pd preferences. Once apply and saved in Pd preferences those paths will allow to access the externals from the pd patch.

Remember as well :
pigpio Daemon
The GPIO externals use pigpiod.
Enable at boot:

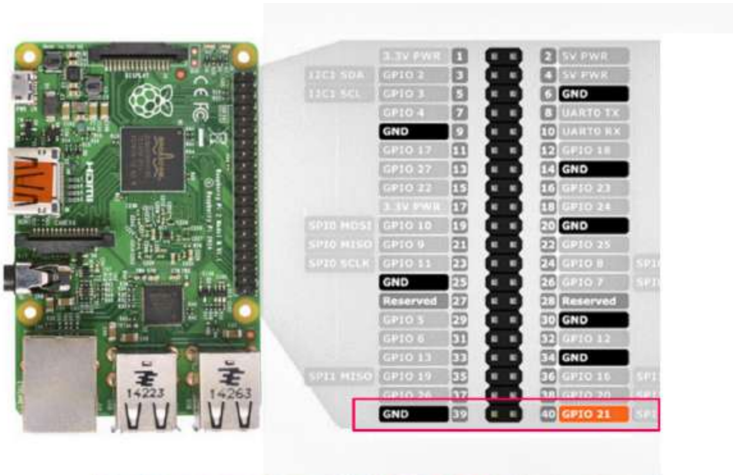
terminal @ RPi3
sudo systemctl enable pigpiod
sudo systemctl start pigpiod

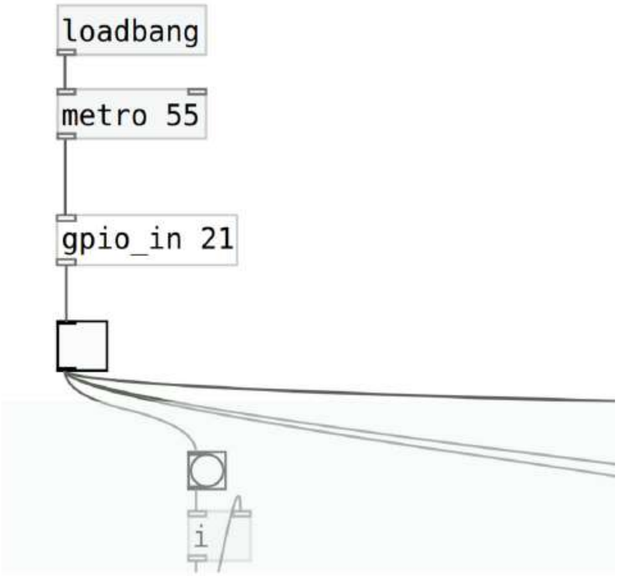
optionally
Check status:
terminal @ RPi3
systemctl status pigpiod

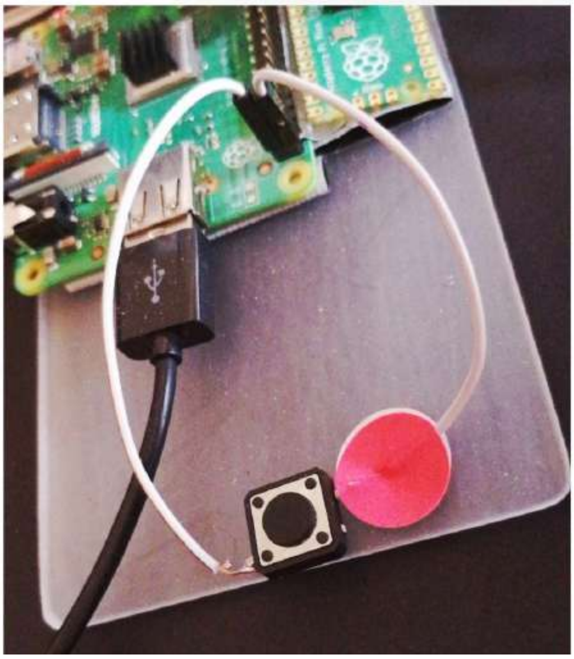
Test GPIO communication:
terminal @ RPi3
pigs r 21

GPIO_IN

controlling input digital Rpi 3A gpio ports







Create **gpio_in** folder for example */home/pi/Desktop/gpio/gpio_in*
copy **gpio_in.c** file (following pages) inside the folder
copy **Makefile** file (following pages) inside the folder

terminal @ RPi3: change directory >
cd */home/pi/Desktop/gpio/gpio_in*

then
make

(alternative option)
make clean
make

A couple of files will be created after compiling :
gpio_in.o
gpio_in.pd_linux

*the file *.pd_linux will be the executable to run the external inside a pd patch.*
Youll need to setup the path & startup of 'gpio_in' in pd preferences
*(remind apply execution and reading permissions to the **gpio_in** folder and **gpio_in.pd_linux**)*
chmod -R 755 gpio_in

CODE

gpio_in.c file

```
#include "m_pd.h"
#include <pigpiod_if2.h>

static t_class *gpio_in_class;

typedef struct _gpio_in {
    t_object x_obj;

    int pi;
    int gpio_pin;
    int last_value;

    t_outlet *out;
} t_gpio_in;


// READ (bang)
void gpio_in_bang(t_gpio_in *x)
{
    if (x == NULL) return;
    if (x->pi < 0) return;

    int val = gpio_read(x->pi, x->gpio_pin);

    if (val != x->last_value)
    {
        x->last_value = val;
        outlet_float(x->out, val);
    }
}


// CONSTRUCTOR
void *gpio_in_new(t_floatarg f)
{
    t_gpio_in *x = (t_gpio_in *)pd_new(gpio_in_class);

    x->gpio_pin = (int)f;
    x->pi = -1;
    x->last_value = 0;

    // connect to pigpiod daemon
    x->pi = pigpio_start(NULL, NULL);

    if (x->pi < 0)
    {
        post("gpio_in: ERROR connecting to pigpiod");
        return NULL;
    }

    set_mode(x->pi, x->gpio_pin, PI_INPUT);
    set_pull_up_down(x->pi, x->gpio_pin, PI_PUD_UP);

    x->last_value = gpio_read(x->pi, x->gpio_pin);

    x->out = outlet_new(&x->x_obj, &s_float);
```

```
    outlet_float(x->out, x->last_value);

    post("gpio_in ready GPIO %d (daemon mode)", x->gpio_pin);

    return (void *)x;
}


// FREE
void gpio_in_free(t_gpio_in *x)
{
    if (!x) return;

    if (x->pi >= 0)
    {
        pigpio_stop(x->pi);
    }
}


// SETUP
void gpio_in_setup(void)
{
    gpio_in_class = class_new(
        gensym("gpio_in"),
        (t_newmethod)gpio_in_new,
        (t_method)gpio_in_free,
        sizeof(t_gpio_in),
        CLASS_DEFAULT,
        A_DEFFLOAT,
        0
    );

    class_addbang(gpio_in_class, gpio_in_bang);

    post("gpio_in loaded (pigpiod_if2 stable version)");
}
```

CODE

Makefile file

```
PD_INCLUDE = /usr/include/pd

CFLAGS = -fPIC -O3 -Wall -Wextra -I$(PD_INCLUDE)

LIBS = -lpigpiod_if2 -lrt -lpthread

TARGET = gpio_in.pd_linux
SRC = gpio_in.c
OBJ = gpio_in.o

all: $(TARGET)

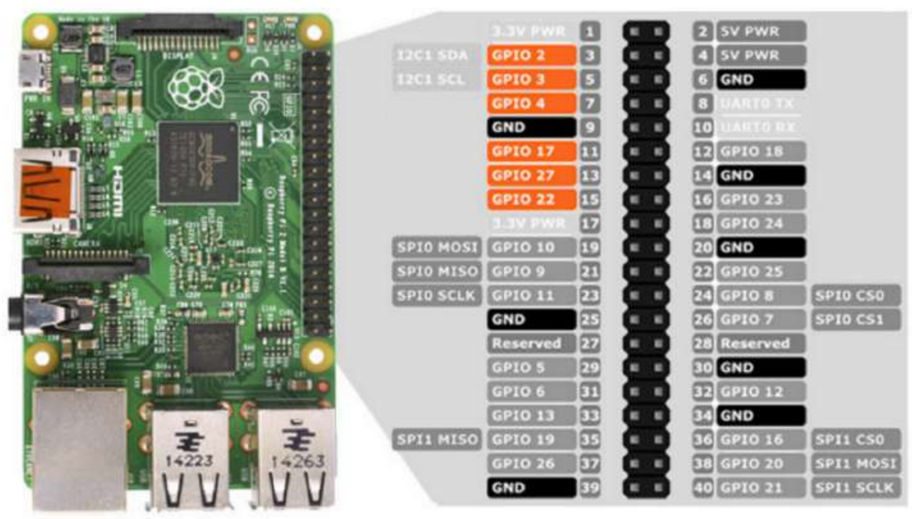
$(TARGET): $(OBJ)
	cc -o $(TARGET) $(OBJ) -shared $(LIBS)

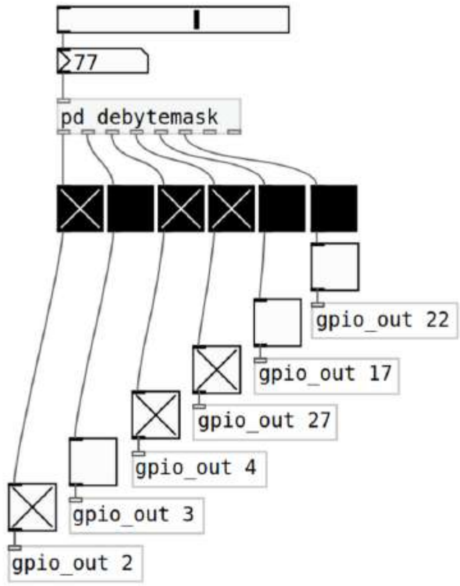
$(OBJ): $(SRC)
	cc $(CFLAGS) -c $(SRC) -o $(OBJ)

clean:
	rm -f $(OBJ) $(TARGET)
```

GPIO_OUT

controlling output digital Rpi 3A gpio ports





Create ***gpio_out*** *folder for example /home/pi/Desktop/gpio/gpio_out*
copy ***gpio_out.c*** *file (following pages) inside the folder*
copy ***Makefile*** *file (following pages) inside the folder*

terminal @ RPi3: change directory >
cd /home/pi/Desktop/gpio/gpio_out

then
make

(alternative option)
make clean
make

If compilation went ok, a file will be created after compiling :
gpio_out.pd_linux

*the file *.pd_linux will be the executable to run the external inside a pd patch.*
Youll need to setup the path & startup of 'gpio_out' in pd preferences

(remind apply execution and reading permissions to the ***gpio_out*** *folder and* ***gpio_out.pd_linux*** *)*
chmod -R 755 gpio_out

CODE

gpio_out.c file

```
#include "m_pd.h"
#include <stdlib.h>

static t_class *gpio_out_class;

typedef struct _gpio_out {
    t_object x_obj;
    int gpio_pin;
    int last_value;
} t_gpio_out;


// envia comanda a pigpiod via pigs
void gpio_out_float(t_gpio_out *x, t_floatarg f)
{
    int val = (f != 0);

    if (val == x->last_value)
        return;

    x->last_value = val;

    char cmd[64];
    sprintf(cmd, "pigs w %d %d", x->gpio_pin, val);
    system(cmd);
}


// constructor
void *gpio_out_new(t_symbol *s, int argc, t_atom *argv)
{
    t_gpio_out *x = (t_gpio_out *)pd_new(gpio_out_class);

    x->gpio_pin = (argc > 0) ? atom_getint(argv) : 17;
    x->last_value = -1;

    return (void *)x;
}


// setup
void gpio_out_setup(void)
{
    gpio_out_class = class_new(
        gensym("gpio_out"),
        (t_newmethod)gpio_out_new,
        0,
        sizeof(t_gpio_out),
        CLASS_DEFAULT,
        A_GIMME,
        0
    );

    class_addfloat(gpio_out_class, gpio_out_float);
}
}
```

CODE

Makefile file

```
PD_INCLUDE = /usr/include/pd

CFLAGS = -Wall -Wextra -O2 -fPIC -I$(PD_INCLUDE)
LDFLAGS = -shared -lpigpio -lrt -lpthread

TARGET = gpio_out.pd_linux

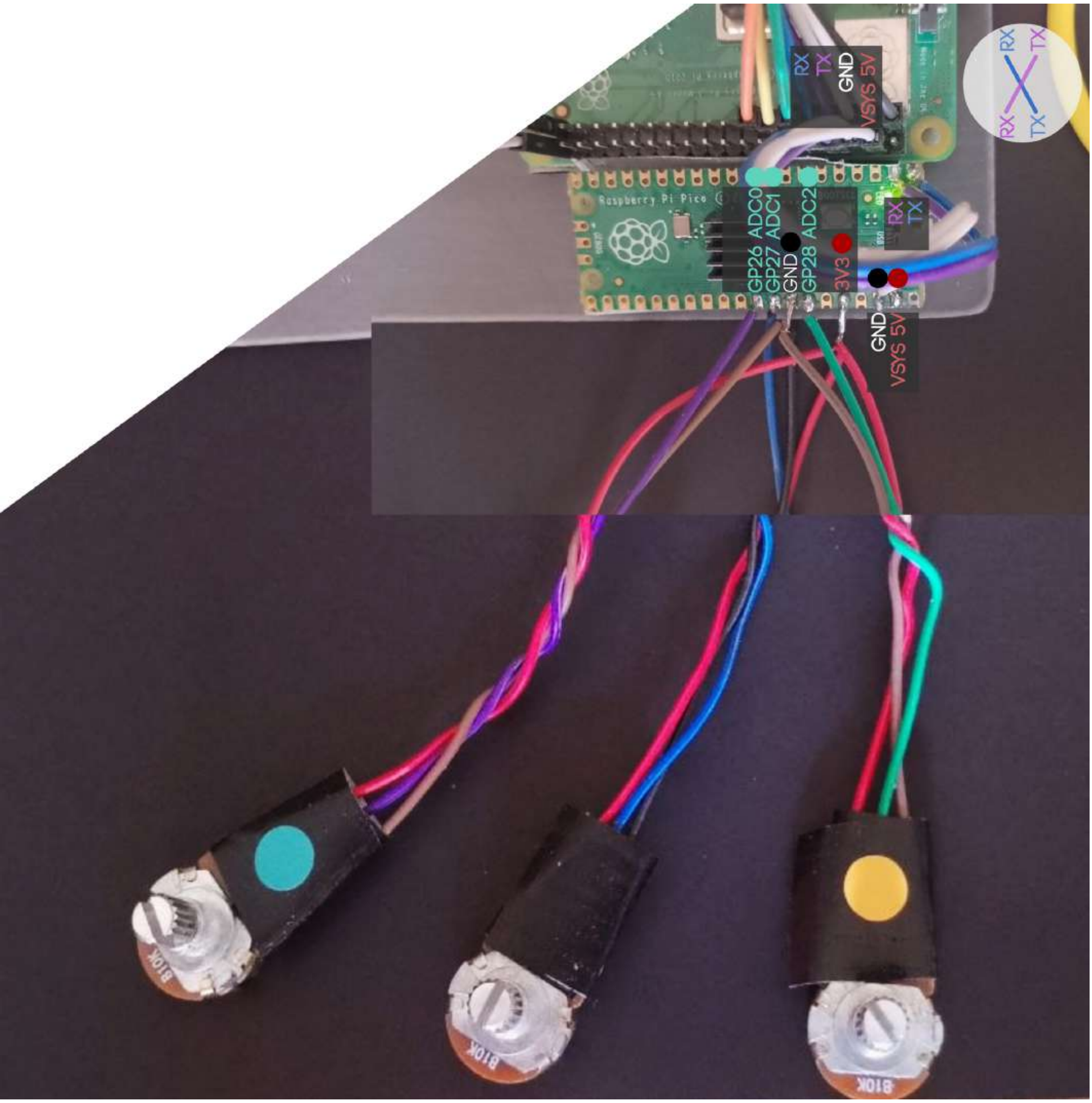
all: $(TARGET)

$(TARGET): gpio_out.c
	$(CC) $(CFLAGS) -o $@ $^ $(LDFLAGS)

clean:
	rm -f *.o *.pd_linux
```

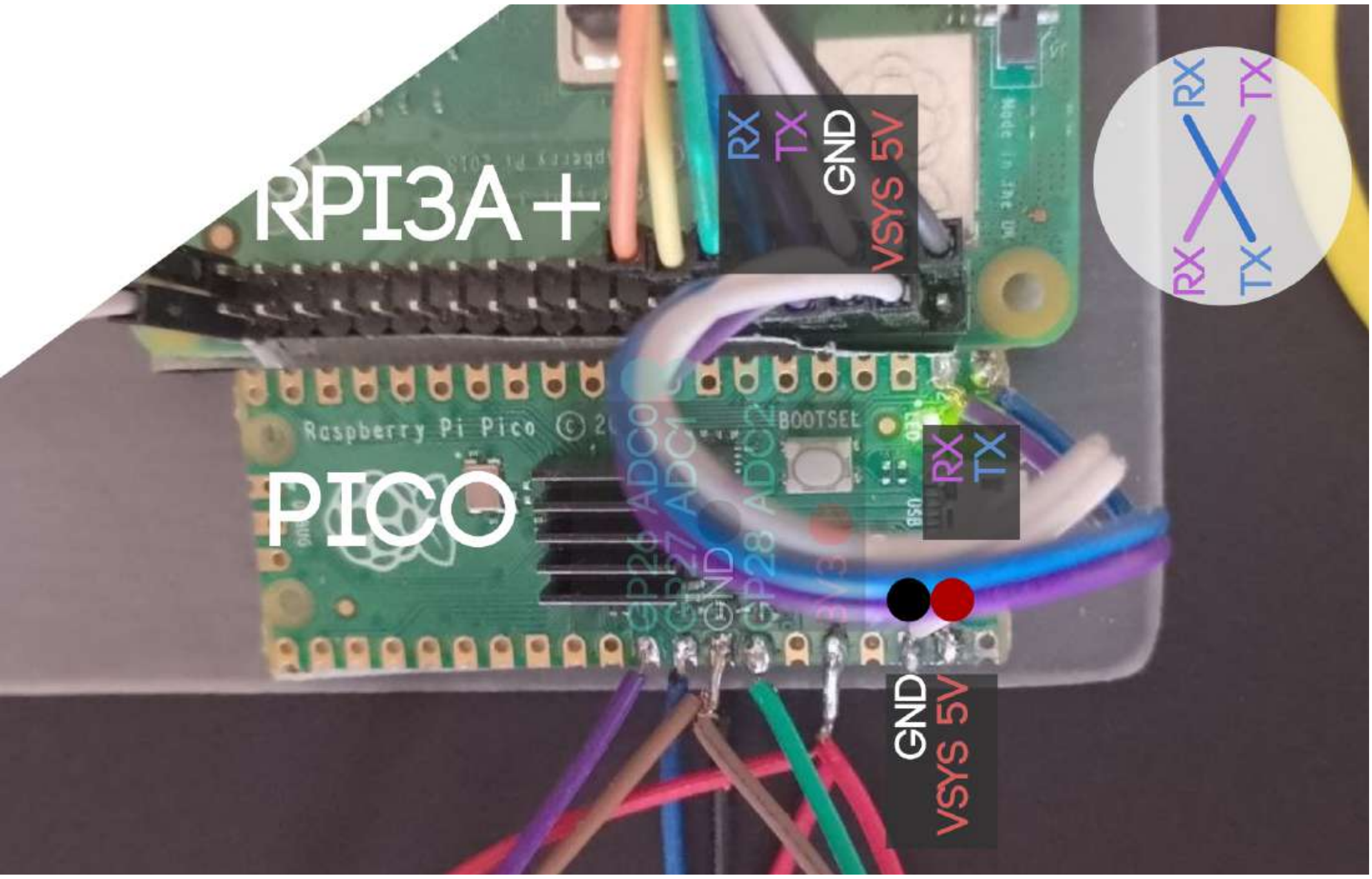
SERIALADC

external to listen the serial communication feed from the Knobs (Picos analog inputs) via RX TX ports



SERIALADC

rxtx wiring must be crossed :
Rx from pico to Tx Pi3A
Tx from pico to Rx Pi3A



Create **serialadc** folder for example */home/pi/Desktop/gpio/serialadc*
copy **serialadc.c** file (following pages) inside the folder
copy **Makefile** file (following pages) inside the folder

terminal @ RPi3: change directory >
cd */home/pi/Desktop/gpio/serialadc*

then
make

(alternative option)
make clean
make

A couple of files will be created after compiling :
serialadc.o
serialadc.pd_linux

*the file *.pd_linux will be the executable to run the external inside a pd patch.*
Youll need to setup the path & startup of 'serialadc' in pd preferences
*(remind apply execution and reading permissions to the **serialadc** folder and **serialadc.pd_linux**)*
chmod -R 755 serialadc

CODE

serialadc.c file

```c
#include "m_pd.h"
#include <stdio.h>
#include <string.h>
#include <unistd.h>
#include <fcntl.h>
#include <termios.h>

typedef struct _serialadc {
    t_object x_obj;
    int fd;
    t_outlet *out1;
    t_outlet *out2;
    t_outlet *out3;
    t_clock *clock;
    char buffer[256];
    int buf_pos;
} t_serialadc;

static t_class *serialadc_class;

// ---------------- SERIAL OPEN ----------------
int open_serial(const char *device)
{
    int fd = open(device, O_RDWR | O_NOCTTY | O_NONBLOCK);

    if (fd < 0) {
        perror("open_serial");
        return -1;
    }

    struct termios tty;

    if (tcgetattr(fd, &tty) != 0) {
        perror("tcgetattr");
        close(fd);
        return -1;
    }

    cfmakeraw(&tty);

    cfsetispeed(&tty, B115200);
    cfsetospeed(&tty, B115200);

    tty.c_cflag |= (CLOCAL | CREAD);
    tty.c_cflag &= ~PARENB;
    tty.c_cflag &= ~CSTOPB;
    tty.c_cflag &= ~CSIZE;
    tty.c_cflag |= CS8;

    tty.c_cc[VMIN] = 0;
    tty.c_cc[VTIME] = 1;

    if (tcsetattr(fd, TCSANOW, &tty) != 0) {
        perror("tcsetattr");
        close(fd);
        return -1;
    }

    tcflush(fd, TCIOFLUSH);

    return fd;
}
```

```c
// ---------------- PARSE ----------------
void process_line(t_serialadc *x, char *line)
{
    int id, value;

    if (sscanf(line, "%d:%d", &id, &value) == 2) {
        if (id == 1) outlet_float(x->out1, value);
        else if (id == 2) outlet_float(x->out2, value);
        else if (id == 3) outlet_float(x->out3, value);
    }
}

// ---------------- LOOP ----------------
void serialadc_tick(t_serialadc *x)
{
    char buf[64];

    int n = read(x->fd, buf, sizeof(buf));

    if (n <= 0) {
        clock_delay(x->clock, 20);
        return;
    }

    for (int i = 0; i < n; i++) {

        char c = buf[i];

        // només ASCII útil
        if (c < 32 && c != '\n' && c != '\r')
            continue;

        if (x->buf_pos < 255)
            x->buffer[x->buf_pos++] = c;

        if (c == '\n' || c == '\r') {

            x->buffer[x->buf_pos] = '\0';

            process_line(x, x->buffer);

            x->buf_pos = 0;
        }
    }

    clock_delay(x->clock, 20);
}

// ---------------- NEW ----------------
void *serialadc_new(void)
{
    t_serialadc *x = (t_serialadc *)pd_new(serialadc_class);

    x->fd = open_serial("/dev/serial0");
    x->buf_pos = 0;

    x->out1 = outlet_new(&x->x_obj, &s_float);
    x->out2 = outlet_new(&x->x_obj, &s_float);
    x->out3 = outlet_new(&x->x_obj, &s_float);

    x->clock = clock_new(x, (t_method)serialadc_tick);
    clock_delay(x->clock, 20);

    return (void *)x;
}
```

```
// ---------------- FREE ----------------
void serialadc_free(t_serialadc *x)
{
    if (x->fd > 0) close(x->fd);
    if (x->clock) clock_free(x->clock);
}

// ---------------- SETUP ----------------
void serialadc_setup(void)
{
    serialadc_class = class_new(
        gensym("serialadc"),
        (t_newmethod)serialadc_new,
        (t_method)serialadc_free,
        sizeof(t_serialadc),
        CLASS_DEFAULT,
        0
    );
}
```

CODE

Makefile file

```
CC = gcc

PD_DIR = /usr/include/pd

CFLAGS = -fPIC -O2 -Wall -Wextra
LDFLAGS = -shared -lpthread

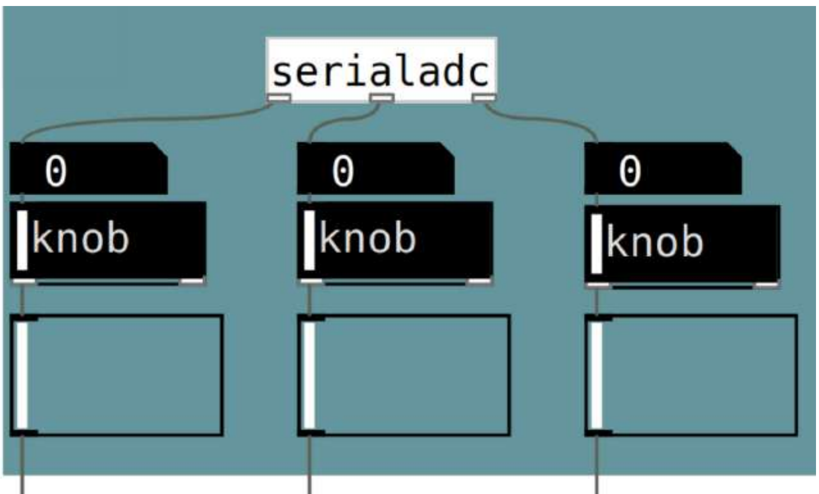
TARGET = serialadc

all: $(TARGET).pd_linux

$(TARGET).pd_linux: $(TARGET).o
	$(CC) -o $@ $^ $(LDFLAGS)

$(TARGET).o: $(TARGET).c
	$(CC) $(CFLAGS) -I$(PD_DIR) -c $< -o $@

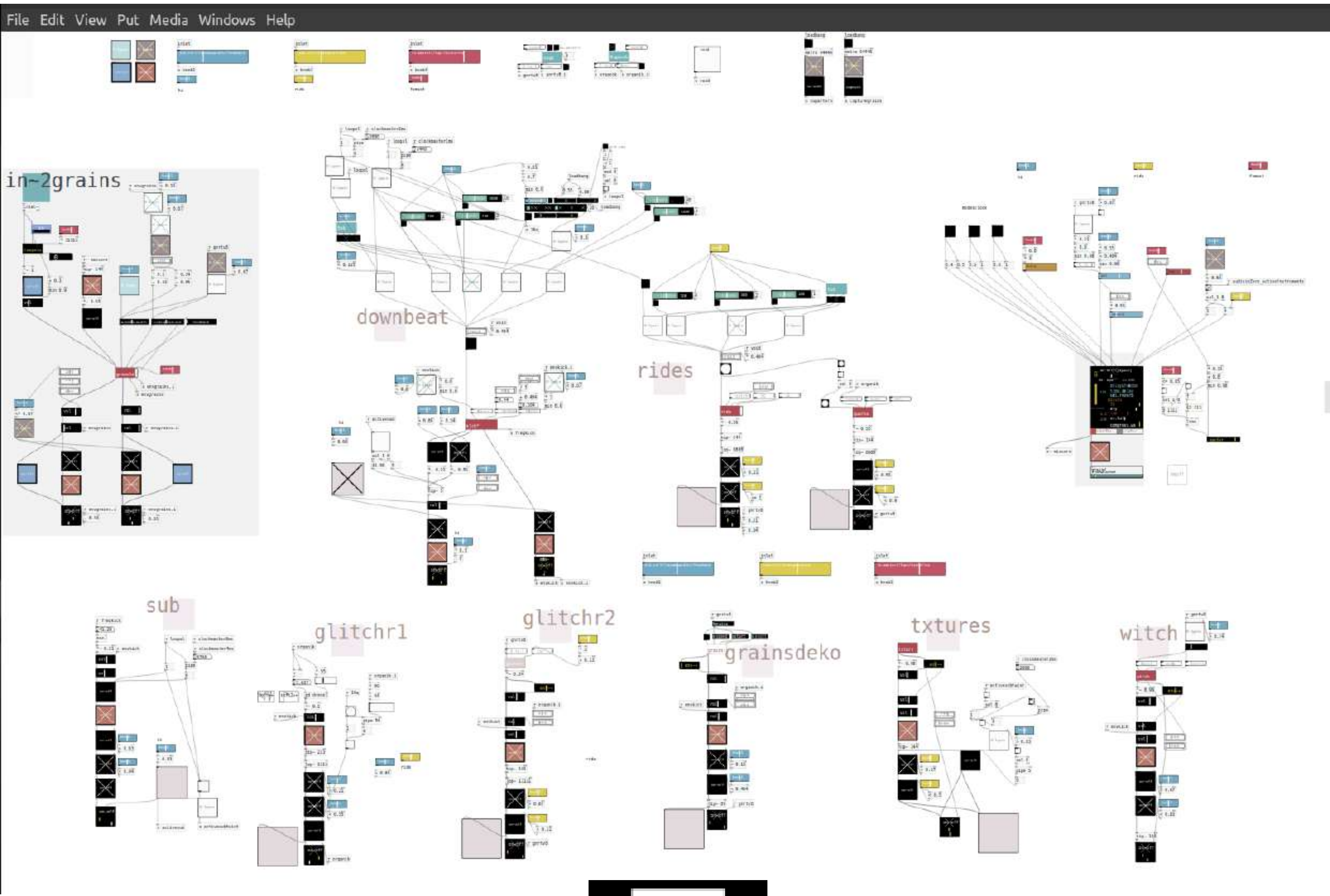
clean:
	rm -f *.o *.pd_linux
```



3

PD / GNRTV.CELLS
ALGORYTHMIC PROGRAMMING

PD SONIC ALGORYTHM



pd code described as txt in an annex file.

PD · EXTERNALS

compile the externals GPIO_IN GPIO_OUT and SERIALADC described in the previous pages.
Result:
gpio_in.pd_linux
gpio_out.pd_linux
serial_adc.pd_linux

Youll need to setup the path & startup of 'gpio_in' 'gpio_out' 'serialadc' in pd preferences
(remind apply execution and reading permissions to those folders
terminal @ RPi3:
chmod -R 755 /home/pi/Desktop/gpio/gpio_in
chmod -R 755 /home/pi/Desktop/gpio/gpio_out
chmod -R 755 /home/pi/Desktop/gpio/serialadc

SETUP REQUIREMENTS

Additionaly to the hardware DIY described previously, in order to improve the sound quality we reccomend Soundblaster PLAY! 3 USB which is a lowcost and tiny extension for your Raspberry 3A+ which even has a default minijack output socket it renders Audio at 12bits.

Once soundcard is connected is required to setup the device at pd preferences in your pi3A

input
Soundblaster USB   channels 1 channel

output
Soundblaster USB channels 2 channels

OS UPDATES

Once tested the algorythm

pigpio Daemon
The GPIO externals use pigpiod.
Enable at boot:
terminal @ RPi3:
sudo systemctl enable pigpiod
sudo systemctl start pigpiod

Operating System

Development and testing were performed on:
Operating System:
Raspbian GNU/Linux 13 (Trixie) /// 13,4 /// 32 bits

ATOMATION BOOT

Following the previous lines, if you want this kind of instruments behave like automats, we'll need that after launching the OS in this case Raspbian, automatically will launch the pd sonic algorythm. (notice that system allows any application you are coding with several languages depending on your project.)
Tere arte several methods to automatic launch an application. Here is a simple method. In case it does not work search on the internet or with your favorite transformer.

In your Pi3A move to */home/pi/.config/autostart*
terminal @ RPi3:
cd /home/pi/.config/autostart

in case autostart does not appear create it as
mkdir cd /home/pi/.config/autostart

then open a raspbian editor for example mousepad (that appears in the top left rapsbian menú > accessories)
*copy the following code and save it as **pd.desktop** inside **/home/pi/.config/autostart****

```
[Desktop Entry]
Type=Application
Name=pd.desktop
Exec=/home/pi/startpd.sh
Terminal=true
```

** /home/pi/.config/autostart/pd.desktop*

*Once created the **pd.desktop** lets create the bash script.*
Create a new file with the same mousepad editor.
*copy the following code and save it as **startpd.sh** inside **/home/pi/***

```
#!/bin/bash
sleep 10
pd /home/wavefunction/Desktop/wavefunction.pd
```

note : wavefunction.pd or whatever name of the app you want to launch

remember to update execution permissions to both files :
terminal

chmod -R 755 /home/pi/.config/autostart/pd.desktop
chmod -R 755 /home/pi/startpd.sh

*Once done the previous steps every time you are rebooting the instrument, the pd algorythm / app will be launched **automatically.***

ANOTHER NON-ANTROPOCENTRIC ARTIFICIAL INTELLIGENCES

This project belongs to the ***Latent Spaces Research*** by ***Xavi Manzanares*** and ***OnesHaptiques Lab***.
As usual we are focused in build <u>generative automats</u> in order to explore another kind of artificial intelligences (non antropocentric) and algorythmic-emergence systems in the sonic and electronic arts domain.

If you have a development like featured in this document, one important thing is to automate all the dataflow in order to build in this case a sort of *generative audio robot*. This will mean that any time you connect to the power youll have a computerized algorythmic instrument without any log in as usual OS access.

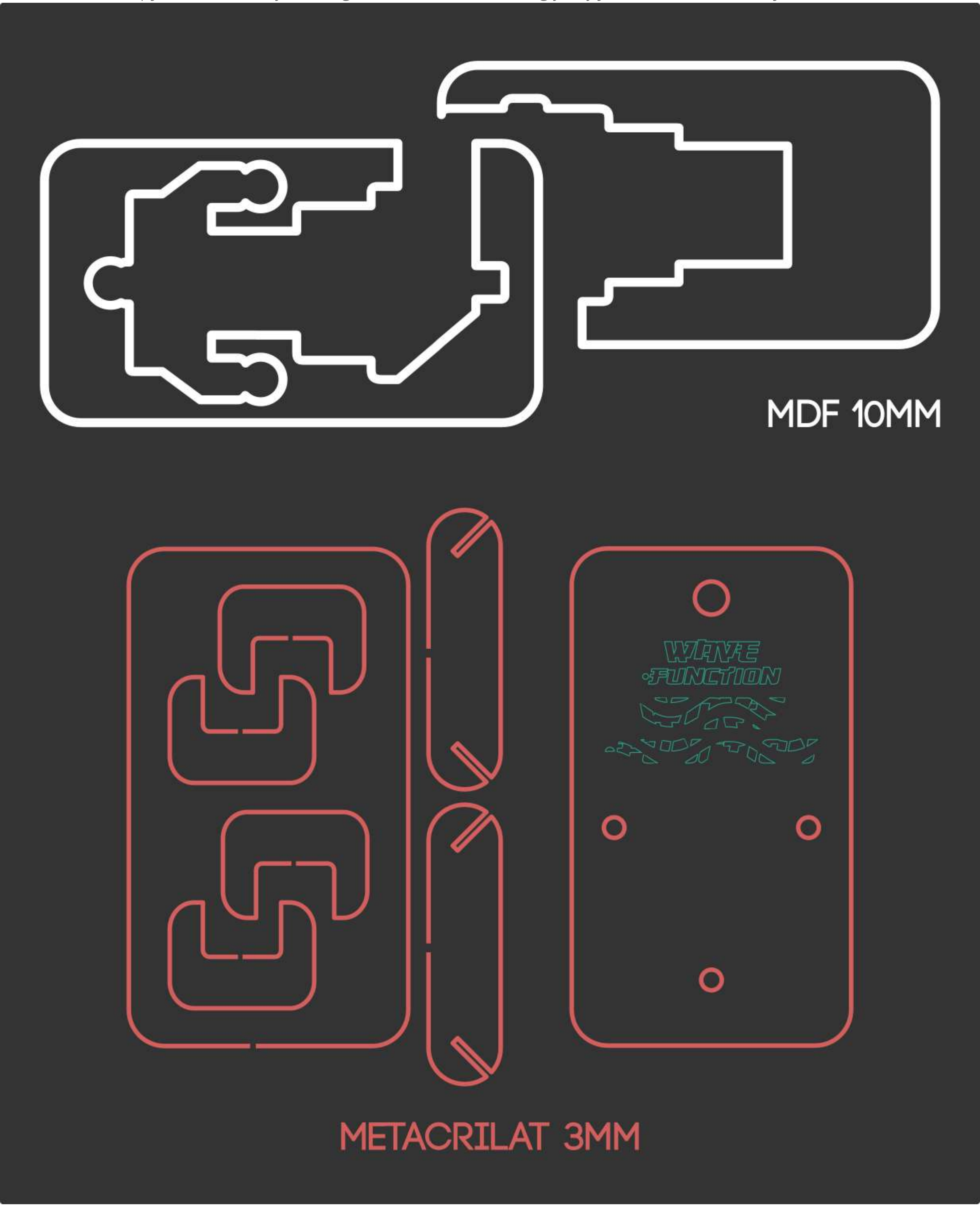
With this kind of instruments, we as performers don't just play music, <u>rather we drift states and phases of many sonic combinations at once</u>. For example, a single knob can control the velocity complexity of patterns and sonic ingredients depending on which particular range you are in the knob.

This multiparametric approach also means that if you dont touch any knob at all,
<u>the system is not static whatsoever</u> :
It focuses on particular ranges of <u>sonic ingredients that are evolving in time</u>.
Somehow emergence states appears and others vanishes.
Somehow a certain level of animated agent behaviour appears.

VECTOR 2CNC CUT

copy and save the following code and save it to .svg if you want to CNC cut your case

copy and save the following code and save it to .svg file if you want to CNC cut your case



```
<?xml version="1.0" encoding="UTF-8" standalone="no"?>
<!-- Created with Inkscape (http://www.inkscape.org/) -->

<svg
  width="400mm"
  height="400mm"
  viewBox="0 0 400 400"
  version="1.1"
  id="svg1"
  sodipodi:docname="vector2kutALL.svg"
  inkscape:version="1.4.2 (ebf0e940d0, 2025-05-08)"
  inkscape:export-filename="000.testScaleMatxembrat+2xTapesOH.png"
  inkscape:export-xdpi="96"
  inkscape:export-ydpi="96"
  xml:space="preserve"
  xmlns:inkscape="http://www.inkscape.org/namespaces/inkscape"
  xmlns:sodipodi="http://sodipodi.sourceforge.net/DTD/sodipodi-0.dtd"
  xmlns="http://www.w3.org/2000/svg"
  xmlns:svg="http://www.w3.org/2000/svg"><sodipodi:namedview
    id="namedview1"
    pagecolor="#ffffff"
    bordercolor="#000000"
    borderopacity="0.25"
    inkscape:showpageshadow="2"
    inkscape:pageopacity="0.0"
    inkscape:pagecheckerboard="0"
    inkscape:deskcolor="#fafafa"
    inkscape:document-units="mm"
    inkscape:zoom="0.36747685"
    inkscape:cx="832.70551"
    inkscape:cy="619.08661"
    inkscape:window-width="1920"
    inkscape:window-height="1052"
    inkscape:window-x="1600"
    inkscape:window-y="0"
    inkscape:window-maximized="1"
    inkscape:current-layer="layer1"
    inkscape:export-bgcolor="#ffffffff"
    showguides="true" /><defs
    id="defs1"><inkscape:path-effect
      effect="simplify"
      id="path-effect5"
      is_visible="true"
      lpeversion="1.3"
      threshold="17,241379"
      steps="1"
      smooth_angles="360"
      helper_size="0"
      simplify_individual_paths="false"
      simplify_just_coalesce="false" /><inkscape:path-effect
      effect="fillet_chamfer"
      id="path-effect23790"
      is_visible="true"
      lpeversion="1"
      nodesatellites_param="F,0,0,1,0,0,0,1 @ F,0,0,1,0,0,0,1 @ F,0,0,1,0,0,0,1 @ F,0,0,1,0,0,0,1 @ F,0,0,1,0,0,0,1 @ F,0,0,1,0,0,0,1 @
F,0,0,1,0,0,0,1 @ F,0,0,1,0,0,0,1 @ F,0,0,1,0,0,0,1 @ F,0,0,1,0,0,0,1 @ F,0,0,1,0,0,0,1 @ F,0,0,1,0,0,0,1 @ F,0,0,1,0,0,0,1 @ F,0,0,1,0,0,0,1 @ F,0,0,1,0,0,0,1 @
F,0,0,1,0,16.809554,0,1 @ F,0,0,1,0,0,0,1 @ F,0,0,1,0,0,0,1"
      radius="0"
      unit="px"
      method="auto"
      mode="F"
      chamfer_steps="1"
      flexible="false"
      use_knot_distance="true"
      apply_no_radius="true"
      apply_with_radius="true"
      only_selected="false"
      hide_knots="false" /><inkscape:path-effect
      effect="fillet_chamfer"
      id="path-effect24919"
      is_visible="true"
      lpeversion="1"
      nodesatellites_param="F,0,0,1,0,0,0,1 @ F,0,0,1,0,0,0,1 @ F,0,0,1,0,0,0,1 @ F,0,0,1,0,0,0,1 @ F,0,0,1,0,0,0,1 @ F,0,0,1,0,0,0,1 @
F,0,0,1,0,0,0,1"
      radius="0"
      unit="px"
      method="auto"
```

mode="F"
chamfer_steps="1"
flexible="false"
use_knot_distance="true"
apply_no_radius="true"
apply_with_radius="true"
only_selected="false"
hide_knots="false" /></defs><g
inkscape:label="Capa 1"
inkscape:groupmode="layer"
id="layer1"><rect
style="fill:#333333;stroke:none;stroke-width:1.5;stroke-linejoin:round;stroke-dasharray:none"
id="rect8"
width="326.16"
height="400"
x="36.919998"
y="0"
rx="1"
ry="1" /><g
id="g6"
transform="translate(-20.45232,18.022543)"><g
id="g3"
transform="translate(-37.8727,46.41285)"><path
id="path1"
style="fill:none;stroke:#d35f5f;stroke-width:1.5;stroke-linejoin:round;stroke-dasharray:none;stroke-opacity:1"
d="m 237.8727,251.70382 v 31.85412 c 0,6.648 5.3523,12.0003 12.0003,12.00029 2.7352,0 5.2512,-0.906 7.2652,-2.4355 l -11.2066,-11.2065 1.6263,-1.62629 11.2711,11.2712 c 1.8949,-2.1205 3.0443,-4.9226 3.0443,-8.0032 v -68.9999 c 0,-3.0804 -1.149,-5.8828 -3.0437,-8.0031 l -11.2717,11.2717 -1.6263,-1.6268 11.2066,-11.2066 c -2.0141,-1.5295 -4.5299,-2.435 -7.2652,-2.435 -6.648,0 -12.0003,5.3518 -12.0003,11.9998 v 34.18332"
sodipodi:nodetypes="csscccsscccccssc" /><path
id="path180"
style="fill:none;stroke:#d35f5f;stroke-width:1.5;stroke-linejoin:round;stroke-dasharray:none;stroke-opacity:1"
d="M 180.39372,282.9011 H 219.127 c 6.648,0 11.9998,-5.3523 11.9998,-12.0003 V 129.40091 c 0,-6.648 -5.3518,-11.99978 -11.9998,-11.99978 h -68.9999 c -6.648,0 -12.0003,5.35178 -12.0003,11.99978 V 270.9008 c 0,6.648 5.3523,12.0003 12.0003,12.0003 h 28.36248 m 13.73672,-60.76631 h -9.7637 c -1.2103,0 -2.0004,1.04831 -2.0004,2.34921 v 15.74322 h -2.9766 c -4.4563,0 -8.0445,-3.95239 -8.0445,-8.86199 v -17.50074 c 0,-4.9098 3.5882,-8.86199 8.0445,-8.86199 h 31.9743 c 4.4564,0 8.0444,3.95219 8.0444,8.86199 v 17.50074 c 0,4.9096 -3.588,8.86199 -8.0444,8.86199 h -2.9766 V 224.484 c 0,-1.3009 -0.9746,-2.34921 -2.1849,-2.34921 H 193.73577 M 191.168,147.09647 h -9.7638 c -1.2103,0 -2.0004,1.04831 -2.0004,2.34921 v 15.74323 h -2.9765 c -4.4563,0 -8.045,-3.95187 -8.045,-8.86147 v -17.50126 c 0,-4.9098 3.5887,-8.86199 8.045,-8.86199 h 31.9742 c 4.4564,0 8.0445,3.95219 8.0445,8.86199 v 17.50126 c 0,4.9096 -3.5881,8.86147 -8.0445,8.86147 h -2.9765 v -15.74323 c 0,-1.3009 -0.9746,-2.34921 -2.1849,-2.34921 h -10.56263 m -16.7086,31.07098 H 186.791 c 1.2102,0 2.0004,-1.04831 2.0004,-2.34921 v -15.74323 h 2.9765 c 4.4564,0 8.045,3.95239 8.045,8.86199 v 17.50074 c 0,4.9098 -3.5886,8.86199 -8.045,8.86199 h -31.9742 c -4.4564,0 -8.0445,-3.95219 -8.0445,-8.86199 V 168.937 c 0,-4.9096 3.5881,-8.86199 8.0445,-8.86199 h 2.9765 v 15.74323 c 0,1.3009 0.9745,2.34921 2.1849,2.34921 h 9.5043 m 1.50947,75.03831 H 186.791 c 1.2102,0 2.0004,-1.04831 2.0004,-2.34921 v -15.74323 h 2.9765 c 4.4564,0 8.045,3.95188 8.045,8.86148 v 17.50126 c 0,4.9098 -3.5886,8.86147 -8.045,8.86147 h -31.9742 c -4.4564,0 -8.0445,-3.95167 -8.0445,-8.86147 V 243.9748 c 0,-4.9096 3.5881,-8.86148 8.0445,-8.86148 h 2.9765 v 15.74323 c 0,1.3009 0.9745,2.34921 2.1849,2.34921 h 9.5043"
sodipodi:nodetypes="csssssssscсsscsssssssscsscсsscsssssssscsscсsscsssssssscsscсsscsssssssssсssc" /><path
id="path181"
style="fill:none;stroke:#d35f5f;stroke-width:1.5;stroke-linejoin:round;stroke-dasharray:none;stroke-opacity:1"
d="m 237.8727,153.58715 v 31.85412 c 0,6.648 5.3523,12.0003 12.0003,12.00029 2.7352,0 5.2512,-0.906 7.2652,-2.4355 l -11.2066,-11.2065 1.6263,-1.62629 11.2711,11.2712 c 1.8949,-2.1205 3.0443,-4.9226 3.0443,-8.0032 v -68.9999 c 0,-3.0804 -1.149,-5.8828 -3.0437,-8.0031 l -11.2717,11.2717 -1.6263,-1.6268 11.2066,-11.2066 c -2.0141,-1.5295 -4.5299,-2.435 -7.2652,-2.435 -6.648,0 -12.0003,5.3518 -12.0003,11.9998 v 34.18332"
sodipodi:nodetypes="csscccsscccccssc" /></g><g
id="g4"
transform="translate(-227.2327,77.917129)"><path
id="path186"
style="display:inline;fill:none;stroke:#2ca089;stroke-width:0.3;stroke-dasharray:none;stroke-opacity:1"
d="m 516.9648,116.16309 -1.3218,6.33611 h 2.56822 l 0.95191,-4.52472 h 1.95508 l -0.51101,2.43298 h -1.32882 l -0.26207,1.23796 h 1.34151 l -0.37646,1.82364 h 2.44025 l 1.10486,-5.2668 1.08645,5.28739 h 2.4534 l -0.1854,-1.39654 h 1.43751 l 1.9827,-4.96278 -0.17399,0.86824 h 1.28411 l -0.26208,1.23795 h -1.27754 l -0.24937,1.21343 h 1.25869 l -0.38961,1.83591 c -0.1662,0.80091 0.39581,1.20117 1.68643,1.20117 h 3.82735 l 0.23622,-1.1889 -2.50467,0.0158 c -0.46813,-0.0119 -0.65515,-0.22154 -0.56185,-0.62818 l 0.26164,-1.23183 h 3.1945 l 0.24937,-1.21955 h -3.18179 l 0.24937,-1.21343 h 3.20721 l 0.2555,-1.23183 h -7.04068 l -0.003,0.0109 h -3.37812 l -0.41504,1.20116 1.04745,0.0162 -1.24555,2.64062 -0.71568,-3.88471 h -2.42798 l 0.0596,0.28956 c -0.27005,-0.20116 -0.71588,-0.30183 -1.33715,-0.30183 h -2.53009 l 0.1271,-0.59751 z m -11.20068,0.60978 1.38053,6.72291 h 2.45296 l -0.24894,-1.3904 h 0.99047 l 0.27436,1.38427 h 2.45339 l -0.18494,-1.39654 h 1.4375 l 2.11461,-5.29308 h -3.44386 l -0.41504,1.20117 1.04789,0.0162 -1.24598,2.64062 -0.71569,-3.88471 h -2.42754 l 0.37034,2.20738 -0.6964,1.67733 -0.71569,-3.88471 z m 7.67792,8.5194 -1.18681,5.58834 h 1.95027 l 0.62277,-2.91135 1.08996,2.90697 h 0.7297 c 0.95666,0.0127 1.51079,-0.29715 1.66322,-0.92869 l 0.87564,-4.18348 -1.97481,0.0162 -0.47683,2.3353 -1.32794,-2.82286 z m 13.91834,0 -0.10605,0.50158 c -0.0616,0.28907 -0.24945,0.42842 -0.56405,0.41791 h -2.42271 l -0.21387,0.95629 h 1.45898 l -0.79238,3.71913 h 1.9358 l 0.788,-3.70512 0.0188,-0.0162 h 0.004 c 0.97938,0 1.53233,-0.30784 1.65882,-0.92387 l 0.19941,-0.94709 z m 11.34925,0 -1.18725,5.58834 h 1.95071 l 0.62277,-2.91135 1.08953,2.90697 h 0.7297 c 0.95666,0.0127 1.51119,-0.29715 1.66365,-0.92869 l 0.87564,-4.18348 -1.97481,0.0158 -0.47683,2.33574 -1.32793,-2.82286 z m -17.68564,0.47618 c -0.9534,0 -1.49654,0.31093 -1.62946,0.93306 l -0.68107,3.25523 c -0.12974,0.61288 0.2851,0.91949 1.24511,0.91949 h 3.16689 l 0.18495,-0.89627 h -2.68524 l 0.68588,-3.26399 h 2.72424 l 0.20423,-0.94752 z m 8.96203,0.01 -1.07024,5.10296 h 1.95071 l 1.07024,-5.10296 z m -19.16435,0.009 -0.88047,4.17909 h -1.43969 l 0.76828,-3.72395 h -2.42228 l -0.19985,0.95628 h 0.48691 l -0.57894,2.76723 c -0.1395,0.60543 0.28217,0.91336 1.26482,0.92387 h 2.43192 c 0.96963,-0.008 1.52259,-0.30758 1.65883,-0.90504 l 0.86118,-4.19748 z m 23.31689,0

c -0.97299,0 -1.52275,0.30483 -1.64918,0.91466 l -0.68106,3.26881 c -0.12323,0.60678 0.29845,0.91029 1.26482,0.91029 h 2.42272 c 0.97938,0 1.53549,-0.30953 1.66846,-0.92869 l 0.67624,-3.24558 c 0.12974,-0.61289 -0.29193,-0.91949 -1.26482,-0.91949 z m -32.65013,0.0109 -0.18012,0.91117 h 0.97776 l -0.22351,0.92868 h -0.93876 l -0.19503,0.93789 0.9633,0.0162 -0.19459,0.91905 c -0.0616,0.28915 -0.24948,0.42842 -0.56404,0.41791 h -0.48165 l -0.19941,0.94753 1.45941,0.0158 c 0.97939,0 1.53235,-0.30785 1.65882,-0.92388 l 0.29145,-1.38821 h 2.42754 l 0.19458,-0.9427 h -2.44199 l 0.19459,-0.91467 h 2.442 l 0.1994,-0.92387 z m 32.95166,0.91948 h 1.45941 l -0.68589,3.24998 h -1.45459 z m -34.9738,0.71492 c -0.16544,0 -0.32601,0.0408 -0.48165,0.12135 -0.15524,0.0805 -0.27671,0.19536 -0.36464,0.34387 -0.088,0.14852 -0.13191,0.30461 -0.13191,0.46873 0,0.16089 0.0422,0.31583 0.12709,0.46434 0.0876,0.14521 0.209,0.25987 0.36464,0.34388 0.15564,0.0805 0.31782,0.1209 0.48647,0.1209 0.16861,0 0.33116,-0.0403 0.48691,-0.1209 0.15518,-0.0839 0.27509,-0.19833 0.35981,-0.34388 0.088,-0.14851 0.13192,-0.30345 0.13192,-0.46434 0,-0.16412 -0.0439,-0.32021 -0.13192,-0.46873 -0.0875,-0.14818 -0.20943,-0.26292 -0.36507,-0.34387 -0.15565,-0.0805 -0.31621,-0.12135 -0.48165,-0.12135 z m 21.34076,8.03403 -0.41197,0.22692 c -0.65104,0.35823 -1.61488,1.03216 -2.50116,1.63045 h 1.44582 1.50324 l -0.0259,0.14763 c -0.0141,0.0812 -0.11138,0.55736 -0.2165,1.05835 l -0.19109,0.91073 0.41898,-0.26546 c 2.39869,-1.52086 4.26049,-2.12678 6.53538,-2.12678 0.6894,0 0.74242,-0.005 0.72444,-0.0687 -0.0107,-0.0379 -0.077,-0.39209 -0.14769,-0.7872 l -0.12885,-0.71842 h -1.85429 -1.85429 l 0.0263,0.14763 c 0.0145,0.0812 0.0361,0.18053 0.0482,0.22121 0.012,0.0399 0.006,0.0663 -0.0123,0.057 -0.0186,-0.009 -0.15094,-0.0738 -0.29407,-0.14324 -0.45028,-0.21956 -0.68282,-0.25169 -1.93537,-0.27204 z m -2.91313,1.85737 h -0.0574 l -0.009,0.0434 c 0.0239,-0.0162 0.043,-0.0275 0.0666,-0.0434 z m -0.0666,0.0434 c -0.44858,0.30382 -0.64161,0.38527 -1.24729,0.81305 -1.35454,0.95663 -2.19674,1.5236 -2.81935,1.89811 -0.22816,0.13725 -0.43597,0.26845 -0.46193,0.29219 -0.0259,0.0236 -0.21779,0.8665 -0.42642,1.87314 -0.20864,1.00664 -0.38968,1.87039 -0.40189,1.91871 -0.0123,0.0487 -0.004,0.0885 0.0166,0.0885 0.0212,0 0.25688,-0.0894 0.52417,-0.19888 0.97279,-0.39842 2.18898,-1.05609 3.39654,-1.83767 0.24989,-0.16165 0.45593,-0.29963 0.46017,-0.29963 0.003,-0.003 0.20442,-0.95324 0.44703,-2.11145 0.24259,-1.15821 0.45951,-2.18999 0.48165,-2.29281 z m -15.21735,-1.89417 0.0271,0.18705 c 0.0522,0.36095 0.0455,0.35234 0.3826,0.48187 0.94908,0.36464 2.10075,0.6549 3.17477,0.79946 h 4.4e-4 c 0.29104,0.0398 0.37516,0.0295 0.37516,-0.0425 0,-0.0386 -0.0535,-0.36114 -0.11877,-0.71667 -0.0651,-0.35552 -0.11833,-0.66072 -0.11833,-0.67767 0,-0.017 -0.83753,-0.0315 -1.86131,-0.0315 z m 5.7579,0.0398 -0.24763,0.73813 c -0.13619,0.40588 -0.24757,0.75187 -0.24718,0.76967 10e-6,4.7e-4 2.6e-4,0.002 4.5e-4,0.002 1.3e-4,2.2e-4 6.9e-4,4.8e-4 8.4e-4,0.002 0.0556,0.0521 1.45538,-0.0625 2.13303,-0.17654 0.97437,-0.16394 2.11667,-0.499 3.06872,-0.90021 0.41741,-0.17598 0.48641,-0.21811 0.52152,-0.31847 l 0.0404,-0.11565 h -2.63484 z m 25.52441,0 -2.01557,4.4e-4 -2.01601,4.3e-4 -0.29101,0.84809 c -0.16008,0.46632 -0.28502,0.84969 -0.27654,0.85816 0.008,0.006 0.14775,0.0406 0.31073,0.0771 0.1898,0.0424 0.4647,0.0665 0.76477,0.0661 0.43074,0 0.46622,0.005 0.44001,0.0683 -0.0782,0.18929 -0.0773,0.19064 0.323,0.34738 0.51807,0.20294 1.42044,0.65547 2.02784,1.01718 0.26313,0.15668 0.49395,0.28585 0.5246,0.28605 2.2e-4,-1e-5 8.5e-4,3e-5 0.002,0 l 4.3e-4,-4.3e-4 h 4.4e-4 c 0.0366,-0.01 1.22469,-2.97786 1.20128,-3.0016 -0.008,-0.008 -0.23526,-0.13896 -0.50488,-0.29087 z m 6.71462,0.002 0.18013,0.0937 c 0.32515,0.16877 1.19283,0.52326 1.72544,0.70484 0.81596,0.27813 1.77534,0.50001 2.71722,0.62818 v 0.002 c 0.32482,0.0442 0.30064,0.0802 0.46149,-0.68205 0.0799,-0.37884 0.14644,-0.70231 0.14814,-0.71842 0.002,-0.0161 -1.17508,-0.0288 -2.61468,-0.0281 z m -5.65315,0.62467 c -0.0216,0.002 -0.0283,0.0209 -0.036,0.053 -0.0529,0.22379 -0.21739,1.06958 -0.21739,1.11705 0,0.0432 0.17748,0.0534 0.96813,0.0534 0.54529,0 0.96768,0.0156 0.96768,0.0359 0,0.042 -0.35054,1.72457 -0.37997,1.82409 -0.0187,0.0602 -0.10133,0.0687 -1.00538,0.0687 h -0.98521 v 0.0955 c 0,0.0773 0.0851,0.15374 0.44484,0.40126 0.24446,0.16819 0.92442,0.64461 1.51112,1.0588 1.18233,0.83472 1.8008,1.25322 2.56778,1.73735 0.27588,0.17416 0.51897,0.31544 0.54915,0.31408 h 4.4e-4 c 1.9e-4,-5e-5 8.5e-4,6e-5 0.002,0 0.0204,-0.006 0.11836,-0.39905 0.21825,-0.87393 l 0.18188,-0.86342 1.56854,-0.0197 1.56898,-0.0192 -0.47333,-0.21158 c -1.02007,-0.45615 -2.28054,-1.2428 -4.42513,-2.76154 -1.13093,-0.8009 -2.11044,-1.4655 -2.71548,-1.84248 -0.19651,-0.12223 -0.27435,-0.16982 -0.31029,-0.16734 z m -37.29133,1.11048 c -0.008,0.008 0.19349,1.03315 0.44791,2.27791 l 0.46236,2.26303 0.45273,0.30707 c 1.03317,0.70046 2.57483,1.55138 3.48857,1.92572 0.33377,0.13648 0.34834,0.13918 0.32782,0.057 -0.0117,-0.0472 -0.0335,-0.1616 -0.0482,-0.25408 l -0.0267,-0.16822 0.76565,0.0109 0.76563,0.0109 0.10431,0.50377 c 0.0572,0.27703 0.10985,0.50975 0.11834,0.51822 0.008,0.008 0.28147,0.0683 0.60742,0.13362 0.57434,0.11571 1.39162,0.22908 1.83676,0.25451 v -0.002 c 0.11951,0.007 0.48938,0.0108 0.82175,0.0192 l 0.60436,0.009 -0.0215,-0.12791 c -0.0117,-0.0704 -0.0485,-0.34041 -0.0815,-0.60015 -0.033,-0.25982 -0.0692,-0.52934 -0.0806,-0.5997 l -0.021,-0.12748 1.11275,-0.0101 1.11275,-0.0101 0.46894,-1.18102 c 0.25792,-0.6495 0.46909,-1.1832 0.46938,-1.1832 2.2e-4,-0.002 -0.14636,0.0467 -0.32563,0.10602 -0.48241,0.16038 -1.05392,0.30218 -1.61018,0.39907 -0.42601,0.0743 -0.64269,0.086 -1.57994,0.0867 -0.70307,0 -1.20528,-0.0177 -1.42259,-0.0508 -1.19606,-0.18361 -2.30571,-0.54428 -3.47937,-1.13064 l -0.42381,-0.21158 -0.19151,0.47048 c -0.10534,0.25888 -0.20212,0.47047 -0.21475,0.47047 -0.0126,0 -0.0748,-0.29435 -0.13806,-0.65446 -0.0935,-0.53176 -0.12977,-0.66473 -0.19414,-0.70966 -0.0437,-0.0302 -0.47964,-0.32058 -0.96856,-0.64482 -0.48893,-0.32433 -1.39196,-0.94583 -2.00681,-1.38121 -0.61485,-0.43538 -1.12399,-0.78515 -1.13247,-0.77668 z m 29.02877,4.68505 c -0.68528,0.0194 -1.23591,0.072 -1.23591,0.14632 0,0.0529 0.62654,3.08166 0.63943,3.09183 0.007,0.006 0.15486,-0.003 0.32869,-0.0202 0.86166,-0.0822 2.116,-0.007 2.94337,0.17698 v -0.002 c 0.57381,0.12784 0.56125,0.12846 0.53512,-0.0254 -0.0204,-0.12054 -0.21913,-1.59603 -0.21913,-1.62739 0,-0.008 0.49969,-0.016 1.11056,-0.0245 l 1.111,-0.0101 0.15251,-0.39206 c 0.11387,-0.29254 0.13983,-0.40027 0.10387,-0.42316 -0.0858,-0.0553 -1.22183,-0.41641 -1.62858,-0.51779 -0.59755,-0.14911 -1.16295,-0.24984 -1.79732,-0.31935 -0.53817,-0.059 -1.35835,-0.0728 -2.04361,-0.0535 z m -3.2037,0.55152 c -0.24598,-8.4e-4 -1.85466,0.64325 -2.73389,1.10698 -0.39458,0.20802 -1.29729,0.71926 -1.33757,0.75741 -0.008,0.008 -0.081,0.33359 -0.16304,0.72455 l -0.14901,0.71054 h 1.87182 c 1.0296,0 1.87182,-0.0109 1.87182,-0.0245 0,-0.0136 0.15144,-0.74616 0.33615,-1.62827 0.18471,-0.88212 0.3357,-1.61652 0.3357,-1.63178 0,-0.01 -0.0111,-0.0147 -0.032,-0.0149 z m 11.76471,1.93054 c -0.0243,8.4e-4 -0.0279,0.0222 -0.0315,0.0525 -0.008,0.0603 0.009,0.18726 0.0355,0.28212 0.13803,0.49115 0.64369,0.82621 1.47256,0.97599 v 0.003 c 0.15572,0.0288 0.3544,0.0539 0.44133,0.0539 h 0.15778 l -0.0986,-0.0784 c -0.0543,-0.0434 -0.5255,-0.36397 -1.04701,-0.71229 -0.6719,-0.4488 -0.86808,-0.57897 -0.92999,-0.57648 z m -23.00484,4.57247 c -0.004,-0.004
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2.6059,1.44253 0,-0.0347 0.19933,-0.99489 0.22045,-1.06185 0.02,-0.0602 0.10907,-0.0683 1.12327,-0.0683 0.60569,0
1.10771,0.0132 1.1167,0.0293 0.009,0.0163 -0.0693,0.42846 -0.17356,0.91555 -0.10431,0.48709 -0.1908,0.90629 -0.19152,0.93087 -
0.003,0.0822 1.26434,0.35716 2.13215,0.46303 v 8.5e-4 c 0.24989,0.0305 0.51543,0.0495 0.58989,0.0495 l 0.13499,-0.009 0.25419,-
1.20029 c 0.26364,-1.2451 0.31057,-1.66081 0.20818,-1.84993 -0.0756,-0.13978 -0.11321,-0.1577 -0.35763,-0.1577 -0.54796,-0.003 -
1.61986,-0.20653 -2.41745,-0.45821 z m -44.81893,0.0482 c -0.3086,0 -0.48235,0.0105 -0.49918,0.0303 -1.2e-4,1.8e-4 -7.8e-4,8.5e-4 -
8.4e-4,0.002 l -4.4e-4,4.4e-4 v 4.4e-4 4.3e-4 c 0,1.9e-4 -3e-5,8.5e-4 0,0.002 v 4.4e-4 4.4e-4 c 9e-5,1.9e-4 3.1e-4,8.5e-4 4.4e-4,0.002
2.8e-4,3.7e-4 0.002,0.002 0.002,0.002 0.0211,0.021 0.21571,0.16155 0.43257,0.31278 0.21348,0.14887 0.40522,0.26896
0.43081,0.26459 h 4.4e-4 4.4e-4 l 4.4e-4,-4.4e-4 c 0.0211,-0.01 0.0673,-0.14897 0.10343,-0.31278 l 0.0666,-0.30225 z m
6.56386,0.20983 c -0.007,0.007 -0.15823,0.70739 -0.33658,1.55687 -0.26482,1.26688 -0.31458,1.55076 -0.26997,1.57789
0.1326,0.0817 1.15349,0.28077 1.90513,0.37148 0.24579,0.0297 0.55751,0.0582 0.69289,0.0582 0.22451,0.003 0.24835,-0.004
0.26734,-0.0845 0.0611,-0.26016 0.61269,-2.99516 0.61269,-3.03839 0,-0.0373 -0.1326,-0.0576 -0.50356,-0.078 -0.62615,-0.0337 -
1.44715,-0.14991 -1.96298,-0.27816 -0.21583,-0.054 -0.39902,-0.0939 -0.40496,-0.0854 z m 20.90163,0.44551 -0.23404,0.0245 -
0.14769,0.67417 c -0.081,0.37087 -0.14722,0.68743 -0.14682,0.70353 3.5e-4,0.0161 0.49995,0.0298 1.11012,0.0298 h 1.10924 l -
0.0219,0.1082 c -0.0122,0.0596 -0.0937,0.44238 -0.18101,0.85071 -0.0873,0.40826 -0.15865,0.74323 -0.15865,0.75171 0,0.005
0.27122,0.0197 0.60261,0.0324 0.58546,0.0225 1.50759,0.15496 1.94545,0.27949 0.32126,0.0914 0.32492,0.0899 0.37691,-0.16296
0.1467,-0.71402 0.38954,-1.82927 0.40057,-1.84029 0.007,-0.007 0.22669,-0.0324 0.48734,-0.057 0.56169,-0.052 0.85695,-0.12264
1.20391,-0.28692 l 0.25594,-0.12135 -0.17705,-0.0754 c -0.88234,-0.37587 -2.2441,-0.73142 -3.3549,-0.87612 -0.46715,-0.061 -
2.58788,-0.0846 -3.07003,-0.0346 z m -25.10763,0.014 c -1.0728,0 -1.85166,0.0152 -1.85166,0.0355 0,0.0923 1.42941,0.95192
2.46961,1.48546 0.4346,0.22294 0.80326,0.40618 0.81955,0.40652 0.0163,8.5e-4 0.0299,-0.021 0.0307,-0.0481 6.5e-4,-0.027 0.0873,-
0.46089 0.1924,-0.96417 l 0.19108,-0.91511 z m 24.64613,0.0521 c -0.022,-0.022 -0.88822,0.14135 -1.31084,0.24706 -0.46108,0.1153
-1.27823,0.36424 -1.28411,0.39119 0.003,0.009 0.55779,0.0175 1.23634,0.0175 h 1.23634 l 0.0675,-0.3211 c 0.0373,-0.17666
0.0616,-0.3262 0.0548,-0.33467 z m -34.81163,0.0754 c -0.0192,-1.3e-4 -0.0337,0.002 -0.0425,0.005 -0.0271,0.0104 -0.0616,0.10443 -
0.0767,0.20895 -0.015,0.10453 -0.0372,0.22942 -0.0487,0.27774 l -0.0206,0.0881 0.74768,0.0109 0.74723,0.0109 -0.14944,0.62993 c
-0.0823,0.34637 -0.15923,0.66966 -0.17136,0.71798 -0.0217,0.0878 -0.0282,0.0885 -0.74242,0.0885 h -0.7205 l -0.11088,0.53574 c -
0.0611,0.29441 -0.10446,0.54173 -0.096,0.55021 0.008,0.008 0.18894,0.051 0.40101,0.0942 1.05222,0.21574 2.14015,0.62985
3.24226,1.23358 v -0.002 c 0.26009,0.1425 0.47975,0.25248 0.48823,0.244 0.008,-0.008 0.0723,-0.28621 0.142,-0.61766 l 0.12622,-
0.60278 1.83106,-0.0197 1.83106,-0.0192 -0.64775,-0.43631 c -0.35642,-0.23998 -0.83826,-0.55442 -1.07068,-0.6987 l -0.42292,-
0.2624 h -0.61445 c -0.69945,0 -0.65207,0.0316 -0.55528,-0.37235 0.0299,-0.12499 0.0498,-0.23028 0.0455,-0.24443 l -4.4e-4,-4.5e-4
v -4.4e-4 c -0.0526,-0.0524 -1.19205,-0.55976 -1.65093,-0.73506 -0.78696,-0.30064 -2.17366,-0.68281 -2.46084,-0.68469 z m
30.31199,1.40442 c -0.17233,-0.0103 -2.93463,1.66751 -3.25804,1.98573 -0.12171,0.11978 -0.58266,2.30276 -0.55221,2.61522
0.0383,0.39325 0.28904,0.67198 0.75249,0.83626 v -4.4e-4 c 0.27741,0.0988 0.34453,0.0944 0.52198,-0.0333 0.0847,-0.0609
0.4917,-0.34645 0.90457,-0.63475 0.41287,-0.2883 0.75351,-0.52918 0.75775,-0.52918 0.0119,-0.009 0.88179,-4.18465 0.88179,-
4.23297 0,-0.004 -0.003,-0.006 -0.008,-0.007 z m 13.02341,0.12573 -0.34404,1.6414 c -0.32277,1.5401 -0.33958,1.64479 -
0.27479,1.69311 0.038,0.0283 0.36241,0.24218 0.72095,0.4753 0.35854,0.23303 0.90879,0.60367 1.22275,0.82399 0.30168,0.21171
0.56098,0.37429 0.59297,0.37235 2.3e-4,-4e-5 8.4e-4,-3.8e-4 0.002,-4.4e-4 h 4.4e-4 v -4.3e-4 h 4.4e-4 l 4.4e-4,-4.5e-4 c 0.0501,-0.0596
0.71627,-3.36573 0.68413,-3.3954 -0.14671,-0.13716 -1.74909,-1.14225 -2.30658,-1.44691 z m -46.44357,0.20019 c -0.73787,0.0259 -
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0.0856 1.45153,-0.10251 l 0.99704,-0.0228 -0.0249,-0.10338 c -0.17353,-0.72249 -0.66828,-1.13553 -1.40901,-1.17707 -0.0505,-0.003
-0.10026,-0.003 -0.14945,-0.002 z m 50.11972,2.20782 c -0.0144,2e-5 -0.0223,0.007 -0.0267,0.0175 -0.0541,0.13461 -
0.38655,1.82589 -0.38655,1.96602 0,0.52031 0.38204,0.8736 1.09741,1.01542 v -0.005 c 0.19924,0.0399 0.76919,0.057
2.27985,0.057 l 2.01381,0.008 -0.61137,-0.30708 c -1.00426,-0.50447 -1.91027,-1.07396 -3.57491,-2.24726 -0.54491,-0.38405 -
0.72896,-0.50471 -0.79151,-0.50464 z m -20.79994,0.0802 c 0,-0.0267 -0.34251,0.19319 -0.76082,0.48844 -1.05396,0.74402 -
2.12449,1.44248 -2.75229,1.79605 -0.29149,0.1642 -0.54504,0.30166 -0.56361,0.31014 -8.5e-4,3e-4 -0.002,6.6e-4 -0.003,0.002 -
0.0137,0.0154 1.1e-4,0.10231 0.0338,0.20062 l 0.0653,0.19056 0.93307,-0.0171 c 1.05068,-0.0192 1.3321,-0.0646 1.81178,-0.29131
0.34646,-0.16378 0.62108,-0.42609 0.77529,-0.74075 0.0756,-0.1542 0.16951,-0.5023 0.28618,-1.05924 0.0956,-0.45648 0.17399,-
0.85162 0.17399,-0.87875 z m -18.00602,0.0434 c -0.0493,0.0491 -0.36581,1.67045 -0.36508,1.87008 0.003,0.61933 0.51866,0.97919
1.52034,1.06142 0.39703,0.0365 3.65467,0.0573 3.65467,0.0276 0,-0.009 -0.22047,-0.12166 -0.48998,-0.25144 -0.94516,-0.45521 -
2.08801,-1.16442 -3.59375,-2.22973 -0.38807,-0.27448 -0.71475,-0.48979 -0.7262,-0.47792 z m 56.75544,0.11959 c -0.008,-0.008 -
0.38664,0.24724 -0.84278,0.56597 -1.05005,0.7336 -1.85523,1.24788 -2.59583,1.6585 -0.32052,0.17776 -0.58245,0.3388 -
0.58245,0.35745 0,0.0187 0.0212,0.0808 0.0473,0.13756 l 0.0473,0.10381 0.93043,-0.0158 c 1.01739,-0.0174 1.30757,-0.0617
1.76927,-0.26897 0.32407,-0.14547 0.6509,-0.4562 0.80114,-0.76179 0.0797,-0.16217 0.17015,-0.48601 0.27874,-1.00141 0.0882,-
0.41893 0.1536,-0.76688 0.14681,-0.77536 z" /><g
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5.5,5.5 0 0 0 -5.50044,-5.499999 z m -32.3696,78.484099 a 3.4365389,3.4365389 0 0 0 -3.43648,3.4365 3.4365389,3.4365389 0 0 0
3.43648,3.4365 3.4365389,3.4365389 0 0 0 3.43648,-3.4365 3.4365389,3.4365389 0 0 0 -3.43648,-3.4365 z m 64.68391,0 a
3.4365389,3.4365389 0 0 0 -3.43649,3.4365 3.4365389,3.4365389 0 0 0 3.43649,3.4365 3.4365389,3.4365389 0 0 0 3.43648,-3.4365
3.4365389,3.4365389 0 0 0 -3.43648,-3.4365 z m -32.30088,50.7339 a 3.4365389,3.4365389 0 0 0 -3.43699,3.4365
3.4365389,3.4365389 0 0 0 3.43699,3.437 3.4365389,3.4365389 0 0 0 3.43649,-3.437 3.4365389,3.4365389 0 0 0 -3.43649,-3.4365 z"
/></g></g></g><path
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3.02307,4.44955 -3.02308,-4.44955 h -1.07312 v 9.16083 h 1.32177 z m 9.17393,2.53886 h 4.14855 v -1.33486 h -4.14855 v -2.57812
h 4.24015 v -1.32178 h -5.57501 v 9.14774 h 5.5881 v -1.32177 h -4.25324 z m 8.1924,3.95224 v -7.96992 h 2.76134 v -1.23017 h -
6.85753 v 1.23017 h 2.77442 v 7.96992 z m 5.53577,-7.21088 1.12547,3.23247 h -2.27712 z m 2.73516,7.17162 h 1.41339 l -
3.67742,-9.27861 h -0.96843 l -3.67742,9.27861 h 1.42647 c 0.28791,-0.75904 0.74596,-1.85834 1.0993,-2.6959 h 3.28481 z m
0.96843,-4.55424 c 0,1.26943 0.52348,2.42108 1.30869,3.25864 0.7983,0.83756 1.92378,1.36104 3.16703,1.36104 2.13317,0
```

3.80829,-1.41339 4.33176,-3.4942 h -1.2956 c -0.47113,1.38721 -1.64895,2.18551 -3.03616,2.18551 -0.87682,0 -1.67512,-0.37952 -2.25095,-0.98152 -0.57582,-0.602 -0.94225,-1.41338 -0.94225,-2.32947 0,-0.94225 0.36643,-1.75364 0.94225,-2.35564 0.57583,-0.602 1.37413,-0.98152 2.25095,-0.98152 1.34795,0 2.47343,0.69361 2.95764,2.06773 h 1.30869 c -0.52348,-2.06773 -2.17243,-3.37642 -4.26633,-3.37642 -1.24325,0 -2.36873,0.53657 -3.16703,1.36104 -0.78521,0.83756 -1.30869,1.98921 -1.30869,3.28481 z m 11.26781,0.53657 3.33716,4.01767 h 1.71438 l -2.8922,-3.44185 c 1.57043,-0.27483 2.36873,-1.50499 2.36873,-2.8006 0,-1.45264 -0.99461,-2.91838 -2.98382,-2.91838 h -3.33716 v 9.16083 h 1.30869 v -7.82596 h 2.02847 c 1.11239,0 1.66204,0.7983 1.66204,1.58351 0,0.7983 -0.54965,1.5966 -1.66204,1.5966 h -1.54425 z m 9.99839,-5.13007 h -4.72437 v 1.30869 h 1.70129 v 6.51728 h -1.70129 v 1.32177 h 4.72437 v -1.32177 h -1.7013 v -6.51728 h 1.7013 z m 1.97613,0 h -1.32178 v 9.14774 h 5.62737 v -1.32177 h -4.30559 z m 8.54575,1.97612 1.12548,3.23247 h -2.27712 z m 2.73517,7.17162 h 1.41338 l -3.67742,-9.27861 h -0.96843 l -3.67742,9.27861 h 1.42647 c 0.28792,-0.75904 0.74596,-1.85834 1.0993,-2.6959 h 3.28482 z m 4.09618,0.0393 v -7.96992 h 2.76134 v -1.23017 h -6.85754 v 1.23017 h 2.77443 v 7.96992 z m 9.76283,-4.01767 h 1.37412 c 1.19091,0 1.60969,0.77212 1.66204,1.30869 0.0654,0.75904 -0.61509,1.55734 -2.06773,1.55734 -1.15165,0 -1.75365,-0.65435 -1.87143,-1.38721 l -1.23017,0.22247 c 0.20939,1.25635 1.13856,2.39491 3.1016,2.39491 2.1986,0 3.2979,-1.33487 3.31098,-2.65664 0.0131,-0.903 -0.35334,-1.64895 -1.17782,-2.13317 0.66743,-0.44495 0.83756,-1.19091 0.83756,-1.91069 -0.0131,-1.77981 -1.57042,-2.5912 -3.04924,-2.5912 -1.21709,0 -2.43417,0.68052 -2.76134,2.06773 l 1.19091,0.28791 c 0.17013,-0.70669 0.87682,-1.0993 1.57043,-1.0993 0.86373,0 1.74055,0.41878 1.74055,1.33486 0,0.602 -0.34026,1.36104 -1.25634,1.36104 h -1.37412 z m 6.36022,-2.47343 2.3949,3.55964 h 0.74596 l 2.3949,-3.55964 v 6.45184 h 1.30869 v -9.16083 h -1.04695 l -3.02308,4.44955 -3.02307,-4.44955 h -1.07313 v 9.16083 h 1.32178 z m 9.52727,0 2.3949,3.55964 h 0.74595 l 2.39491,-3.55964 v 6.45184 h 1.30869 v -9.16083 h -1.04696 l -3.02307,4.44955 -3.02307,-4.44955 h -1.07313 v 9.16083 h 1.32178 z"
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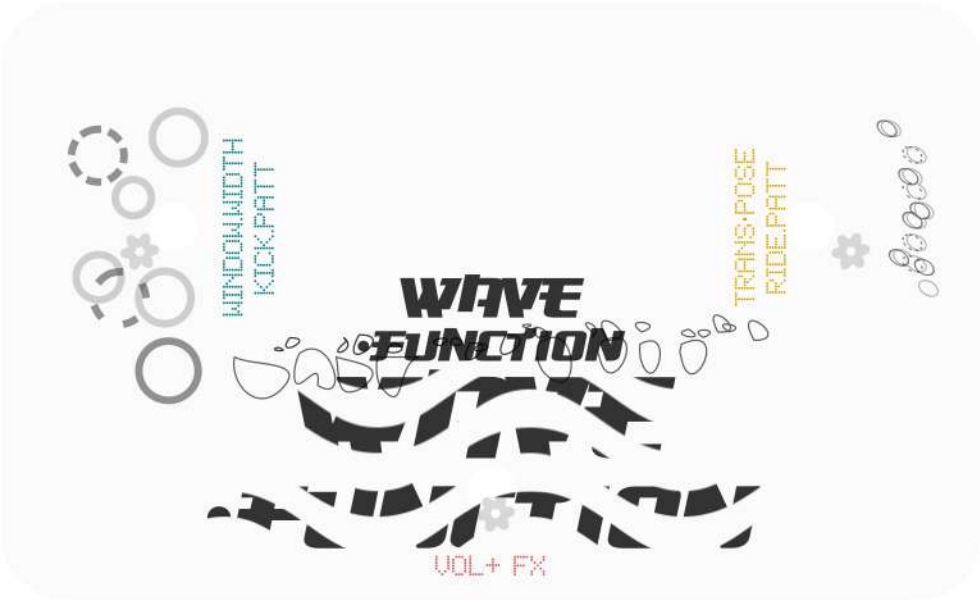
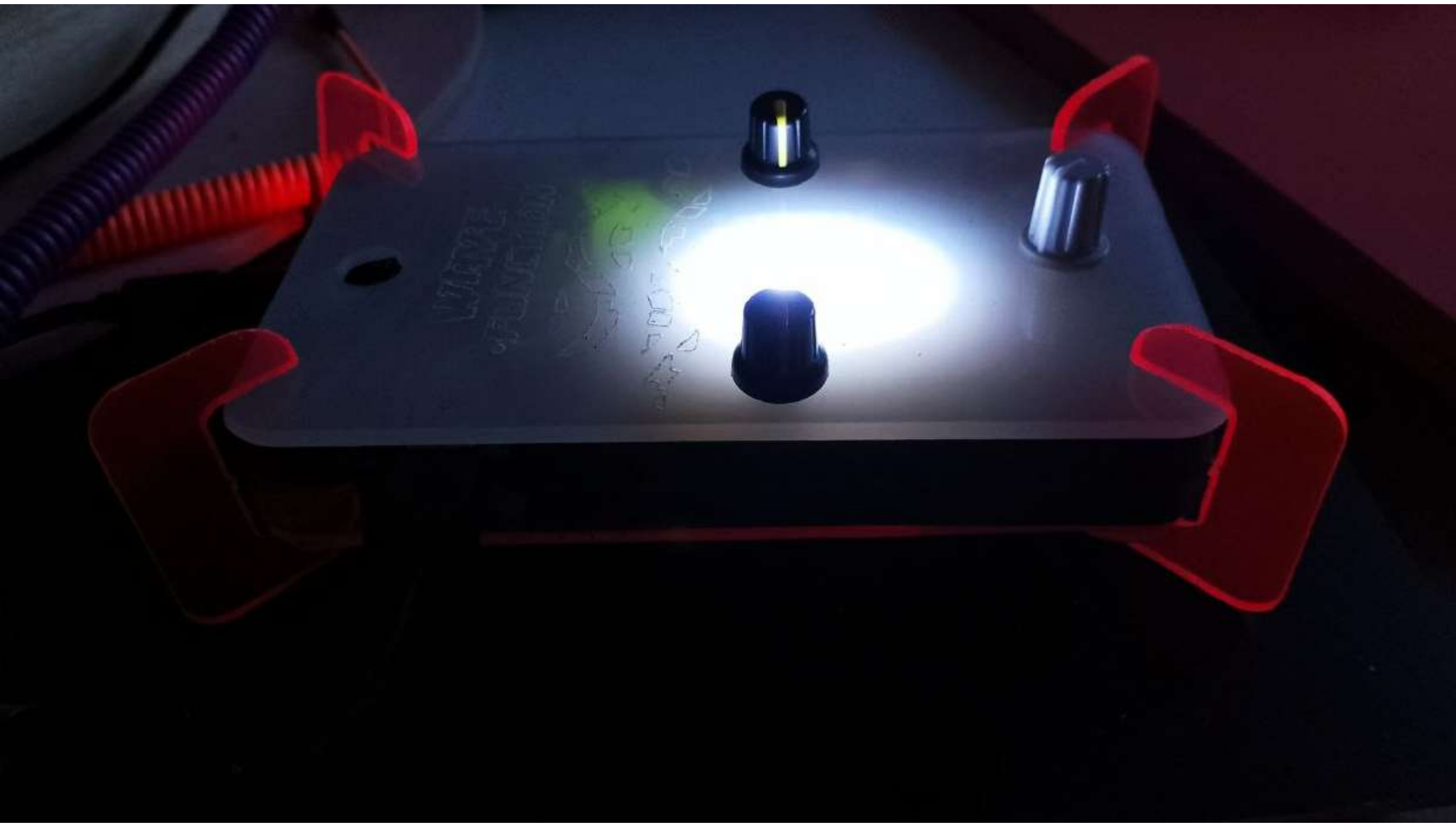
WAVE
FUNCTION

STICKER



100 MM



WAVE •FUNCTION

Wave Function is the name of this unconventional instrument.
All coding elements to build it (from scratch) are documented in this file.
The type of algorythmic language used (pd/gnrtv.cells) allows
a high sonic complexity in a very tiny and compact computation platform,
therefore a nice permacompùting / low consumption platform.

It works as a 'pedal' or a post FX processor unit that deconstructs and texturizes
the incoming audio signal with granular synthesis techniques.

In addition when no audio input is connected, it behaves as a generative pattern synthesizer
creating minimal steps and patterns crossing over musical genres
like (minimal/deep/dub)techno, glitch and soundscaping.

The instrument approach drifts among generative states
which are emerging with several sonic ingredients.
Therefore as performers we do not control one parameter of a certain sonic event :
rather we drift and flow about growing sonic structures combining
texturization and nonconventional sounds and beats
from chill states to other more oversized hi intensity states..

Wave Function is a borrowed term from Quantum Mechanics.
In the quantum domain, any particle has its own wavefunction
that describes multiple superpositions / probabilities of states
(position / velocity / spin etc of the particle) until its measured or observed,
where this wavefunction (with multiple states) collapses in a single 'definite' one.

This weird and shocking behaviour of nature at microtinyscales creates the quantum theory
tested many times that however in some elements still creates too many questions
respect to to Einsteins Relativistic Large scales physical laws..
Definitely a nice moment in which human knowledge
is still figuring out about a full comprehension of the fundamentals of nature.

. ..具.. .

Wave·Function
R+D OH·Lab
Code + Design + Engineering + Sonic Design
by Xa.Manzanares & OH·Lab
jul 2026

. ..具.. .

Espais Latents

https://oneshaptiques.space/espaislatents

Open Sourced Code available at :
https://github.com/xamanza/espais.latents

Agraïments

A les comunitats d'arts electròniques que participo i que tant inspiren <3 :
Algoritmics Barcelona / Toplap, Xarxa d'Artistes Digitals, Befaco, Telenoika, ODD ...

Espais latents és un procés de recerca
fruit de les beques Crea25 / Institut de Cultura de Barcelona

Coproduït per B.Crea'25 & Ones Hàptiques LAB

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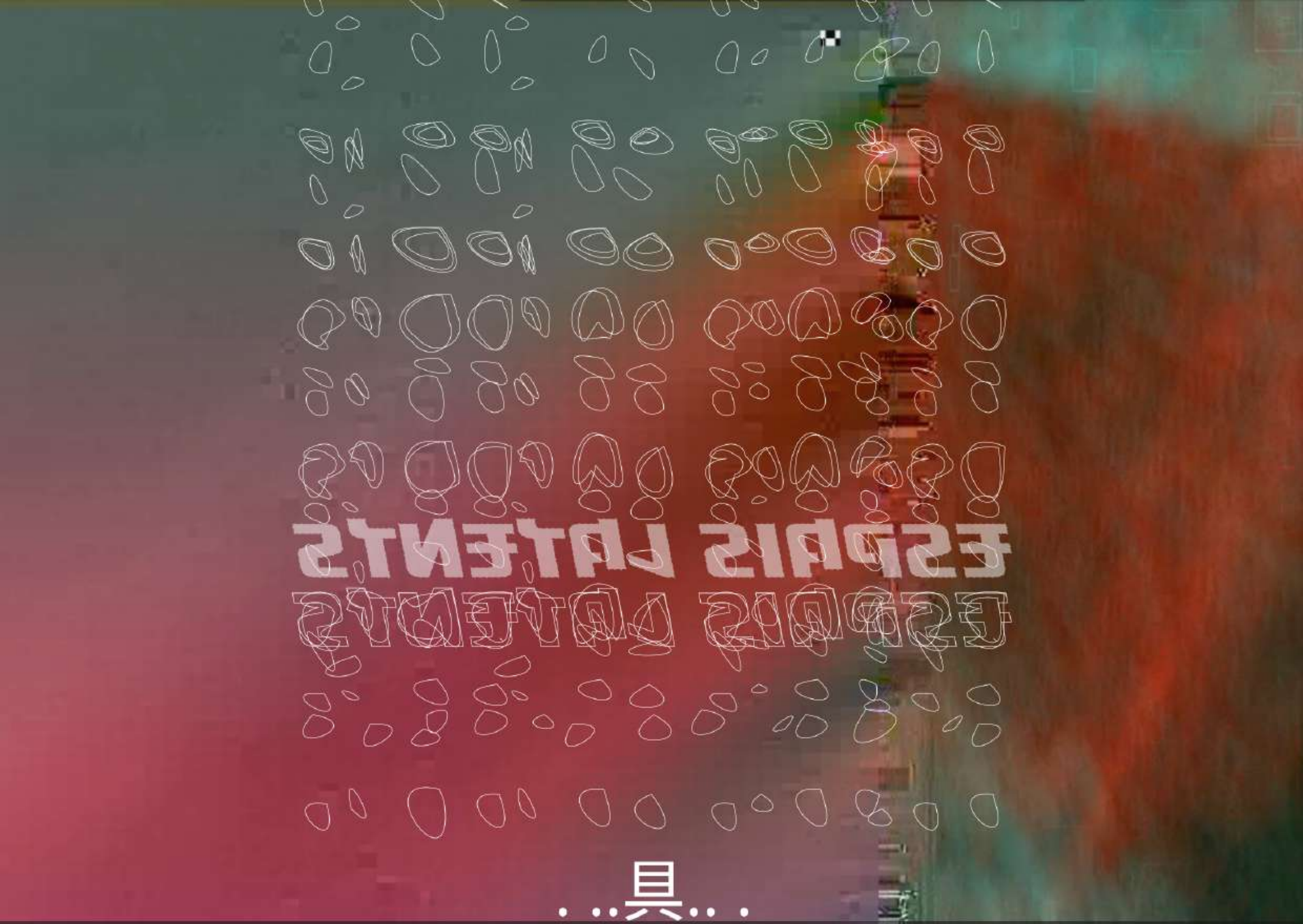
Transdisciplinary Researcher
Expertise in Sonic Algorythmic Design and Generative Automats for installations
Algorythmic music maker as Xaniah and dAAX
Ones Hàptiques Lab founder



En una narrativa mesclant disseny especulatiu, ciènci(es) i arts electrònques,
Espais Latents s'aproxima a diverses àrees del coneixement,
per a generar preguntes en vers la fenomenologia de la intel·ligència i la consciència :
des dels organismes unicel·lulars, als organismes complexos com l'ésser humà,
així com altres formes d'intel·ligències sintètiques
que poden emergir del context algorítmic i del darrer impacte IA.

Xa Manzanares